



MS-A1004

4 CHANNEL Digital Signal Processing Amplifier

SERVICE MANUAL



JBL Consumer Products
8500 Balboa Blvd.
Northridge, CA. 91329

Rev1 5/2011

Released 2011
Discontinued XXXX

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MS-A1004 Specifications

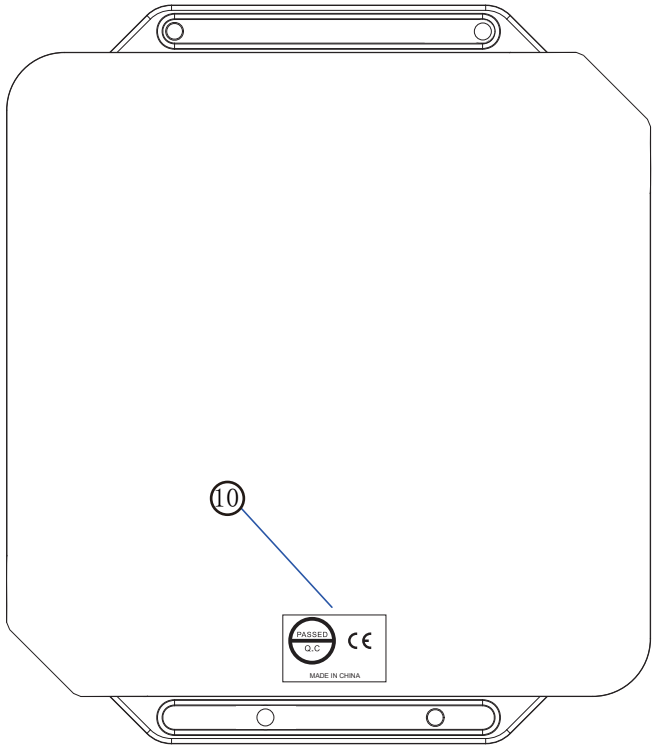
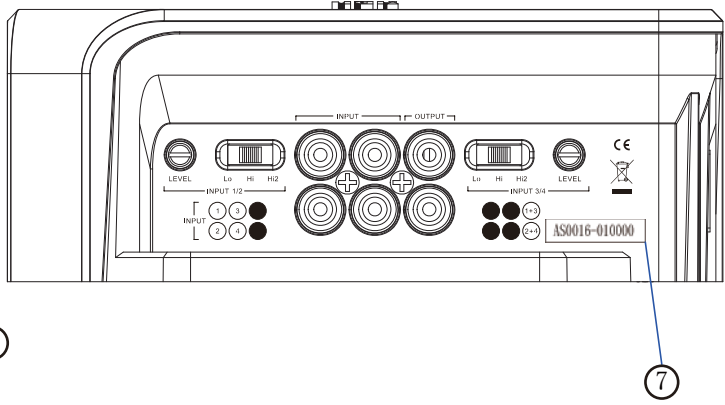
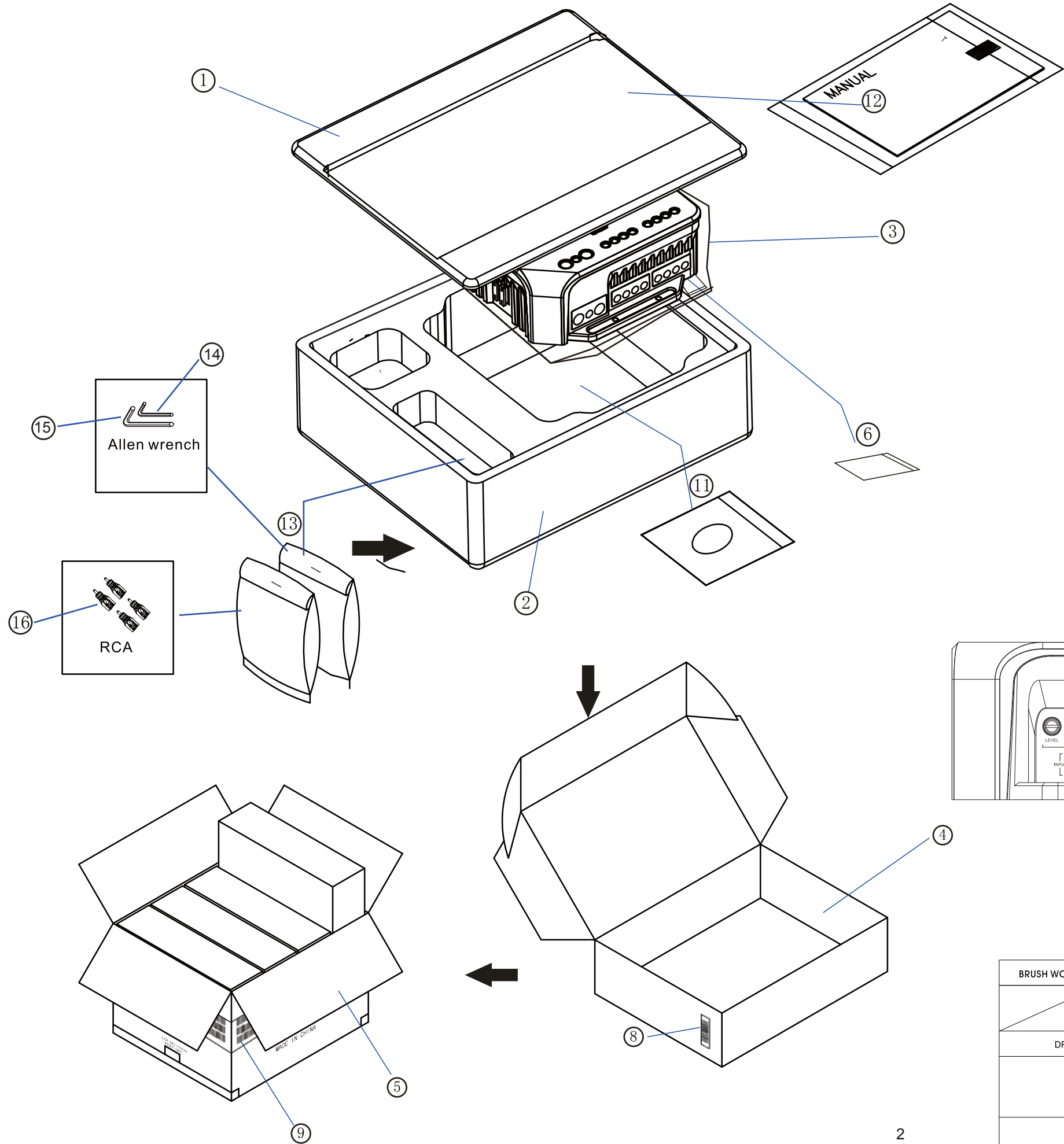
Output Power:	100W RMS x 4 channels @ 4 ohms; ≤1% THD + N
(14.4V supply)	100W RMS x 4 channels @ 2 ohms; ≤1% THD + N
	200W RMS x 2 channel (bridged) @ 4 ohms; ≤1% THD + N
Signal-to-noise ratio:	80dBA (reference 1W into 4 ohms)
	90dBA (reference rated power into 4 ohms)
Frequency response:	10Hz-27kHz (-3dB)
THD+N 1KHz LPF=22KHz	<0.5% (rated power @ 4 ohms)
Channel Separation	>60db
Input Impedance	10 KΩ (Hi2 = 120)
Maximum input signal:	20.0V
Maximum sensitivity:	100mV
Bass Boost @ 80Hz	10dB
DC Offset	<50mV
Idle Current @ 4 ohms	<1.5A
Max Current Draw	38A (rated power @ 4 ohms)
Remote Operating Voltage	5-16VDC
Turn-on delay time	1 sec
Circuit Protection	Temperature (85±5C), Short circuit, Operating voltage range (9-16V)
Dimensions:	7-3/16" x 8-1/4" x 2-3/4" (183mm x 210mm x 70mm)
Weight	5.57lb (2.53kg)
Fuse:	2 x 30A

JBL continually strives to update and improve existing products, as well as create new ones. The specifications and details in this and related JBL publications are therefore subject to change without notice.

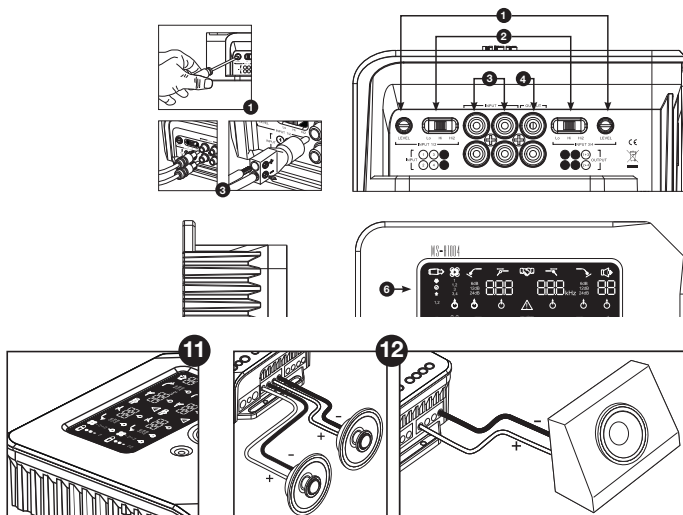
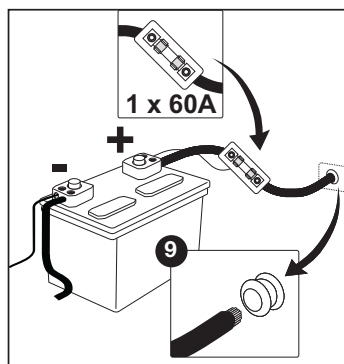
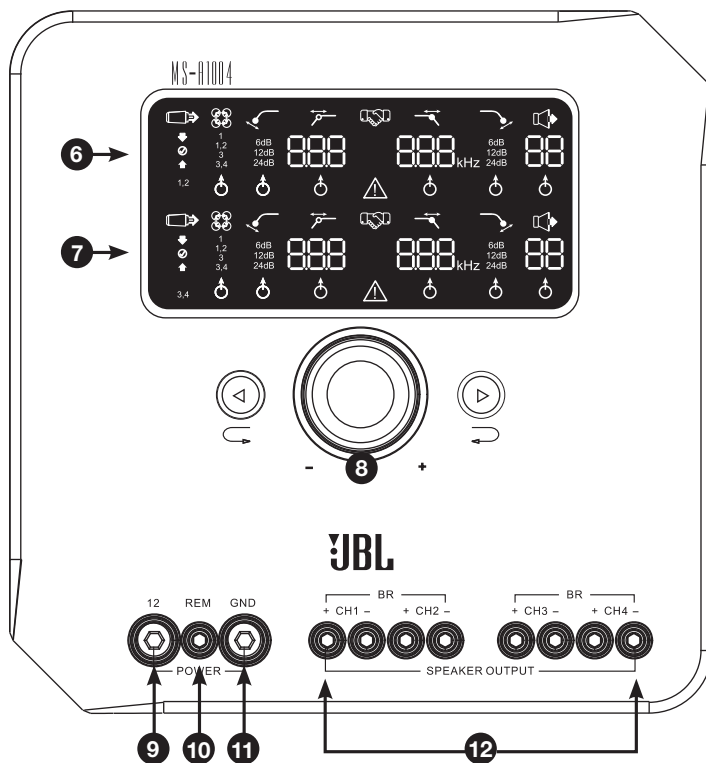
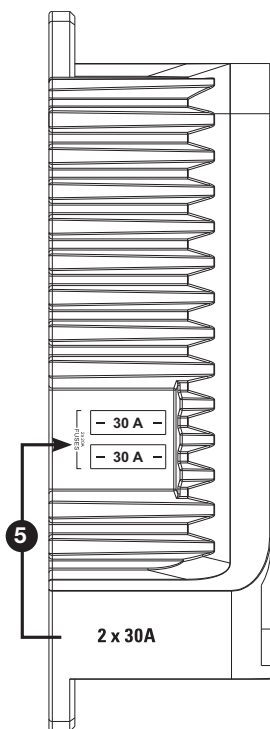
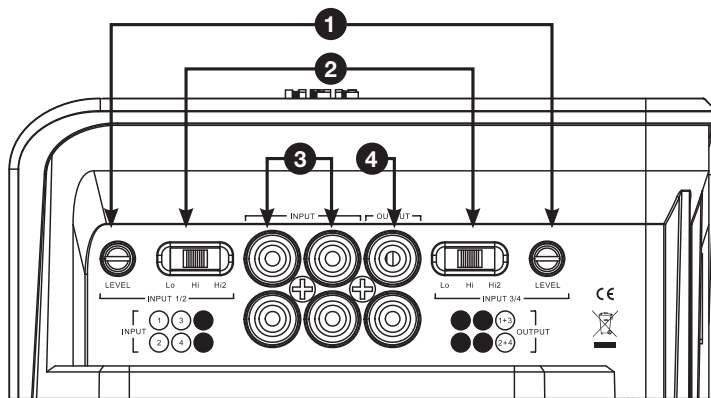
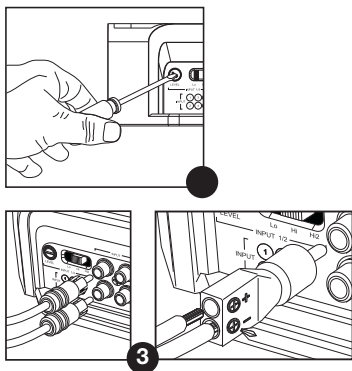


REVISIONS		
REV	DESCRIPTION	DATE
2		
3		

NO	PART NUMBER	DESCRIPTION	QTY
1	57.12.TINY4R1000	FOAM PACKING -TOP	1
2	57.12.TINY4R1001	FOAM PACKING-BOTTOM	1
3		PLASTIC BAG	1
4	59.11.TINY4R100	OUTER CARTON	1
5		MASTER CARTON	1
6		DESICCANT	1
7		SERIAL LABEL	2
8		GIFT LABEL	1
9		CARTON LABEL	1
10		CE LABEL	1
11	38.112.TINY0	SETUP CD	1
12	visit www.jbl.com	OWNER'S MANUAL	1
13		PLASTIC BAG	2
14	65.1.2R520	2.5mm ALLEN WRENCH	1
15	65.1.0420	4.0mm ALLEN WRENCH	1
16	26.121.112101	RCA to bare wire Adapters	4



BRUSH WORK	UNIT	SCALE	DATE	MODEL	REMARK
	MM	Ns	2010.06.24	DRAW NO	
DRAW	CHECK	APPRO	Packing Drawing		
				CODE NO.	ISSUE
					1



MS-A1004 FEATURES

1 Input-Level Control

Used to match the input sensitivity to the signal voltage for proper analog-to-digital conversion. See "Setting the Input Level, and Enabling or Disabling Signal-Sensing Turn-On" for details. DO NOT use these controls for setting the relative output level of amplifier channels!

2 Input Signal Selector

Lo/Hi/Hi2 sets the input voltage and impedance range. See "Setting the Input Level, and Enabling or Disabling Signal-Sensing Turn-On" and "The factory-installed system in my car shows a 'speaker disconnected' message, or it fails to play when a speaker is disconnected or when an amplifier is connected to its output. What should I do?" for details.

3**Audio Inputs**

Use RCA audio cables for preamp-level connections or the included RCA-to-bare-wire adapters for speaker-level input connections.

4 Summed Pass-Through Outputs

Input channels 1 and 3 are combined and sent to one output. Inputs 2 and 4 are combined and sent to the other output.

5 Onboard Fuses

2 x 30A ATC type.

6 Channels 1 and 2 Display Panel

Displays the settings of channels 1 and 2 of the amplifier.

7 Channels 3 and 4 Display Panel

Displays the settings of channels 3 and 4 of the amplifier.

8 User Controls

Allows adjustment of amplifier settings. See "MS-A1004 User Controls" for details.

9 +12V Power Input

Connect to vehicle battery with a 60A fuse within 18 inches (45.7cm) of the positive battery terminal.

10 Remote Turn-On Input

Connect switched +5V to +12V. NOTE: The MS-A1004 also includes signal-sensing turn-on. You may choose the turn-on method during setup. See "How does the digital input mixer work?" and "MS-A1004 Connections" for details.

11 Chassis Ground Input

Connect to a paint-free spot on the vehicle chassis.

12 Speaker Outputs

MS-A1004

DIGITAL SIGNAL PROCESSING AMPLIFIER

PLEASE READ THIS BEFORE YOU BEGIN!

JBL® MS Series amplifiers include many features not found on conventional car audio amplifiers. Also, the setup procedure for MS Series amplifiers is different from that of conventional car audio amplifiers. The following overview of features and functions will help you plan a great system and make the best use of the MS-A1004's innovative features.

About the digital signal processing (DSP) included in MS Series amplifiers:

All of the signal processing in MS Series amplifiers is digital. Digital signal processing, along with the intuitive controls and display included in MS Series amplifiers, makes precise setup easy. Only the input-level controls are analog.

Will my settings be lost if I disconnect the amplifier or the car's battery?

No. All of the DSP settings are stored in nonvolatile memory, so no settings will be lost if power is removed from the amplifier.

Why are the input-level controls analog?

In order to provide the best signal-to-noise ratio and to maximize the resolution of the digital-to-analog conversion, the maximum input signal level to the analog-to-digital (A/D) converters must be precisely set. This must be an analog control. The included setup CD and the procedure described in this manual make setting the level simple and precise. Once the input-level control is set, the control should not be used to "tune" the system. Use the digital output-level control to adjust the relative level between amplifier channels to tune the system.

NOTE: Although some typical connections and controls are shown, JBL® MS Series amplifiers include many features and digital controls not found on conventional car audio amplifiers. Also, the setup (or testing) procedure for MS Series amplifiers is different (more complicated) from that of conventional car audio amplifiers.

Details can be found in the MS-A1004 owner's manual, which can be downloaded here:

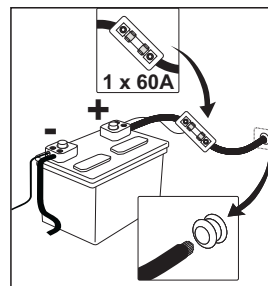
<http://www.jbl.com/EN-US/Support/Pages/ProductSelector.aspx>

MS-A1004 CONNECTIONS

Power Inputs

1. +12V Power Input

Connect the unit to the vehicle battery using a minimum size of 8 AWG (8mm²) wire with a 60A fuse placed within 18 inches (45.7cm) of the positive battery terminal. Be sure to use an insulation grommet at every location where the power wire will pass through metal.



2. Remote Turn-On Input (Optional to Connect)

No discrete remote-turn-on connection to the MS-A1004 is necessary. If your head unit includes a remote-turn-on lead and you wish to connect it, connect the unit to this terminal.

3. Chassis Ground Input

Using at least 8 AWG (8mm²) wire, connect this terminal to a nearby point of the vehicle's chassis (sheet metal). Scrape away the paint from the area to ensure a good connection. Do not ground the amplifier to the vehicle's frame.

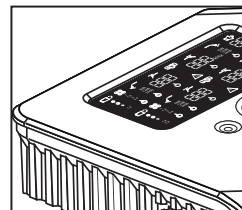
Audio Inputs

1. Using RCA Outputs

If the unit that precedes this amplifier in the signal chain includes RCA-type output connectors, connect them directly to the RCA inputs of this amplifier.

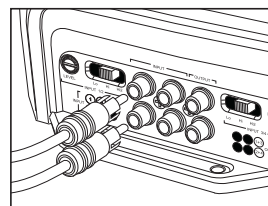
2. Using Speaker-Level Signals

If the preceding equipment doesn't have RCA-type connectors, use the RCA-to-bare-wire adapters included with the MS-A1004. Connect the + signal to the terminal marked + and the - signal to the terminal marked -.



Pass-Through Audio Outputs (Full-Range)

Input channels 1 and 3 are summed and sent to the top RCA output. Inputs 2 and 4 are summed and sent to the bottom RCA output. Using these outputs, additional amplifiers can be easily added. For example: When using the MS-A1004 for front and rear speakers, you could use these outputs for a subwoofer amplifier. The outputs are full-range. No high-pass or low-pass filters are applied in the MS-A1004.



Speaker Outputs

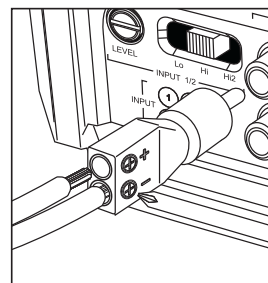
Connect each speaker to the amplifier channel that will correspond to the input used or how it's assigned. See "Setting the Input Mixer" to learn how to assign the input signals to the output channel pair. Be sure to observe proper polarity when connecting the speakers to the outputs.

1. Stereo Connection

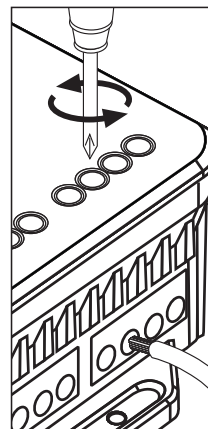
Connect the left-channel loudspeaker(s) to the output terminals corresponding to the left-channel input. Connect the right-channel speaker(s) to the output terminals corresponding to the right-channel input.

2. Bridged Connection

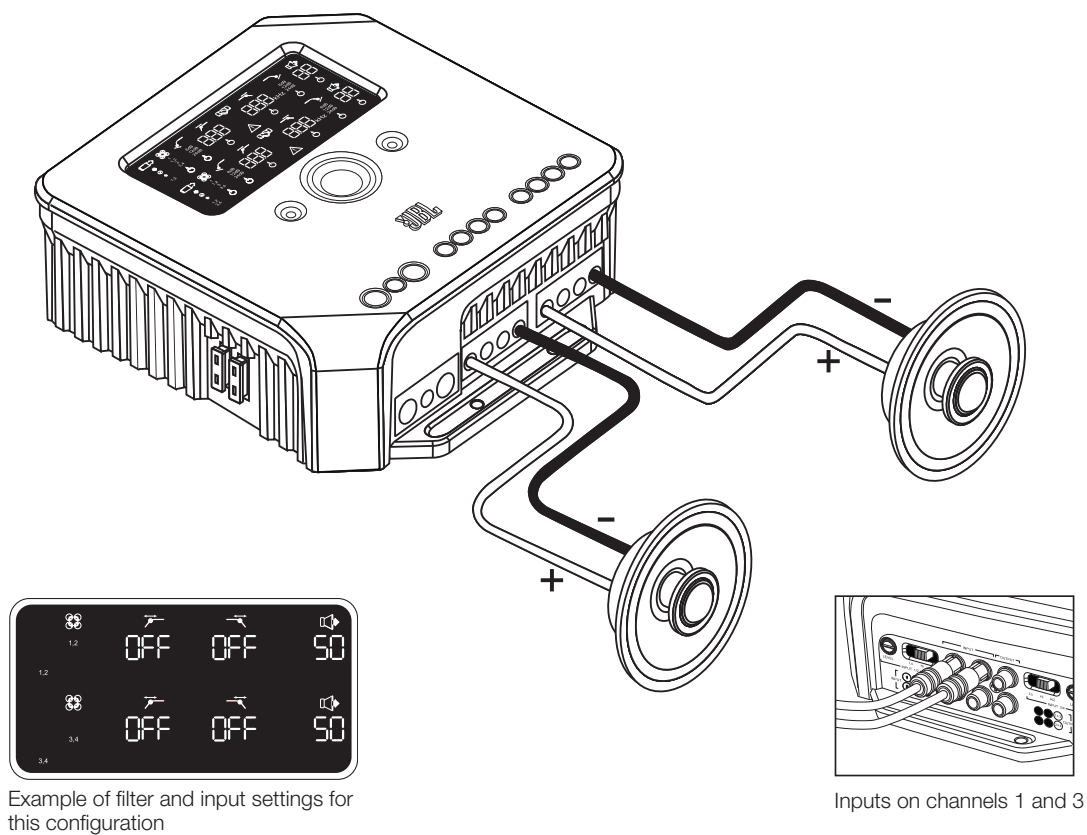
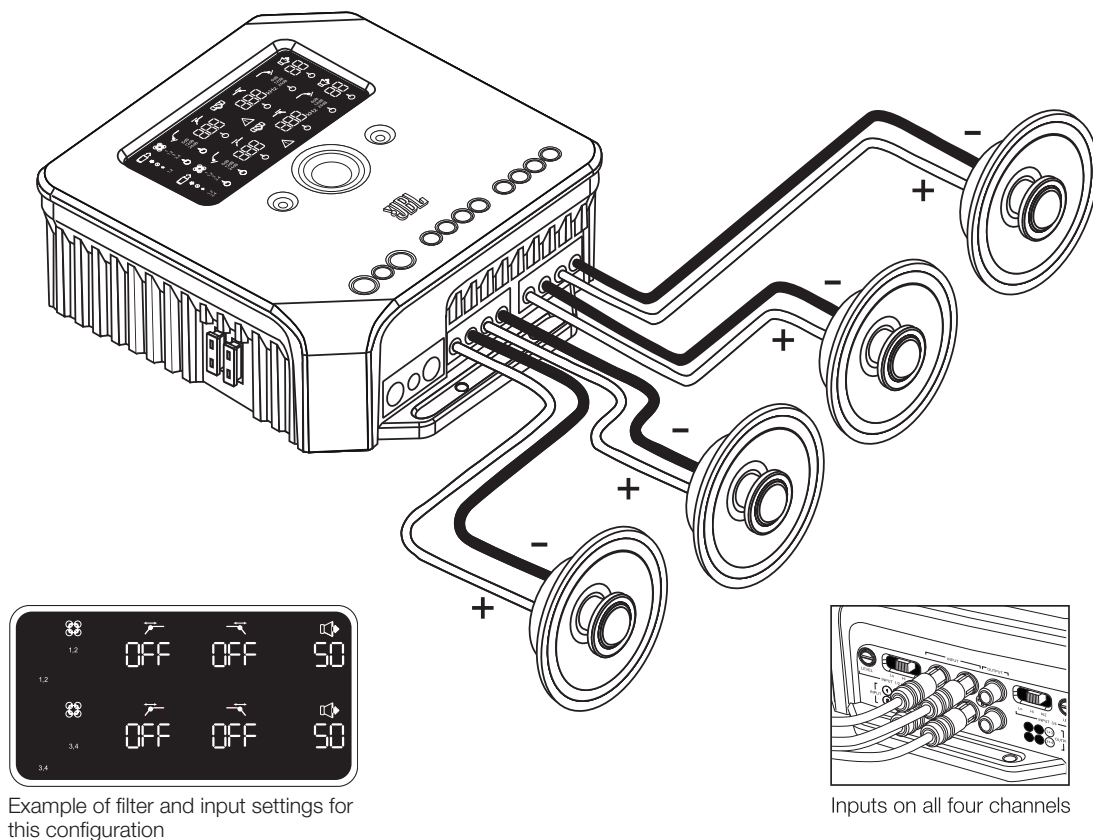
Connect the positive terminal on the loudspeaker(s) to the positive (+) terminal on Channel 1 (or 3). Connect the negative terminal on the speaker(s) to the negative (-) terminal on Channel 2 (or 4).



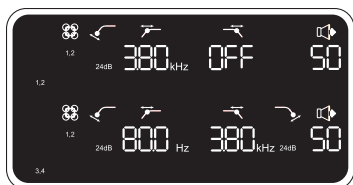
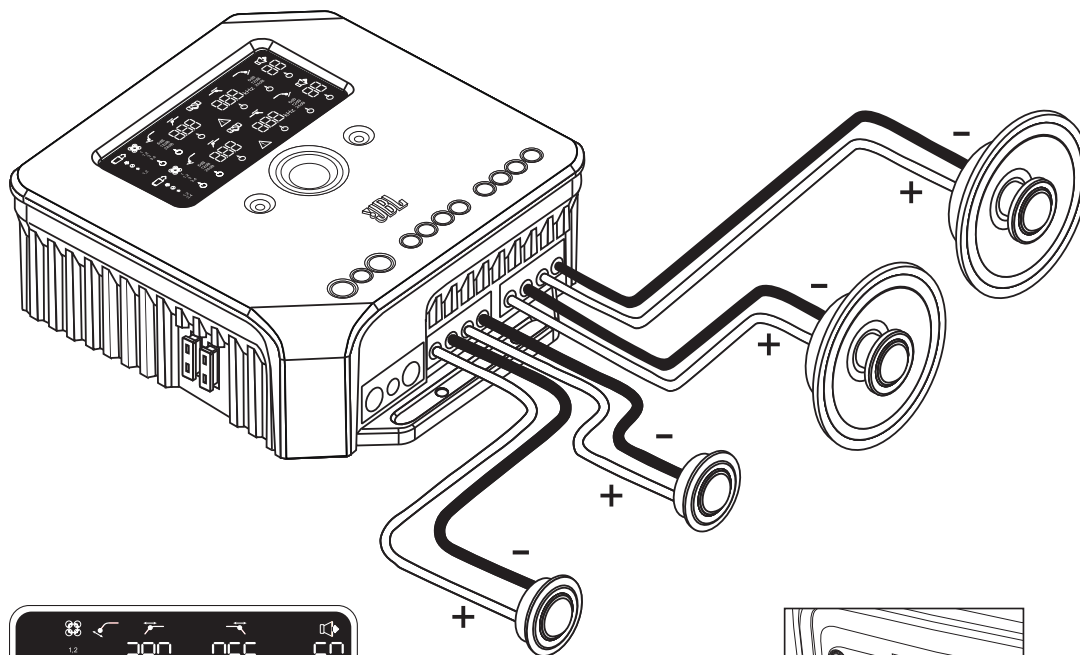
See "System Diagrams" for examples of system configurations in which the amplifier is most commonly used.



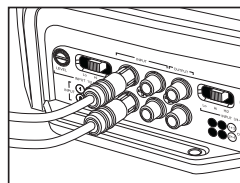
SYSTEM DIAGRAMS



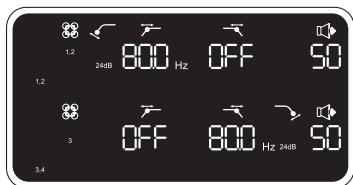
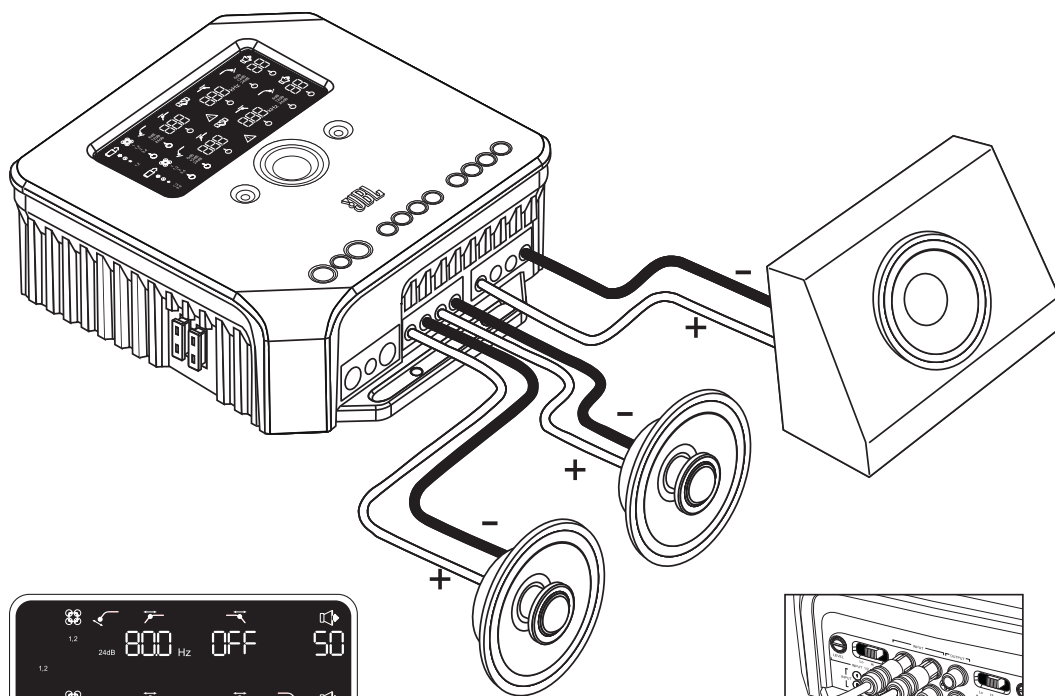
SYSTEM DIAGRAMS



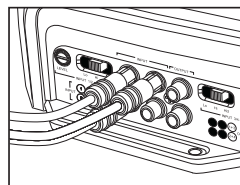
Example of filter and input settings for this configuration



Inputs on channels 1 and 2

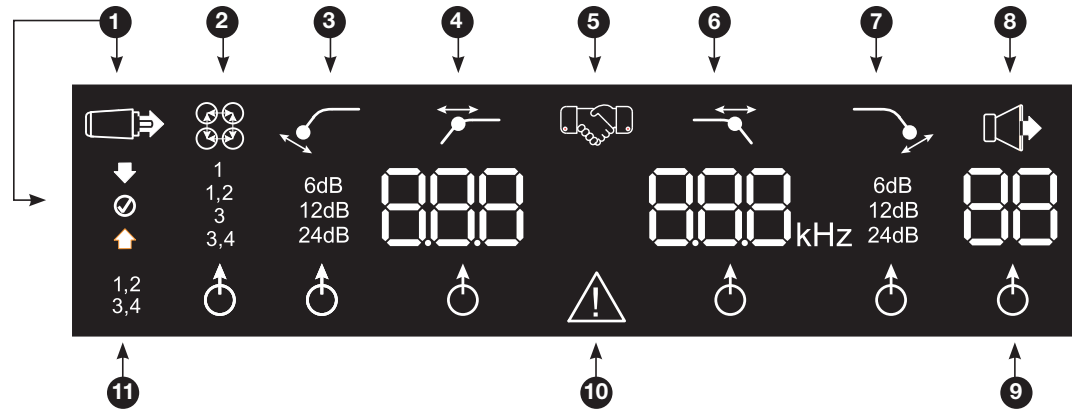
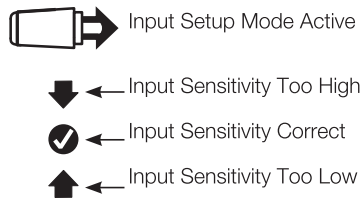
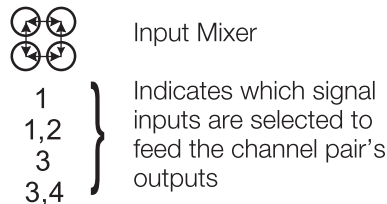
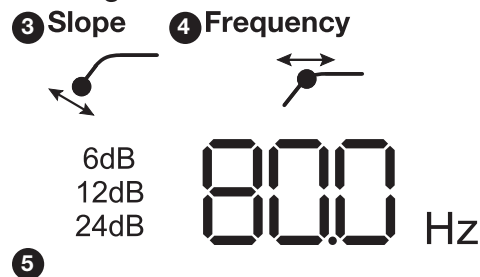


Example of filter and input settings for this configuration



Inputs on channels 1, 2 and 3

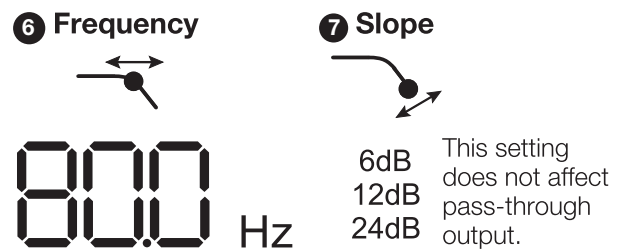
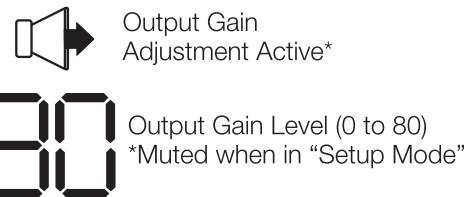
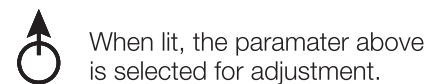
MS-A1004 DISPLAY ICONS

**1 Input-Level Settings****2 Input Channel Mixer****High-Pass Filter****Pairing Confirmation**

Bass-Level Controller Pairing*

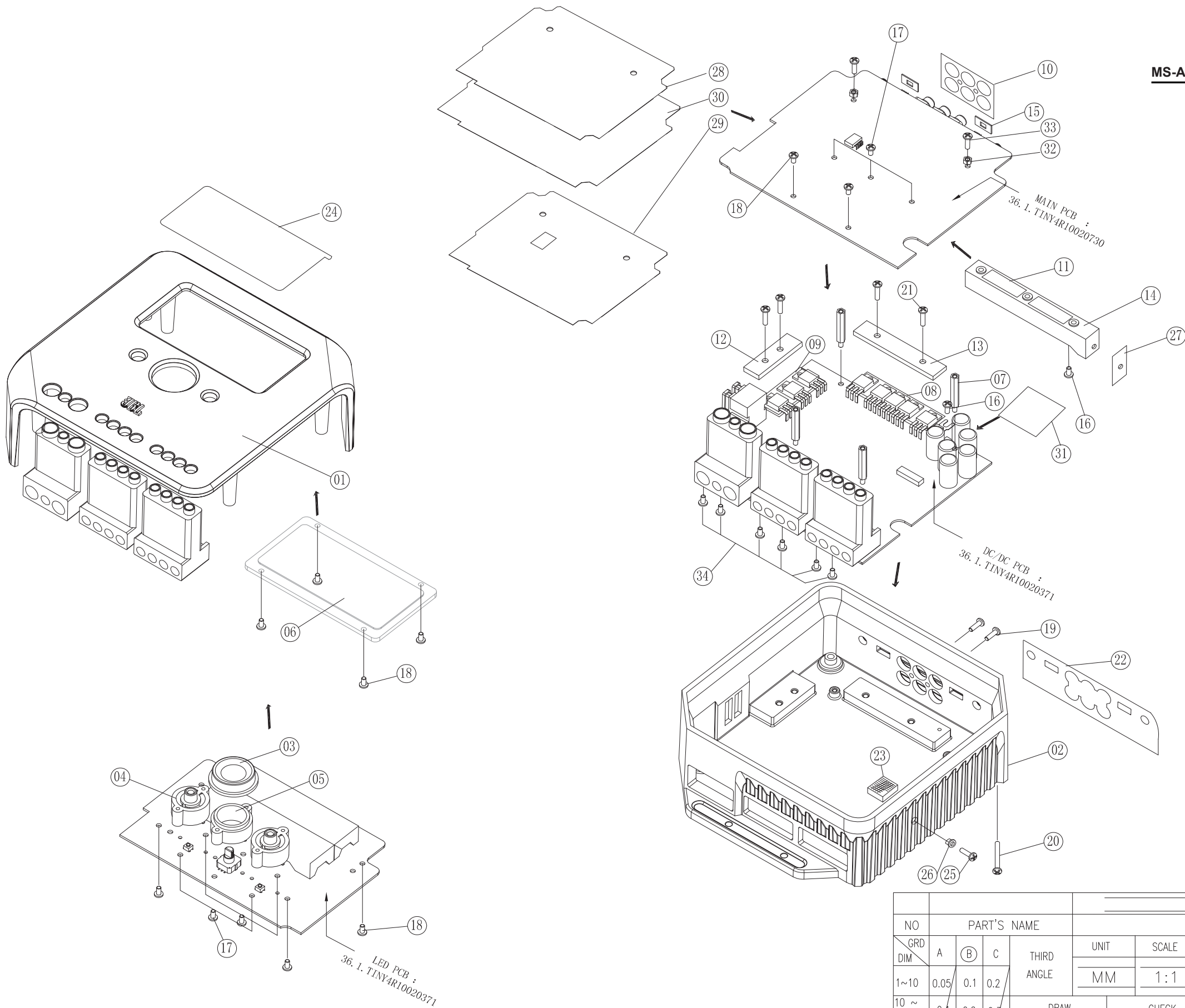


*The MS-WBC wireless bass controller is an optional accessory.

Low-Pass Filter**8 Output Gain Adjustment****9 Adjustment-Selection Indicator****10 Protection****11 Channel ID**

1,2 Indicates the amplifier channels affected by the row of settings to the right in the display. (Channels 1, 2 use the upper display and channels 3, 4 use the lower display.)

3,4



MS-A1004



34	Screw	31. 11. 0213R00610	6PCS
33	Screw	31. 11. 0413R01200	2PCS
32	Support	34. 19. 0030030054	2PCS
31	Paper	48. 152. 0320320R50	1PCS
30	Shield Plate	63. 15. TINY4R10010	1PCS
29	Paper	48. 152. 1541130R51	1PCS
28	Paper	48. 152. 1541130R50	1PCS
27	Silcon Pad	48. 142. 035023R201	1PCS
26	Washer	50. 12. SW30012	1PCS
25	Screw	31. 11. 0213R01400	1PCS
24	Protection Film	48. 192. 1200500R10	1PCS
23	Plasticene	33. 14. 0200200061	1PCS
22	Front Label	56. 15. TINY4R10010	1PCS
21	Screw	31. 11. 0564R01411	4PCS
20	Screw	31. 11. 0514R02000	4PCS
19	Screw	31. 13. 0563R01000	2PCS
18	Screw	31. 11. 0513R00610	14PCS
17	Screw	31. 13. 0563R00610	9PCS
16	Screw	32. 13. 0563R005271	2PCS
15	Nonwovens	48. 171. 0180090020	2PCS
14	Heatsink	53. 11. 101002301070	1PCS
13	Heatsink	53. 14. 069001500371	1PCS
12	Heatsink	53. 14. 045001500371	1PCS
11	Silcon Pad	48. 142. 037015R202	2PCS
10	Label	50. 12. TINY4R10011	1PCS
09	Silcon Pad	48. 142. 045020R202	1PCS
08	Silcon Pad	48. 142. 070018R202	1PCS
07	PCB Support	50. 13. TINY4R10010	4PCS
06	Acryl	43. 11. TINY4R1000	1PCS
05	Knob bracket	50. 12. TINY4R10010	1PCS
04	Volume Knob	35. 1. TINY4R100300	2PCS
03	Power Knob	35. 1. TINY4R10030	1PCS
02	Bottom Cover	42. 1. TINY4R10070	1PCS
01	Top Cover	41. 1. TINY4R10070	1PCS
MARK NO.	DESCRIPTION	PART NO.	Q'TY

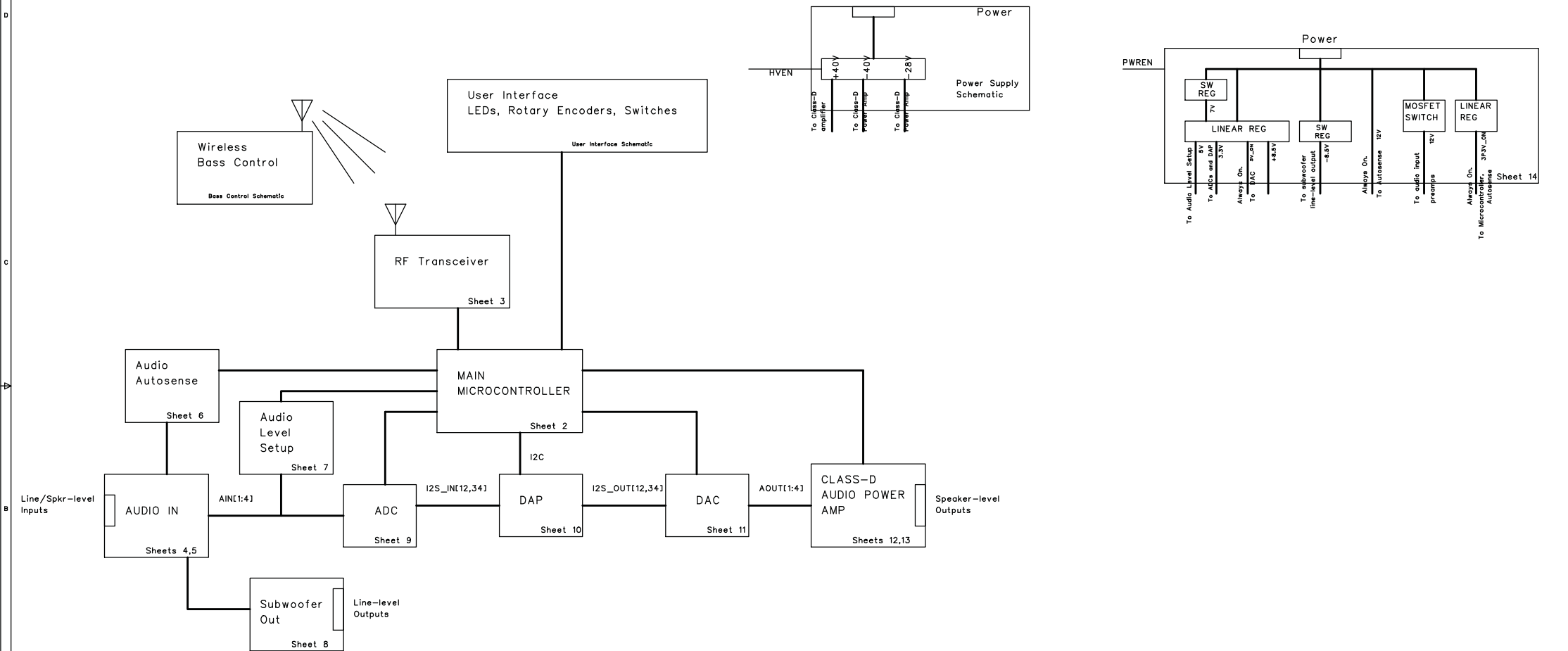
MS-A1004

NO		PART'S NAME							SPECIFICATION		
GRD DIM	A	Ⓑ	C	THIRD ANGLE	UNIT	SCALE			MODEL		
					MM	1:1	2010.04.10		DRAW NO		
	1~10	0.05	0.1	0.2							
10 ~ 100	0.1	0.2	0.3	DRAW		CHECK		APPRO		NA	
100 ~ 500	0.2	0.3	0.5								
500 ~ 1000	0.3	0.5	1.5								
				,LTD.						CODE NO.	ISSUE
											1

MS-A1004



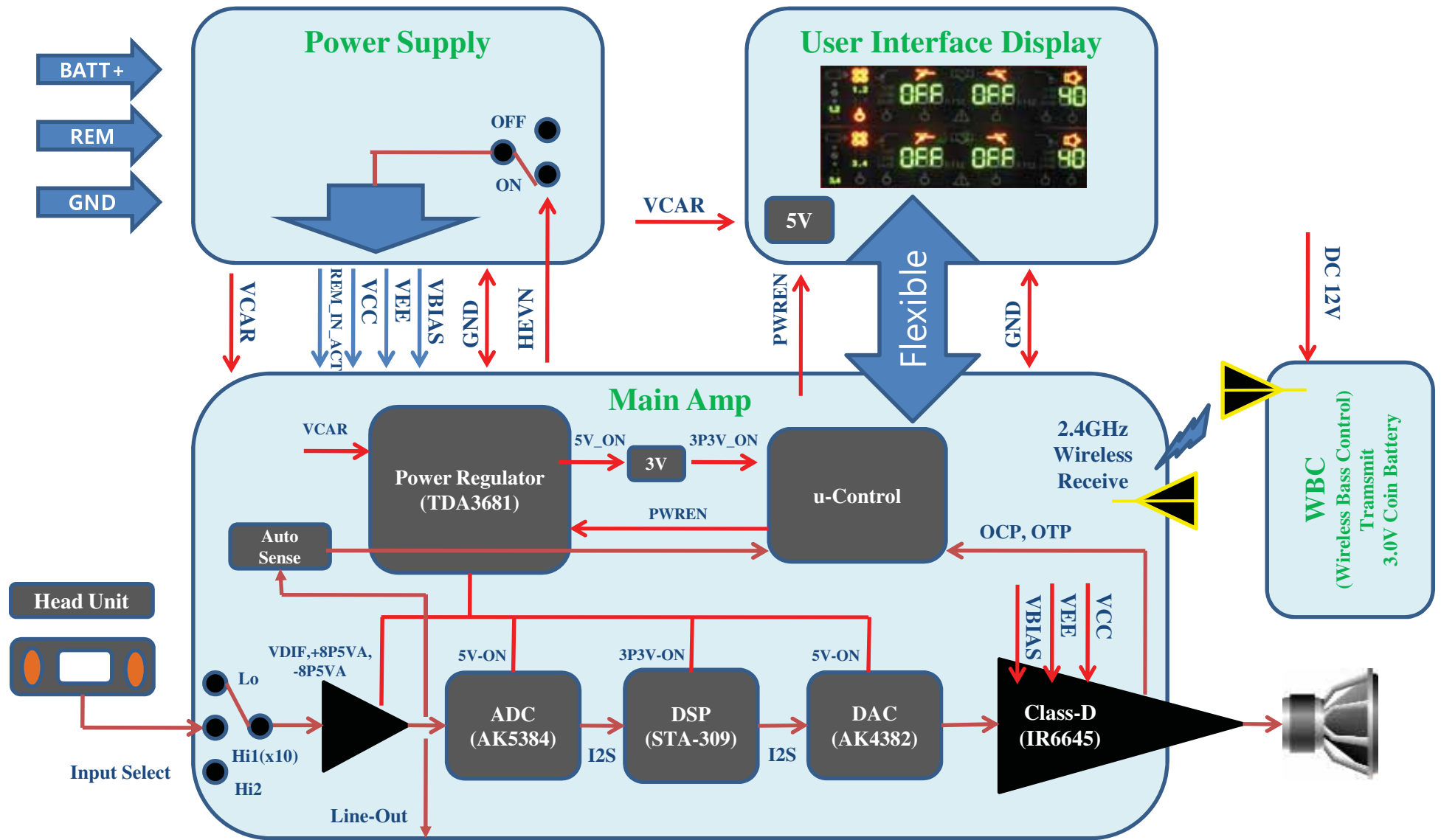
NOTE: Unless otherwise specified all non-polarized capacitors are ceramic, 0603 package, 50V (or higher), and 10% tolerance. Unless otherwise specified, all resistors are 0603 package, 5% tolerance. All components have temperature ranges of -40 to +85 or wider.

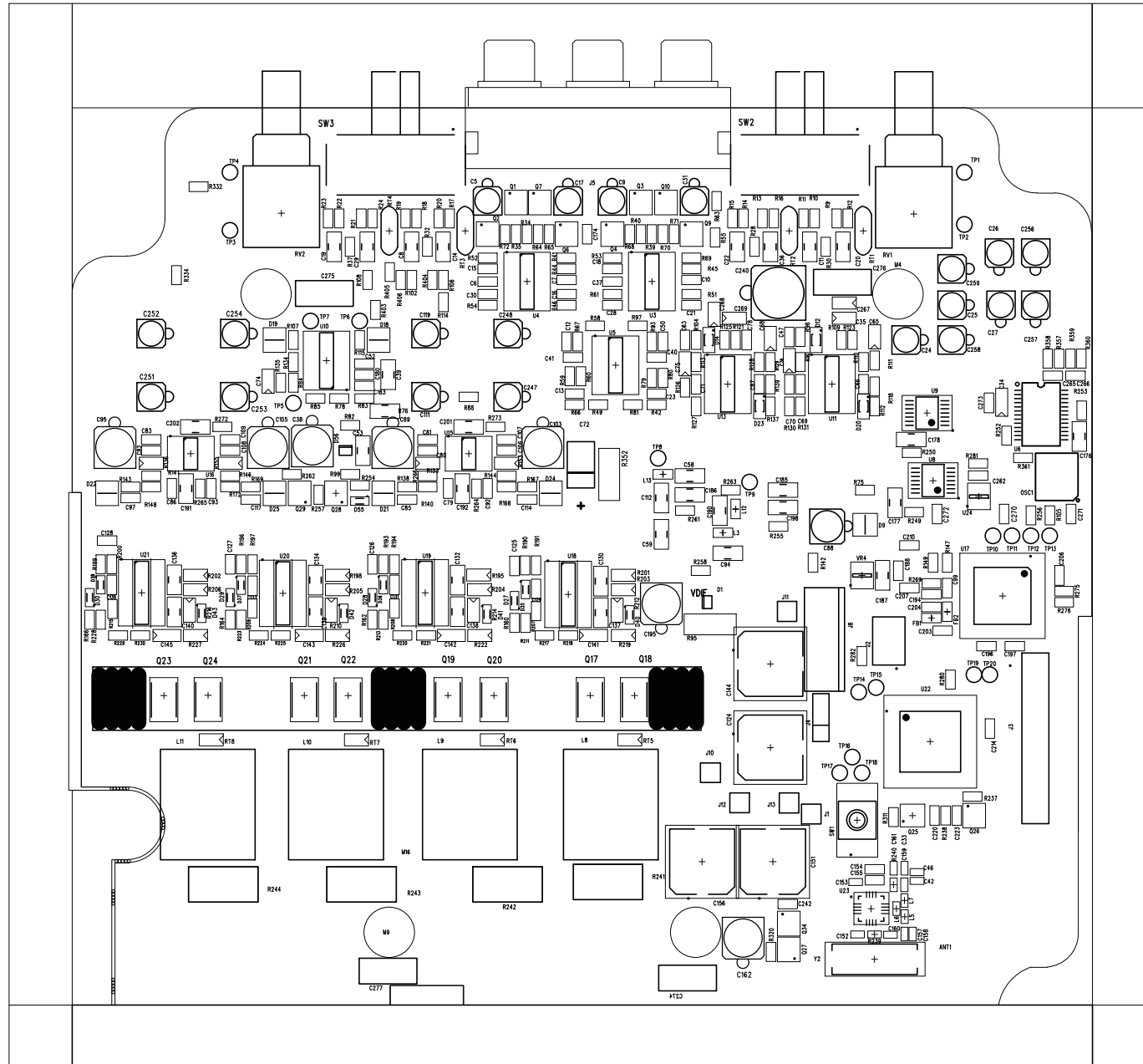


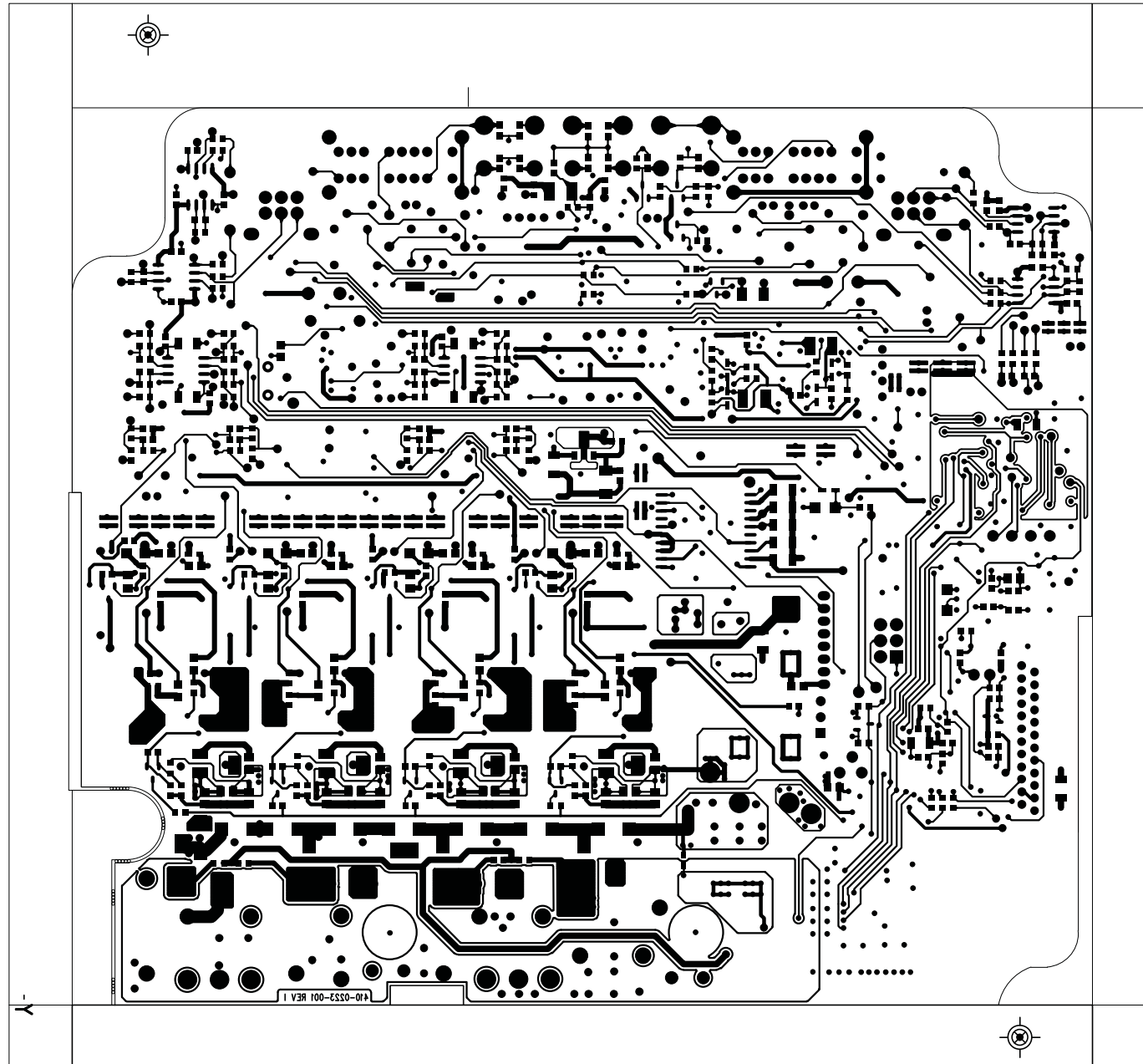
PROPRIETARY INFORMATION - THESE DOCUMENTS AND THE INFORMATION CONTAINED THEREIN ARE PROPRIETARY AND ARE NOT TO BE REPRODUCED OR DISCLOSED TO OTHERS FOR MANUFACTURE OR ANY OTHER PURPOSE EXCEPT THOSE SPECIFICALLY AUTHORIZED IN WRITING BY HARMAN INTERNATIONAL

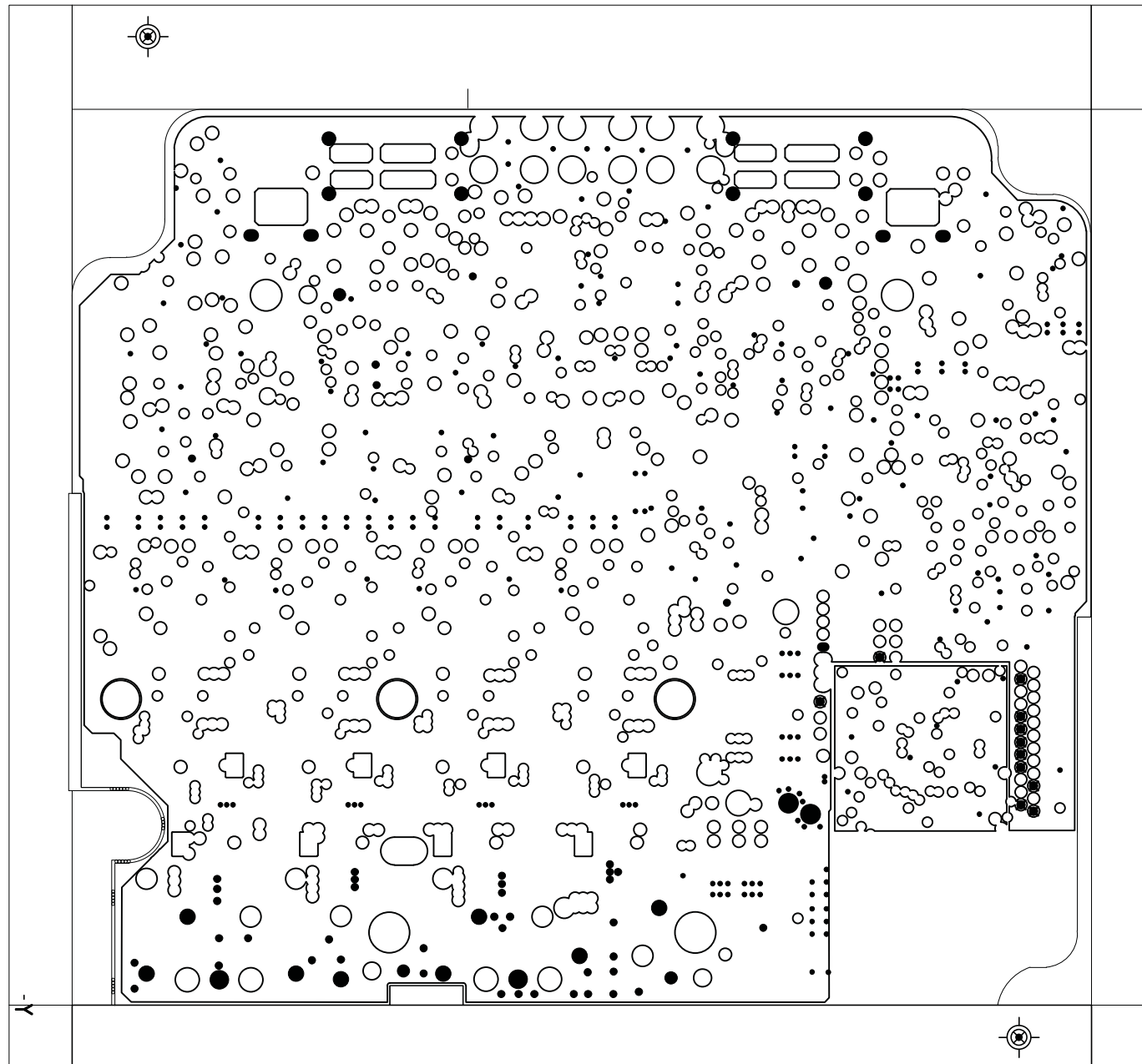
Harman Mobile and Companion Products Group		
Title Block Diagram		
Size B	Document Number Tiny Amp 4 x 100W	Rev H
Date: Jan, 11, 2011		
Sheet 1 of 15		

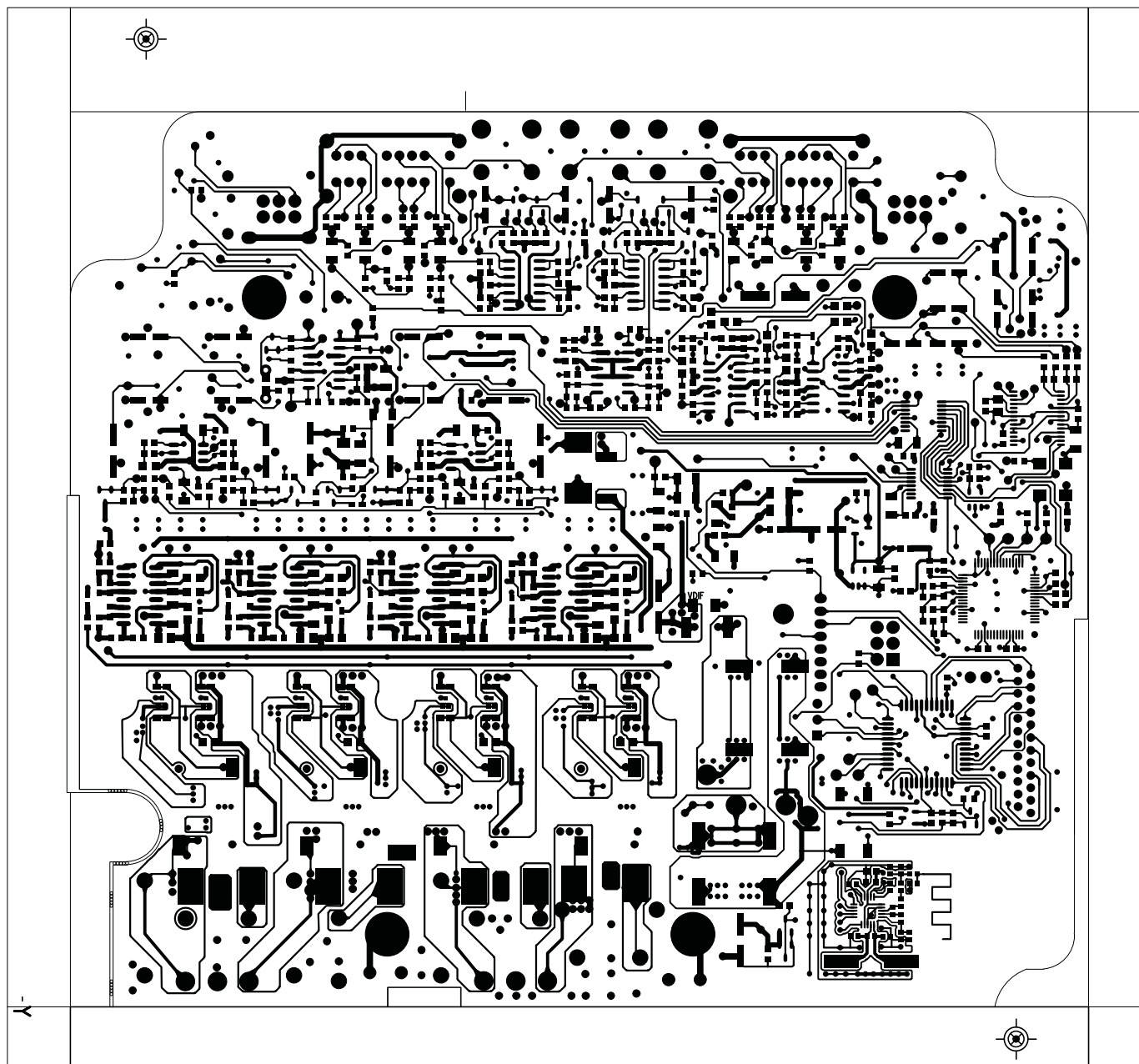
MS-A1004 / MS-A5001 CAR AMPLIFIER BLOCK DIAGRAM

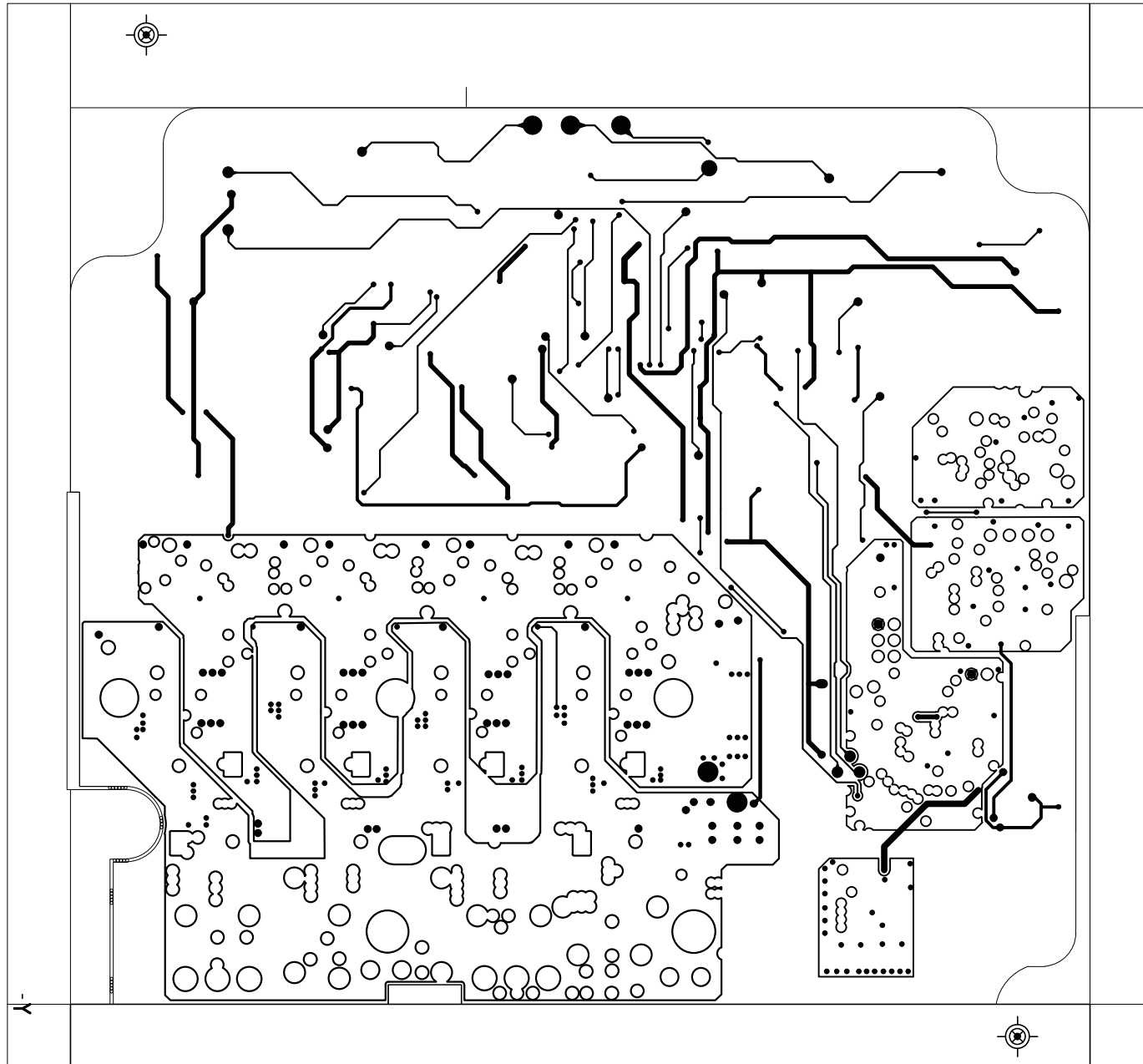


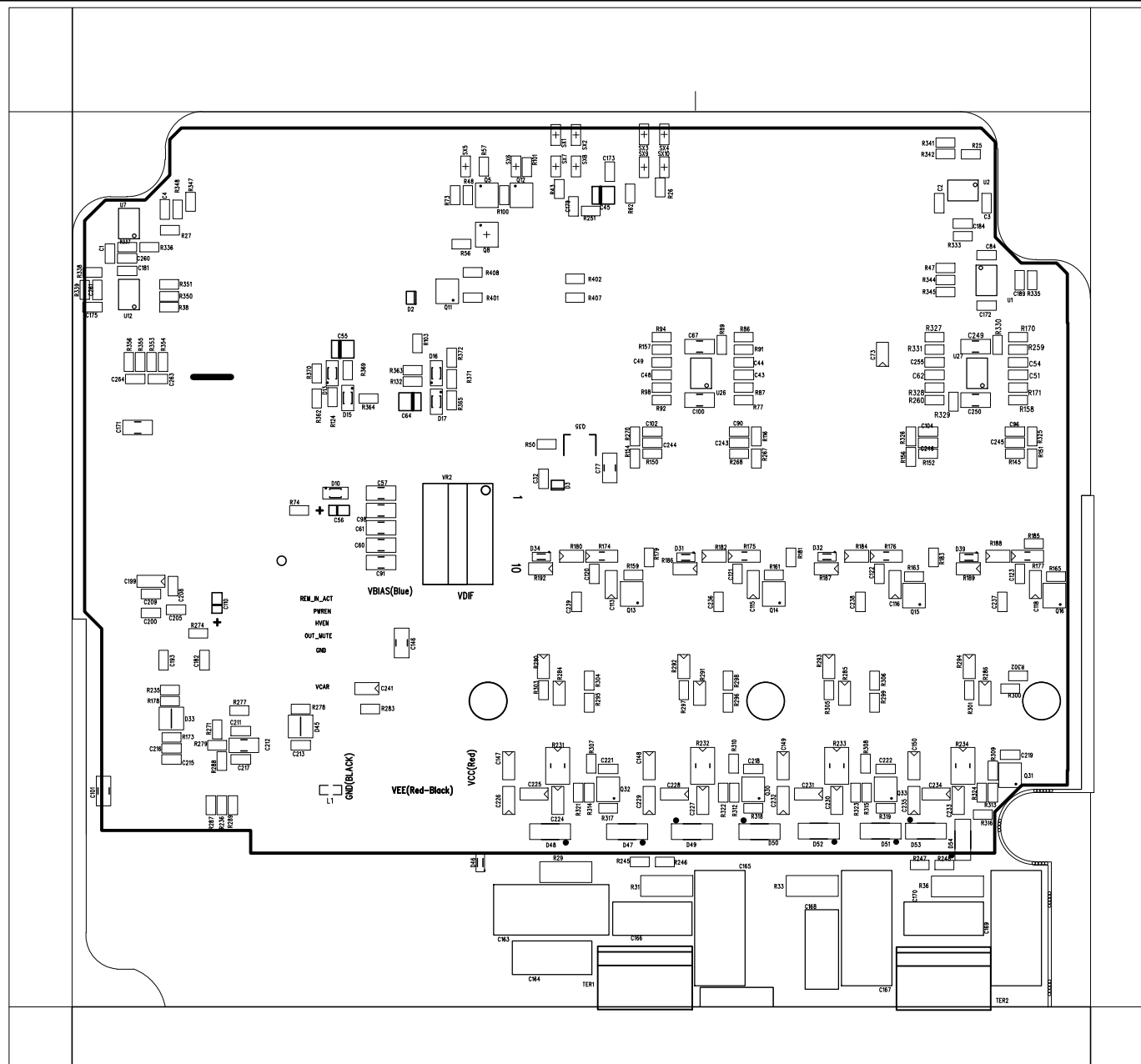












MS-A1004 Electrical Parts List				
Part Number		Description	Qty	Reference Designator or Application
PREAMP/INPUT PCB				
<i>Semiconductors</i>				
IC-TL074ID	SMD IC QUAD OP-AMP	TL074ID	4	U3,4,10,13
IC-LM224	SMD IC QUAD OP-AMP	LM224 SO-14	1	U11
IC-TL074IDTI	SMD IC QUAD OP-AMP	TI074ID TI	1	U5
IC-ADTL082ARZ	SMD IC DUAL OP-AMP	ADTL082ARZ SO-8	2	U15,16
IC-TPS77033	SMD IC	TPS77033 SOT-23 Low-Power 50-mA Low-Dropout (LDO) Linear Regulators	1	VR4
IC-STA309A	SMD IC	STA309A (TQFP64) Multi-channel digital audio processor	1	U17
IC-nRF24L01+	SMD IC RF TRANSCEIVER	nRF24L01+QFN-24	1	U23
IC-74AHC1G04	SMD IC INVERTER	74AHC1G04 TSSOP5	1	U24
IC-MC9S08GT16	SMD IC MICROCONTROLLER	MC9S08GT16ACFBE(QFP44)	1	U22
IC-AK5384VTP	SMD IC 24 BIT 96KHZ 4 CHAN ADC	AK5384VTP (TSSOP-16)	1	U6
IC-AK4382VTP	SMD IC 192kHz 24-BIT SCH DAC	AK4382VTP (TSSOP-16)	2	U8,9
IC-IRS20957S	Protected Digital Audio Driver	IRS20957STRPBF SO-16	4	U18,19,20,21
FET-FDV3O1N	SMD MOSFET	FDV3O1N SOT-23	1	Q25
FET-IRF6645TRBF	SMD MOSFET	IRF6645TRBF	8	Q17,18,19,20,21,22,23,24
TR-BT3904LT1	SMD NPN transistor	MMBT3904 40V SOT-23	6	Q2,4,6,9,27,34
TR-BT3906	SMD PNP transistor	MMBT3906 40V SOT-23	4	Q1,3,7,10
DIODE-LL4148WT	SMD diode	1N4148,1N4148WT SOD-523	12	D27,28,29,30,35,36,37,38,40,41,42,43
DIODE-LL4148WS	SMD diode	1N4148,1N4148W SOD-323	4	D12,14,20,23
DIODE-BAV99	SMD diode	BAV99 SOT23,BAV99	7	D9,18,19,21,22,24,25
DIODE-ZMM20	SMD zener diode	ZENER, 20V/0.5W,MINI MELFZMM20	1	D1
DIODE-MMSZ4689	SMD diode	MMSZ4689	1	D56
IC-TL072ID	SMD IC DUAL OP-AMP	TL072ID SO-8	6	U1,2,7,12,26,27
IC-TDA3681	SMD IC Multiple voltage regulator	TDA3681(SOT418-3),TDA3681TH/N2S	1	VR2
FET-FDV3O1N	SMD MOSFET	FDV3O1N	1	Q8
TR-BT3904LT1	SMD NPN transistor	MMBT3904 SOT-23	2	Q5,12
TR-BT3906	SMD PNP transistor	MMBT3906 SOT-23	5	Q11,13,14,15,16
TR-YKTA1661	SMD transistor	SMALL SIGNAL PNP KTA1661Y	1	Q35
DIODE-MURA120	SMD diode	MURA120	8	D47,48,49,50,51,52,53,54
DIODE-IN4148L	SMD diode	SWITCHING,MINI MELF LL4148/RLS4148	1	D2
DIODE-LL4148WT	SMD diode	1N4148,1N4148WT	5	D31,32,34,39,46
DIODE-LL4148WS	SMD diode	1N4148,1N4148W	5	D13,15,16,17,10
DIODE-BAV99	SMD diode	BAV99 SOT23,BAV99/SOT	2	D33,45
DIODE-ZMM9R1	SMD diode	ZENER, 10V/1W,SOD-106PTZ9.1B	1	D3
<i>Resistors</i>				
RES-2202-0402-F	SMD Resistor	RES SMD 22.0KF 1/16W 0402	1	R240
RES-1R00-0805-F	SMD Resistor	RES SMD 1 F 1/8W 0805	4	R219,222,226,227
RES-10R0-0805-F	SMD Resistor	RES SMD 10 F 1/8W 0805	4	R210,212,214,216,
RES-000-0603-F	SMD Resistor	RES SMD 0 1/10W 0603	8	R11,12,15,16,19,20,23,24
RES-000-0402-F	SMD Resistor	RES, SMD 0R0 OHM 1/16W ±1%	1	C46
RES-1651-0603-F	SMD Resistor	RES SMD 1.65KF1/10W 0603	4	R211,213,223,228
RES-1003-0603-F	SMD Resistor	RES SMD 100KF1/10W 0603	6	R75,111,126,129,130,320
RES-2804-0603-F	SMD Resistor	RES SMD 2.8MF 1/10W 0603	1	R78
RES-100-0603-F	SMD Resistor	RES SMD 10F 1/10W 0603	11	11
RES-1002-0603-F	SMD Resistor	RES SMD 10KF1/10W 0603	14	R28,30,32,37,49,55,63,85,81,281,263,142, 255, 280
RES-1002-0805-F	SMD Resistor	RES SMD 10KF1/8W 0805	4	R195,198,201,202
RES-1202-0603-F	SMD Resistor	RES SMD 12KF1/10W 0603	2	R42,66
RES-1203-0603-F	SMD Resistor	RES SMD 120KF1/10W 0603	4	R90,112,113,137
RES-1803-0603-F	SMD Resistor	RES SMD 180KF1/10W 0603	4	R96,104,118,139
RES-1001-0603-F	SMD Resistor	RES SMD 1KF1/10W 0603	24	R44,45,46,51,52,53,54,58,61,97,140,147, 148,149,168,172,190,191,193,194,196,197, 199, 200
RES-2201-0603-F	SMD Resistor	RES SMD 2K2KF 1/10W 0603	8	R35,40,41,64,68,69,70,72
RES-1003-0603-F	SMD Resistor	RES SMD 100KF 1/10W 0603	4	R110,127,128,131
RES-2002-0603-F	SMD Resistor	RES SMD 20KF 1/10W 0603	3	R121,125,237
RES-2202-0603-F	SMD Resistor	RES SMD 22KF 1/10W 0603	4	R141,144,146,266

Part Number		Description	Qty	Reference Designator or Application
PREAMP/INPUT PCB				
RES-33R0-0603-F	SMD Resistor	RES SMD 33F 1/10W 0603	5	R105,256,275,276,361
RES-3401-0603-F	SMD Resistor	RES SMD 3.4KF 1/10W 0603	1	R269
RES-3300-0603-F	SMD Resistor	RES SMD 330F 1/10W 0603	4	R357,358,359,360
RES-3902-0603-F	SMD Resistor	RES SMD 39KF 1/10W 0603	2	R82,99
RES-4022-0603-F	SMD Resistor	RES SMD 40.2KF 1/10W 0603	2	R134,135
RES-4023-0603-F	SMD Resistor	RES SMD 402KF 1/10W 0603	1	R84
RES-432-0603-J	SMD Resistor	RES SMD 4.3KJ1/10W 0603	5	R34,39,65,71,261
RES-4751-0603-F	SMD Resistor	RES SMD 4.75KF 1/10W 0603	4	R217,220,224,229
RES-4701-0603-F	SMD Resistor	RES SMD 4.7KF 1/10W 0603	2	R332,334
RES-4120-0805-F	SMD Resistor	RES SMD 412 F 1/8W 0805	4	R133,136,153,155
RES-4700-0603-F	SMD Resistor	RES SMD 470F1/10W 0603	4	R160,162,164,166
RES-4753-0603-F	SMD Resistor	RES SMD 475KF 1/10W 0603	5	R83,102,106,108,114
RES-2263-0603-F	SMD Resistor	RES SMD 226KF 1/10W 0603	2	R107,115
RES-4532-0603-F	SMD Resistor	RES SMD 45.3KF1/10W 0603	8	R9,10,13,14,17,18,21,22
RES-5R0-3W-F	SMD Resistor	RES SMD 5 OHM F 3W	4	R241,242,243,244
RES-8251-0805-F	SMD Resistor	RES SMD 8.25KF 1/8W 0805	4	R,203,204,205,206
RES-8251-0603-F	SMD Resistor	RES SMD 8.25KF 1/10W 0603	4	R207,208,209,215
RES-5601-0603-F	SMD Resistor	RES SMD 5.6KF 1/10W 0603	4	R59,60,79,80
RES-5621-0603-F	SMD Resistor	RES SMD 5.62KF 1/10W 0603	4	R218,221,225,230
RES-8252-0603-F	SMD Resistor	RES SMD 82.5KF 1/10W 0603	4	R143,138,167,169
RES-6201-0603-F	SMD Resistor	RES SMD 6.2KF1/10W 0603	2	R258,282
RES-2402-0603-F	SMD Resistor	RES SMD 24KF 1/10W 0603	2	R67,93
RES-7871-0603-F	SMD Resistor	RES SMD 7.87KF 1/10W 0603	1	R123
RES-8450-0603-F	SMD Resistor	RES SMD 845F 1/10W 0603	1	R109
RES-1004-0402-F	SMD Resistor	RES SMD 1.00MF 1/16W 0402	1	R239
RES-4990-2512-F	SMD Resistor	RES SMD 499F 1W 2512	1	R95
RES-2700-2512-F	SMD Resistor	RES SMD 270F 1W 2512	1	R352
RES-000-0603-F	SMD Resistor	RES SMD 0R0 1/10W 0603	1	L1
RES-4753-0603-F	SMD Resistor	RES SMD 475KF 1/10W 0603	5	R73,117,119,120,122
RES-100-0603-F	SMD Resistor	RES SMD 10F 1/10W 0603	6	R89,274,277,279,329,330
RES-1000-0603-F	SMD Resistor	RES SMD 100 F 1/10W 0603	4	R151,154,156,267
RES-1001-0603-F	SMD Resistor	RES SMD 1KF1/10W 0603	8	R116,270,325,326,342,345,348,351
RES-1131-0603-F	SMD Resistor	RES SMD 1.13KF 1/8W 0603	4	R341,344,347,350
RES-1651-0805-F	SMD Resistor	RES SMD 1.65KF 1/8W 0805	4	R186,187,189,192
RES-1002-0603-F	SMD Resistor	RES SMD 10KF1/10W 0603	8	R26,43,50,56,62,271,278,287
RES-1502-0603-F	SMD Resistor	RES SMD 15KF1/10W 0603	1	R283
RES-1003-0603-F	SMD Resistor	RES SMD 100KF 1/10W 0603	22	R124,132,245,246,247,248,295,296,297,299,300,301,303,305,362,363,364,365,369,370,371,372
RES-2002-0603-F	SMD Resistor	RES SMD 20KF 1/10W 0603	3	R57,101,288
RES-2403-0603-F	SMD Resistor	RES SMD 240KF 1/10W 0603	2	R48,100
RES-6491-1206-F	SMD Resistor	RES SMD 6.49KF 1/4W 1206	4	R174,175,176,177
RES-9531-0603-F	SMD Resistor	RES SMD 9.53KF 1/10W 0603	1	R178
RES-4872-0603-F	SMD Resistor	RES SMD 48.7KF 1/10W 0603	1	R173
RES-30R0-0805-F	SMD Resistor	RES SMD 30 F 1/8W 0805	8	R284,285,286,290,291,292,293,294
RES-33R0-0603-F	SMD Resistor	RES SMD 33F 1/10W 0603	1	R289
RES-3300-0603-F	SMD Resistor	RES SMD 330F 1/10W 0603	4	R353,354,355,356
RES-3301-0603-F	SMD Resistor	RES SMD 3.3KF 1/10W 0603	4	R145,150,152,268
RES-3601-0603-F	SMD Resistor	RES SMD 3.6KF 1/10W 0603	4	R25,27,38,47
RES-3002-0805-F	SMD Resistor	RES SMD 30KF1/8W 0805	4	R180,182,184,188
RES-3003-0603-F	SMD Resistor	RES SMD 300KF1/8W 0603	1	R74
RES-47R0-0603-F	SMD Resistor	RES SMD 47F1/10W 0603	4	R179,181,183,185
RES-4700-0603-F	SMD Resistor	RES SMD 470F1/10W 0603	1	R236
RES-4701-0603-F	SMD Resistor	RES SMD 4.7KF1/10W 0603	26	R77,86,87,91,92,94,98,157,158,159,161,163,165,170,171,259,260,327,328,331,333,335,336,337,338,339
RES-4871-0603-F	SMD Resistor	RES SMD 4.87KF1/10W 0603	1	R103
RES-5111-0603-F	SMD Resistor	RES SMD 5.11KF 1/10W 0603	1	R235
RES-7502-0603-F	SMD Resistor	RES SMD 75KF 1/10W 0603	4	R298,302,304,306
RES-152-2512-F	SMD Resistor	RES SMD 1.5KF 1W 2512	4	R29,31,33,36
RES-ESDA-0603	SMD static Resistor	0603 ESDA	10	SX1,2,3,4,5,6,7,8,9,10
RES-1R00-1210	SMD Resistor	RESISTORA,1 ohm 1210 1/3 W,1%	4	R231,232,233,234

Part Number		Description	Qty	Reference Designator or Application
PREAMP/INPUT PCB				
<i>Capacitors</i>				
CAP0161R0-0402	SMD Capacitor	CAP, SM 1PF 16V +/-0.1P NPO 0402	1	C161
CAP050103-0603	SMD Capacitor	CAP, SM 0.01uF 50V 10%NPO 0603	1	C155
CAP016333-0402	SMD Capacitor	CAP, SM 0.033UF 16V 10%NPO 0402	1	C153
CAP050104-0603	SMD Capacitor	CAP, SM 0.1UF 50V 10%NPO 0603	33	C6,C37,C40,C41,C47,C63,C66,C69,C70, C71,C78,C87,C125,C126,C128,C180,C188, C196,C197,C203,C204,C206,C207,C210, C214,C242,C183,C262,C270,C271,C272, C273,C127
CAP050224-1206	SMD Capacitor	CAP, SM 0.22UF 50V 10%NPO 1206	14	C8,11,14,19,20,22,29,36,94,112,186,185, 190,198
CAP0161R5-0402	SMD Capacitor	CAP, SM 1.5PF +/-0.1P 16V NPO 0402	1	C159
CAP050102-0603	SMD Capacitor	CAP, SM 1000PF 50V 5%NPO 0603	11	C52,80,81,82,83,106,107,108,109,154,220
CAP025106-1206	SMD Capacitor	CAP, SM 10UF cer -20%+80% 25V 1206	15	C39,53,58,59,130,132,134,136,137,138, 139,140,177,178,187
CAP050151-0603	SMD Capacitor	CAP, SM 150PF 50V 5%NPO 0603	4	C85 ,97 ,114 ,117
CAP050122-0603	SMD Capacitor	CAP, SM 1200PF 50V 10%NPO 0603	1	C194
CAP050152-0603	SMD Capacitor	CAP, SM 1500PF 50V 10%NPO 0603	2	C265,266
CAP050105-1206	SMD Capacitor	CAP, SM 1UF 50V 10% 1206	5	C191,192,201,202,176
CAP050225-0805	SMD Capacitor	CAP, SM 2.2UF 50V 10%NPO 0805	14	C34,35,65,68,74,75,76,141,142,143,145, 267,268,269
CAP050222-0402	SMD Capacitor	CAP, SM 2200PF 50V 10%NPO 0402	1	C157
CAP050221-0603	SMD Capacitor	CAP, SM 220PF 50V 10%NPO 0603	1	C99
CAP050270-0402	SMD Capacitor	CAP, SM 27PF 50V 2%NPO 0402	2	C152,160
CAP0164R7-0402	SMD Capacitor	CAP, SM 4.7PF 16V 2%NPO 0402	1	C158
CAP050470-0603	SMD Capacitor	CAP, SM 47PF 50V 5%NPO 0603	20	C7,10,12,13,15,16,18,21,23,28,30,50,79,86, 92,93,129,131,133,135 ,
CAP0316100-4	SMD electrolytic Capacitor	10UF/16V 4 20%	20	C5,9,17,24,25,26,27,31,111,119,247,248, 251,252,253,254,256,257,258,259
CAP0316101-6.3	SMD electrolytic Capacitor	CAP, E 100UF 16V 20% SM6X6	1	C38
CAP1016101-D	SMD Ta electrolytic Capacitor	100uf/16V "D" type	1	C72
CAP0616220-6.3	SMD electrolytic Capacitor	22UF/16V 6.3 20% Bipolar-NP	4	C89,95,103,105
CAP0616101-6.3	SMD electrolytic Capacitor	100UF/16V 6.3 20% Bipolar-NP	1	C162
CAP0316221-6.3	SMD electrolytic Capacitor	220UF/16V 6.3 20%	1	C240
CAP0350221-8	SMD electrolytic Capacitor	220UF/50V 8 20%	4	C124,144,151,156
CAP0335330-5	SMD electrolytic Capacitor	33UF/35V 5 20%	1	C195
CAP0310101-6.3	SMD electrolytic Capacitor	CAP 33uF 20% 10V ELECTROLYTIC	1	C88
CAP050225-0805	SMD Capacitor	CAP, SM 2.2UF 50V 10%XR5 0805	2	C73,199
CAP050222-0603	SMD Capacitor	CAP, SM 2200PF 50V 5%NPO 0603	4	C243,244,245,246
CAP050223-0603	SMD Capacitor	CAP, SM 22000PF 50V 10%NPO 0603	4	C90,96,102,104
CAP025106-1206	SMD Capacitor	CAP, SM 10UF cer -20%+80% 25V 1206	8	C57,60,61,77,91.98,101,146
CAP050105-0805	SMD Capacitor	CAP, SM 105PF 50V 5%NPO 0805	1	C241
CAP050152-0603	SMD Capacitor	CAP, SM 1500PF 50V 10%NPO 0603	2	C263,264
CAP050470-0603	SMD Capacitor	CAP, SM 47PF 50V 5%NPO 0603	12	C43,44,48,49,51,54,62,184,189,255,260,26
CAP050104-0603	SMD Capacitor	CAP, SM 0.1UF 50V 10%NPO 0603	25	C1,2,3,4,32,120,121,122,123,175,182,193, 200,205,208,209,211,213,215,216,217,236, 237,238, 239
CAP050104-0805	SMD Capacitor	CAP, SM 0.1UF 100V 10%NPO 0805	16	C147,148,149,150,224,225,226,227,228, 229,230,231,232,233,234,235
CAP050105-1206	SMD Capacitor	CAP, SM 1UF 50V 10%NPO 1206	6	C67,100,171,212,249,250
CAP1020100-B	SMD Ta electrolytic Capacitor	10uf/20V "B" type	2	C55,64
CAP106R3220-A	SMD Ta electrolytic Capacitor	22uf/6.3v "A" type	1	C110
CAP10251R0-A	SMD Ta electrolytic Capacitor	1uf/25v "A" type	1	C56
CAP100102-M	cassette Capacitor	CAP,B 102J 100V ±5%	4	C274,275,276,277
CAP250104-M	cassette Capacitor	0.1uf 250V CBB21B	4	C164,166,168,170
CAP250474-M	cassette Capacitor	0.47uf 250V CBB21B	4	C163,165,167,169
<i>Miscellaneous</i>				
CORE-2R7-0402	SMD Inductor	2.7nH 0402 +/-0.1nH	1	L5
CORE-3R9-0402	SMD Inductor	3.9nH 0402 +/-0.1nH	1	L7
CORE-8R2-0402	SMD Inductor	8.2nH 0402 +/-0.1nH	1	L6
CORE-1R0-0603	SMD Inductor	1nH 0603 +/-0.1nH	3	L3,12,13
CORE-220M	Inductor	22UH 7G17B-220M	4	L8,9,10,11

Part Number		Description	Qty	Reference Designator or Application
PREAMP/INPUT PCB				
CORE-DS80-0603	SMD magnetic ring	600 ohm@100mhz 0603	2	FB1,2
RET- 120-F	thermistor	PTC VISHAY 2381 660 56393 120-ohm nominal	4	RT1,2,3,4
CNCTR-24-1R0	24P Connector	CNCTR, FPC/FFC 24P 1MM VER PCB	1	J3
CY-SMT-16M	SMD crystal oscillator	AS-16.000-15-EXT-SMD-TR(16MHz 15PF)	1	Y2
CY-SMT-12.288M	SMD crystal oscillator	(12.288MHz)贴片 ROHS OSC 12.288MHZ /15PF 50PPM 3.3V SMD	1	OSC1
CNCTR-06-2R0	6P Connector	3*2P common height	1	J2
W-8-120MM	8P harness	AWG #26, CHL0640-8 to CHL0640-8 120m/m	1	J8
RET- 50K-F	thermistor+wire	FTD5-350 50K length : 135	1	TP19,20
3PSK-43D01K	switch LO-HI-HI2	43D01KNOB(10mm)/SD-SS-10P8T-0700-G10	2	SW2,3
5-C50K-90	potentiometer LEVEL	C50K DUAL POT RK09L12B0A31/R0923G1QA1-5-C50K-90	2	RV1,2
RCA JACK*6	RCA connector	RCA JACK*6	1	J5
CNCTR-03-5R0	3P Connector	WF03P	2	TRE1 TER2
W-28-230MM	28P harness	WF28P,24AWG 230mm/ (185mm).	1	(J10 , (J11, (J12) , (J13)
POWER PCB				
<i>Semiconductors</i>				
IC-TL494	SMD IC	PWM IC,SO-16TL494CD	1	U1
IC-KIA7812	SMD IC +12V REGULATOR	EPITAXIAL PLANAR NPN "DPAK",KIA7812	1	U2
DIODE-LL4004	SMD diode	RECTIFIER,1A, LL4004G	2	D10,11
DIODE-IN4148L	SMD diode	SWITCHING SIGNALLL4148	2	D4,5
DIODE-LL4937	SMD diode	FAST RECOVERY,MELFLL4937G	1	D709
DIODE-SMCJ24A	SMD diode	ZD, SMC J24A DO-214AB	1	D9
DIODE-ZMM15	SMD diode	ZENER, 15V/500mW,MINI MELFZMM15	1	D3
TR-YKTA1661	SMD transistor	SMALL SIGNAL PNP "SOT-89",KTA1661Y	2	Q5 ,10
TR-BT3904LT1	SMD transistor	SMALL SIGNAL NPN "SOT-23",MMBT3904	4	Q1,4,17,18
TR-GRKRC102	SMD transistor	EPITAXIAL PLANAR NPN "SOT-23",KRC102S	1	Q8
TR-GRA1504	SMD transistor	SMALL SIGNAL PNP "SOT-23",KTA1504GR	1	Q3
TR-KDS184	SMD diode	EPITAXIAL PLANAR NPN "SOT-23",KDS184	1	D2
FET-IRF3205	MOSFET	N-CH MOSFET,TO-220 IRF3205	6	Q2,9,11,12 ,14,15
TR-FRF1620D	diode	FAST RECOVERY FRF1620D	2	D7,D8
TR-A60	diode	RECTIFIER6A06	1	D1
<i>Resistors</i>				
RES-ESDA-0603	SMD static Resistor	0603 ESDA	1	SX1
RES-1001-0603-F	SMD Resistor	SMD"0603" 1/10W 1K F	2	R33,37
RES-3301-0805-F	SMD Resistor	SMD"0805" 1/8W 3.3K F	1	R4
RES-1502-0805-F	SMD Resistor	SMD"0805" 1/8W 15KF	2	R8,13
RES-1002-0805-F	SMD Resistor	SMD"0805" 1/8W 10KF	2	R18,34
RES-2202-0805-F	SMD Resistor	SMD"0805" 1/8W 22KF	3	R12,14,15
RES-3302-0805-F	SMD Resistor	SMD"0805" 1/8W 33KF	1	R16
RES-4701-0805-F	SMD Resistor	SMD"0805" 1/8W 4.7KF	2	R5,35
RES-4702-0805-F	SMD Resistor	SMD"0805" 1/8W 47KF	1	R17,
RES-6800-0805-F	SMD Resistor	SMD"0805" 1/8W 680 F	1	R9
RES-000-0805-F	SMD Resistor	SMD"0805" 1/8W 0 F	3	R6,20,27
RES-10R0-0805-F	SMD Resistor	SMD"0805" 1/8W 10 F	1	R10
RES-1001-0805-F	SMD Resistor	SMD"0805" 1/8W 1KF	4	R1,R2,R38,R32
RES-221-1206-J	SMD Resistor	SMD"1206" 1/4W 220J	2	R19,26
RES-101-1206-J	SMD Resistor	SMD"1206" 1/4W 100 J	6	R22,23,24,25,29,30
RES-102-1206-J	SMD Resistor	SMD"1206" 1/4W 1KJ	2	R21,R28
RES-2004-0805-F	SMD Resistor	SMD"0805" 1/8W 2MF	1	R3
RES101-2W	Resistor	MOR,2WJ 100 OHM	1	R31
<i>Capacitors</i>				
CAP050105-0805	SMD Capacitor	CAP, SM 1UF 50V 10%CER 0805	1	C2
CAP050104-0805	SMD Capacitor	SMD"0805" 50V 10% 104P	7	C3,10,12,24,25,29,31

Part Number		Description	Qty	Reference Designator or Application
POWER PCB				
CAP050104-1206	SMD Capacitor	SMD"1206" 50V 10% 104P	1	C27
CAP0316100-4	SMD electrolytic Capacitor	ELECTROLYTIC"MVG" 5%, 4 10/16V	1	C6
CAP0335470-6.3	SMD electrolytic Capacitor	ELECTROLYTIC"MVG" 5%,6.3 47/35V	1	C28
CAP0316101-6.3	SMD electrolytic Capacitor	ELECTROLYTIC"MVG" 5%, 6.3 100/16V	3	C1.11.14
CAP100102-M	cassette Capacitor	BOX 100V 5%102J	1	C7
CAP400222-M	cassette Capacitor	BOX 400V 5%222K	1	C15
CAP063105-M	metalized polyester Capacitor	CAP, F 1UF 63V ±5%	1	C5
CAP025332-E	long-pin electrolytic Capacitor	EC,3300uF/25V"WL 25*18 long pin	2	C4,13
CAP050102-E	electrolytic Capacitor	1000/50V SHL 16*20	6	C16,17,18,19,20,21
CAP025102-E	electrolytic Capacitor	1000uF/25V SHL/SD 10×16mm	1	C26
CAP025221-E	electrolytic Capacitor	CAP,E 220UF/25V ±20%11x6 85°C 2R5MMLS	1	C30
<i>Miscellaneous</i>				
CORECL-310A	Inductor	CL-310A	1	L1
CORECL-1700B	Inductor	IRON CORECL-1700B	1	L2
RET- 50K-F	thermistor	FTD5-350 50K	1	TH1
RET- 49K-F	thermistor	DIODE TYPE, 49KOHM +/- 3%KCD493H399G	1	TH2
CORE-TINY4100	transformer	ZF43813 5T(1.2x5):13T(1.5x1)+8T(1.0x1)	1	T1
20PWF-08031	fuse holder	JSF-08031	2	for FH1
AA-PPRC0	relay	SWITCH, RELAY DPST 5A 12V	2	K1,2
	3P	WF03P,16AWG 100mm	2	CON1,2
CNCTR-14-2R0	14P needle	double-row needles 90 degrees, pin distance 2.5MM	1	CON5
CNCTR-8-2R5	8P Connector	WAL-08P one side with harness 2.0mm	1	CON3
SCR-M3*6 MI	screw	BMB PH M3*6 MI-P	6	P1 P2 P3 2PCS soldering after conn locked
TER-JSZ3-79	3P connector	JSZ3-79	1	P1
TER-JSZ4-80-1	4P connector	JSZ4-80-1	1	P3
TER-JSZ4-80-2	4P connector	JSZ4-80-2	1	P2
FIBER7-200	paper pad	200MM*7 *0.4	0.3	Q9,14,15
FIBER6-200	paper pad	200MM*6 *0.3	0.4	Q2,11,12,D7,8
CNCTR-28-1R0	28P Connector	28P LWL0640-28	1	CON6
DISPLAY PCB				
<i>Semiconductors</i>				
IC-BA50D	SMD IC Low Dropout Voltage Regulator	BA50DD0WHFP HRP5	1	Q1
IC-MIC2982YWM	SMD IC	MIC2982YWM High-Voltage High-Current Source Driver Array SSOP-20	2	U503 505
IC-A6276ELW	SMD IC	A6276ELW 16-BIT SERIAL-INPUT, LED DRIVER SSOP-20 (24)	1	U501
IC-74HCT	SMD IC 3-to-8 line decoder/demultiplexer	74HC/HCT238/SO SO16	2	U502 504
TR-FTZ506E	SMD diode	FTZ506E	1	DP501
<i>Resistors</i>				
RES-6202-0603-F	SMD Resistor	RES SMD 62KF 1/10W 0603	4	R502, 505, 511, 512
RES-4302-0603-F	SMD Resistor	RES SMD 43KF 1/10W 0603	4	R503, 504, 507, 508
RES-2700-0805-F	SMD Resistor	RES SMD 270F 1/8W 0805	1	R501
<i>Capacitors</i>				
CAP050104-0603	SMD Capacitor	CAP, SM 104PF 50V 10% 0603	3	C501,502,503
CAP050104-0805	SMD Capacitor	SMD"0805" 50V 10% 104P	2	C1,3
CAP0325220-4	SMD electrolytic Capacitor	ELECTROLYTIC"CHIP"SC/MV"5PHI22uF/25V	1	C2
CAP0310102-10	SMD electrolytic Capacitor	ELECTROLYTIC"MVG/SC" 20%, 10Φ1000uF/10V	1	C6
20F7-12C12P	encoder	ALP5 ENCODER EC12E1220406) /EC1201H-20F7-12C12P	1	ENC501
29.142.04R30	switch	0.5A length of handle is 1MM	2	SW501,SW502
LED-LTG-0732M	LED	LTG-0732M	2	LED501,502

Part Number		Description	Qty	Reference Designator or Application
DISPLAY PCB				
RES220-2W	carbon film Resistor	RES, CF 22R OHM 2W ±5%	1	R3
CNCTR-24-1R0	24P Connector	CNCTR, FPC/FFC 24P 1MM VER PCB horizontal	1	J501
SCR-3x5 NI	screw	STT2 BH, 3x5 NI washer	1	
SCR-M3x6 NI	screw	SMB M3x6 NI-P	3	
HEA-AL 6063	heatsink	(AL 6063,16.0mm×101mm×23mm)	1	
MECHANICAL				
41.1.TINY4R10070	upper cover	AL/ADC12	1	Silver Paint
42.1.TINY4R10070	bottom cover	AL/ADC12	1	Black Sanding Paint/silk
35.1.TINY4R10030	power supply knob	Silver Plated ABS	1	Silver Plated
35.1.TINY4R100300	volume button	Silver Plated ABS	2	Silver Plated
50.12.TINY4R10010	knob bracket	ABS	1	Polishing
43.11.TINY4R1000	acryl	Acryl GRAY	1	
50.13.TINY4R10010	copper column	BSBM 5.0	4	
50.12.TINY4R10011	bracket	material increasing(PC 0.25T)	1	
53.14.045001500371	heatsink	SPCC3.0,L=45mm	1	
53.14.069001500371	heatsink	SPCC3.0,L=69mm	1	
48.171.0180090020	nowoven cloth	FELT 18.0x9.0x2t	2	
48.142.035023R201	silicone film	35mm 23mm 0.2mm	1	USED ON HEATSINK
32.13.0563R005271	silicone film	STT2 BH, 3x5 NI washer	1	
48.142.070018R202	silicone film	HF300P,70mm×18mm×0.2mm	1	
48.142.045020R202	silicone film	HF300P,45mm×20mm×0.2mm	1	
24.11.003052	fuse	FUSE30A LITTEL	2	FH1
48.152.0320320R50	paper pad	32mm 32mm 0.5mm	1	C18,20,21
48.152.1541130R50	paper pad	single glued	1	shield cover
48.152.1541130R51	paper pad	single glued	1	shield cover
48.192.1200500R10	protecting film	protecting film	1	stick on window
50.12.SW30012	spacer		1	between screw and heatsink
31.11.0213R01400	screw	SMP 3x14 BK	1	fix small heatsink
18.1240.0006050	24P harness	soft harness 24P 60MM	1	J3-LED PCB J501
31.11.0913R00610	screw	DF SERIES , M3x6 NI-P	4	DF SERIES USED ON LED PCB AND TOP COVER
31.13.0963R00610	screw	DF SERIES , M3x6 NI-P	6	DF SERIES USED ON LED PCB AND
31.11.0513R00610	screw	SMB M3x6 NI-P	10	
31.13.0563R01000	screw	STT2 BH, 3x10 BLACK	2	for RCA JACK
31.11.0514R02000	screw	SMB M4*20 BLACK	4	
31.11.0564R01411	screw	SMB M4x14 NI-P LOCK HEAD	4	
31.11.0413R01210	screw	SCREW, M3X12 90CS FLAT PHIL 镀镍	2	
53.12.0210016R100	piezo ceramic heatsink	21*16*0.1t(MM)	1	for VR2
34.19.0030030054	AL column	H5 L=3 diameter of the hole is 3	2	
54.11.1080	bunch	80mm	2	
63.15.TINY4R10010	sheild cover	Tinplate 0.2T	1	

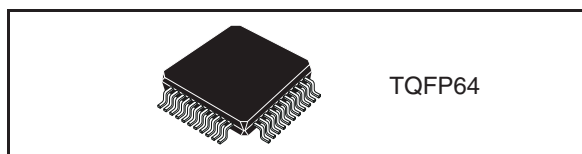


STA309A

Multi-channel digital audio processor with DDX®

Features

- 8 channels of 24-bit DDX®
- >100 dB SNR and dynamic range
- Selectable 32 kHz - 192 kHz input sample rates
- 6 channels of DSD/SACD input
- Digital gain/attenuation +58 dB to -100 dB in 0.5 dB steps
- Soft volume update
- Individual channel and master gain/attenuation plus channel trim (-10 dB to +10 dB)
- Up to 10 independent 32-bit user programmable biquads (EQ) per channel
- Bass/treble tone control
- Pre and post EQ full 8-channel input mix on all 8 channels
- Dual independent limiters/compressors
- Dynamic range compression or anti-clipping modes
- AutoModes:
 - 5-band graphic EQ
 - 32 preset EQ curves (rock, jazz, pop, etc.)
 - Automatic volume controlled loudness
 - 5.1 to 2-channel downmix
 - Simultaneous 5.1- and 2-channel downmix outputs
 - 3 preset volume curves
 - 2 preset anti-clipping modes
 - Preset movie nighttime listening mode
 - Preset TV channel/commercial AGC mode
 - 5.1, 2.1 bass management configurations
 - AM frequency automatic output PWM frequency shifting
 - 8 preset crossover filters
- Individual channel and master soft/hard mute
- Automatic zero-detect and invalid input mute
- Automatic invalid input detect mute



- Advanced PopFree operation
- Advanced AM interference frequency switching and noise suppression modes
- I²S output channel mapping function
- Independent channel volume and DSP bypass
- Channel mapping of any input to any processing/DDX® channel
- DC blocking selectable high-pass filter
- Selectable per-channel DDX® damped ternary or binary PWM output
- Max power correction for lower full-power THD
- Variable per channel DDX® output delay control
- 192 kHz internal processing sample rate, 24-bit to 36-bit precision

Description

The STA309A is a single chip solution for digital audio processing and control in multi-channel applications. It provides output capabilities for DDX® (direct digital amplification). In conjunction with a DDX® power device, it provides high-quality, high-efficiency, all digital amplification. The device is extremely versatile, allowing for input of most digital formats including 6.1/7.1-channel and 192 kHz, 24-bit DVD-audio, DSD/SACD. In 5.1 application the additional 2 channels can be used for audio line-out or headphone drive. In speaker mode, with 8 channel outputs in parallel, the STA309A can deliver more than 1 W.

Table 1. Device summary

Order code	Package
STA309A	TQFP64

1 Block diagram

Figure 1. Block diagram

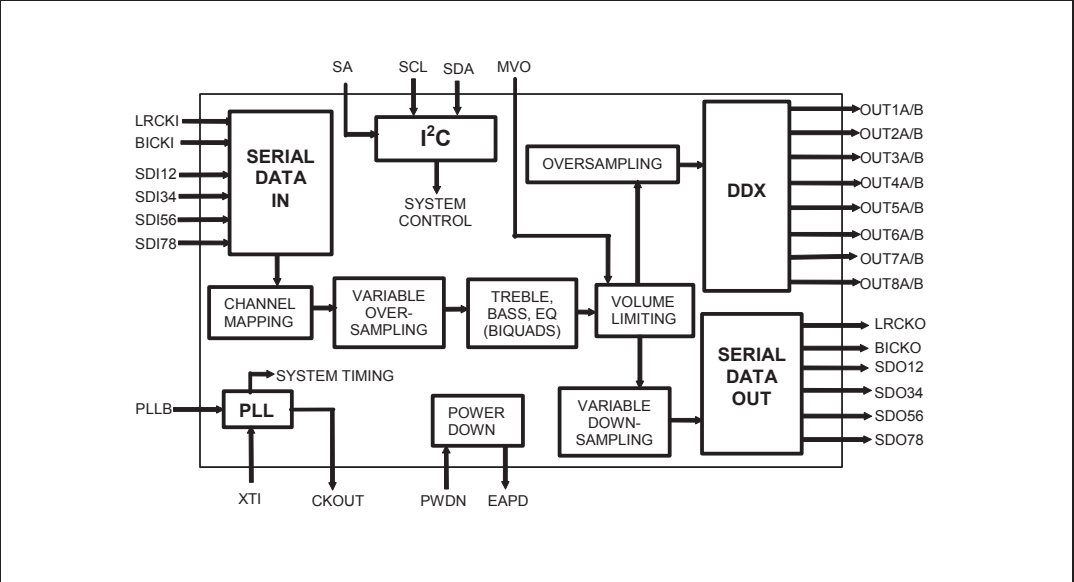
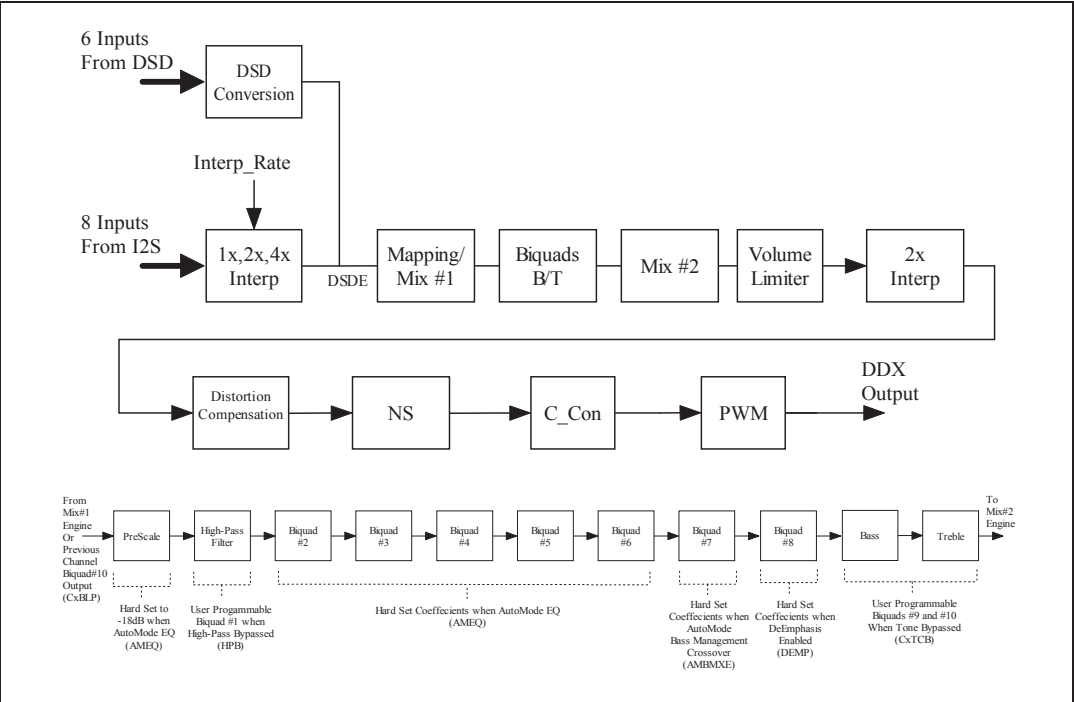


Figure 2. Channel signal flow



2 Pin connections

Figure 3. Pin connection (top view)

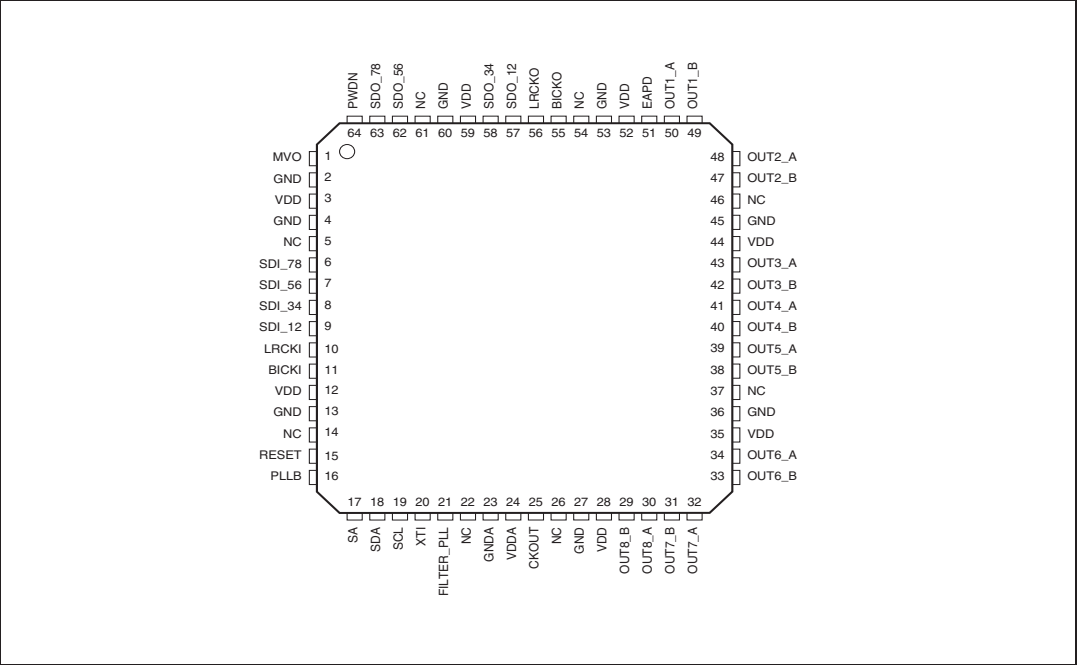


Table 2. Pin description

Pin	Type	Name	Description
1	5-V tolerant TTL input buffer	MVO/DSD_CLK	Master volume override/ DSD input clock
6	5-V tolerant TTL input buffer	SDI_78/DSD_6	Input serial data channels 7 & 8/ DSD input channel 6
7	5-V tolerant TTL input buffer	SDI_56/DSD_5	Input serial data channels 5 & 6/ DSD input channel 5
8	5-V tolerant TTL input buffer	SDI_34/DSD_4	Input serial data channels 3 & 4/ DSD input channel 4
9	5-V tolerant TTL input buffer	SDI_12/DSD_3	Input serial data channels 1 & 2/ DSD input channel 3
10	5-V tolerant TTL input buffer	LRCKI/DSD_2	Input left/right clock/ DSD input channel 2
11	5-V tolerant TTL input buffer	BICKI/DSD_1	Input serial clock/ DSD input channel 1
15	5-V tolerant TTL schmitt trigger input buffer	RESET	Global reset
16	CMOS input buffer with pull-down	PLL_BYPASS	Bypass phase locked loop

Table 2. Pin description (continued)

Pin	Type	Name	Description
17	CMOS input buffer with pull-down	SA	Select address (I ² C)
18	Bidirectional buffer: 5-V tolerant TTL schmitt trigger input; 3.3-V capable 2mA slew-rate controlled output.	SDA	Serial data (I ² C)
19	5-V tolerant TTL schmitt trigger input buffer	SCL	Serial clock (I ² C)
20	5-V tolerant TTL schmitt trigger input buffer	XTI	Crystal oscillator input (clock input)
21	Analog pad	FILTER_PLL	PLL filter
23	Analog ground	GNDA	PLL ground
24	3.3V analog supply voltage	VDDA	PLL supply
25	3.3-V capable TTL tristate 4mA output buffer	CKOUT	Clock output
29	3.3-V capable TTL 16mA output buffer	OUT8B	PWM channel 8 output B
30	3.3-V capable TTL 16mA output buffer	OUT8A	PWM channel 8 output A
31	3.3-V capable TTL 16mA output buffer	OUT7B	PWM channel 7 output B
32	3.3-V capable TTL 16mA output buffer	OUT7A	PWM channel 7 output A
33	3.3-V capable TTL 16mA output buffer	OUT6B	PWM channel 6 output B
34	3.3-V capable TTL 16mA output buffer	OUT6A	PWM channel 6 output A
38	3.3-V capable TTL 16mA output buffer	OUT5B	PWM channel 5 output B
39	3.3-V capable TTL 16mA output buffer	OUT5A	PWM channel 5 output A
40	3.3-V capable TTL 16mA output buffer	OUT4B	PWM channel 4 output B
41	3.3-V capable TTL 16mA output buffer	OUT4A	PWM channel 4 output A
42	3.3-V capable TTL 16mA output buffer	OUT3B	PWM channel 3 output B
43	3.3-V capable TTL 16mA output buffer	OUT3A	PWM channel 3 output A
47	3.3-V capable TTL 16mA output buffer	OUT2B	PWM channel 2 output B

Table 2. Pin description (continued)

Pin	Type	Name	Description
48	3.3-V capable TTL 16mA output buffer	OUT2A	PWM channel 2 output A
49	3.3-V capable TTL 16mA output buffer	OUT1B	PWM channel 1 output B
50	3.3-V capable TTL 16mA output buffer	OUT1A	PWM channel 1 output A
51	3.3-V capable TTL 4mA output buffer	EAPD	Ext. amp power-down
55	3.3-V capable TTL 2mA output buffer	BICKO	Output serial clock
56	3.3-V capable TTL 2mA output buffer	LRCKO	Output left/right clock
57	3.3-V capable TTL 2mA output buffer	SDO_12	Output serial data channels 1&2
58	3.3-V capable TTL 2mA output buffer	SDO_34	Output serial data channels 3&4
62	3.3-V capable TTL 2mA output buffer	SDO_56	Output serial data channels 5&6
63	3.3-V capable TTL 2mA output buffer	SDO_78	Output serial data channels 7&8
64	5-V tolerant TTL schmitt trigger input buffer	PWDN	Device power-down
3,12,28,35, 44,52,59	3.3-V digital supply voltage	VDD	3.3-V supply
2,4,13,27, 36,45,53,60	Digital ground	GND	Ground
5, 14, 22, 26,37,46,54, 61		NC	Not connected

Inverter

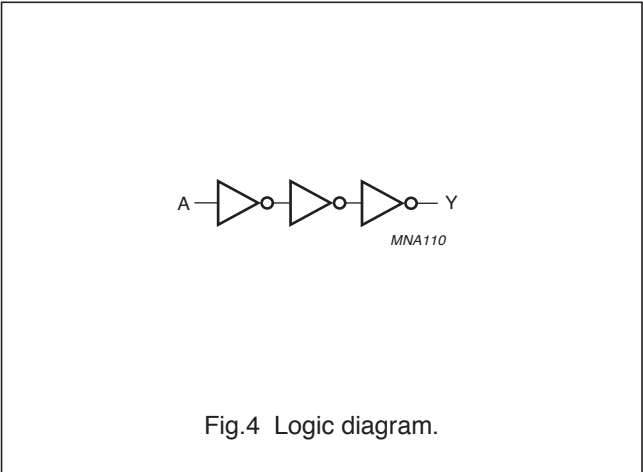
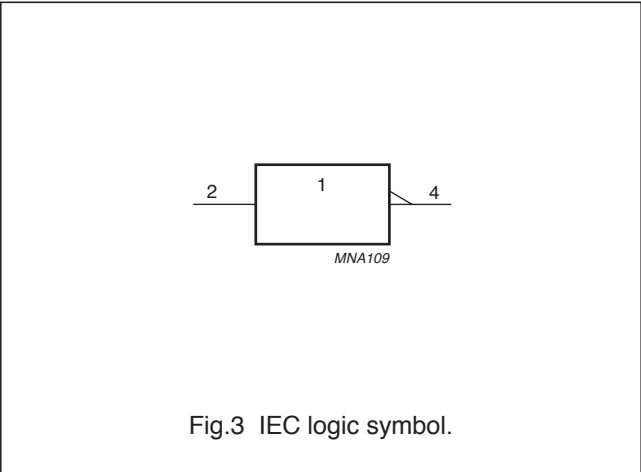
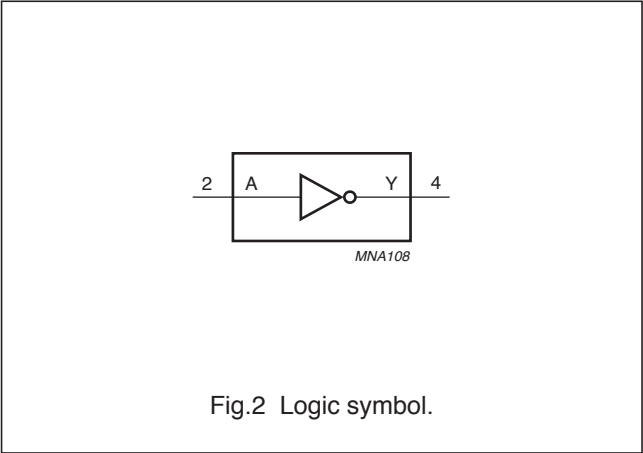
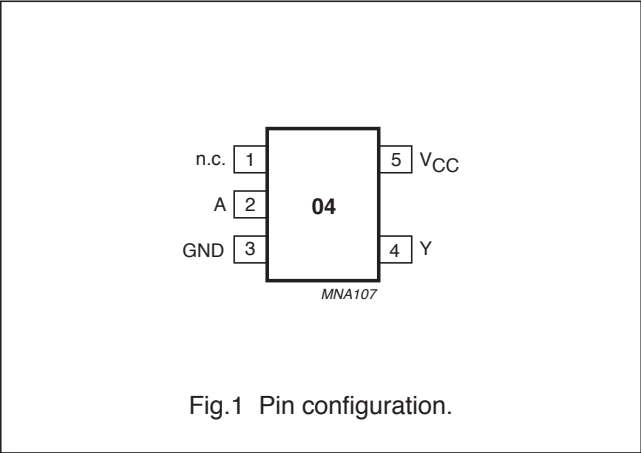
74AHC1G04; 74AHCT1G04

ORDERING AND PACKAGE INFORMATION

TYPE NUMBER	PACKAGES					
	TEMPERATURE RANGE	PINS	PACKAGE	MATERIAL	CODE	MARKING
74AHC1G04GW	−40 to +125 °C	5	SC-88A	plastic	SOT353	AC
74AHCT1G04GW	−40 to +125 °C	5	SC-88A	plastic	SOT353	CC
74AHC1G04GV	−40 to +125 °C	5	SC-74A	plastic	SOT753	A04
74AHCT1G04GV	−40 to +125 °C	5	SC-74A	plastic	SOT753	C04

PINNING

PIN	SYMBOL	DESCRIPTION
1	n.c.	not connected
2	A	data input A
3	GND	ground (0 V)
4	Y	data output Y
5	V _{CC}	supply voltage



MC9S08GT16A/GT8A Features

8-Bit HCS08 Central Processor Unit (CPU)

- 40-MHz HCS08 CPU
- HC08 instruction set with added BGND instruction
- Support for up to 32 interrupt/reset sources

Memory Options

- FLASH read/program/erase down to 1.8 V
- Up to 16K FLASH; up to 2K RAM

Power-Saving Modes

- Three very low power stop modes
- Reduced power wait mode
- Very low power real time interrupt for use in run, wait, and stop

Clock Source Options

- Clock sources to internal hardware frequency locked-loop (FLL): internal, external, crystal, or resonator
- Internal clock with $\pm 0.2\%$ trimming resolution and $\pm 0.5\%$ deviation across voltage or across temperature

System Protection

- Optional watchdog computer operating properly (COP) reset
- Low-voltage detection with reset or interrupt
- Illegal opcode detection with reset
- Illegal address detection with reset
- FLASH block protect and security

Peripherals

- ATD — 8-channel, 10-bit analog-to-digital converter
- SCI — Two serial communications interface modules
- SPI — Serial peripheral interface module
- IIC — Inter-integrated circuit bus module
- Timer — One 3-channel timer PWM module (TPM) plus one 2-channel TPM
- KBI — 8-pin keyboard interrupt module

Input/Output

- 8 high-current pins (20 mA each)

- Software selectable pullups on ports when used as input
- Internal pullup on $\overline{\text{RESET}}$ and IRQ pin to reduce customer system cost
- Up to 38 general-purpose input/output (I/O) pins, plus one output-only pin, depending on package selection

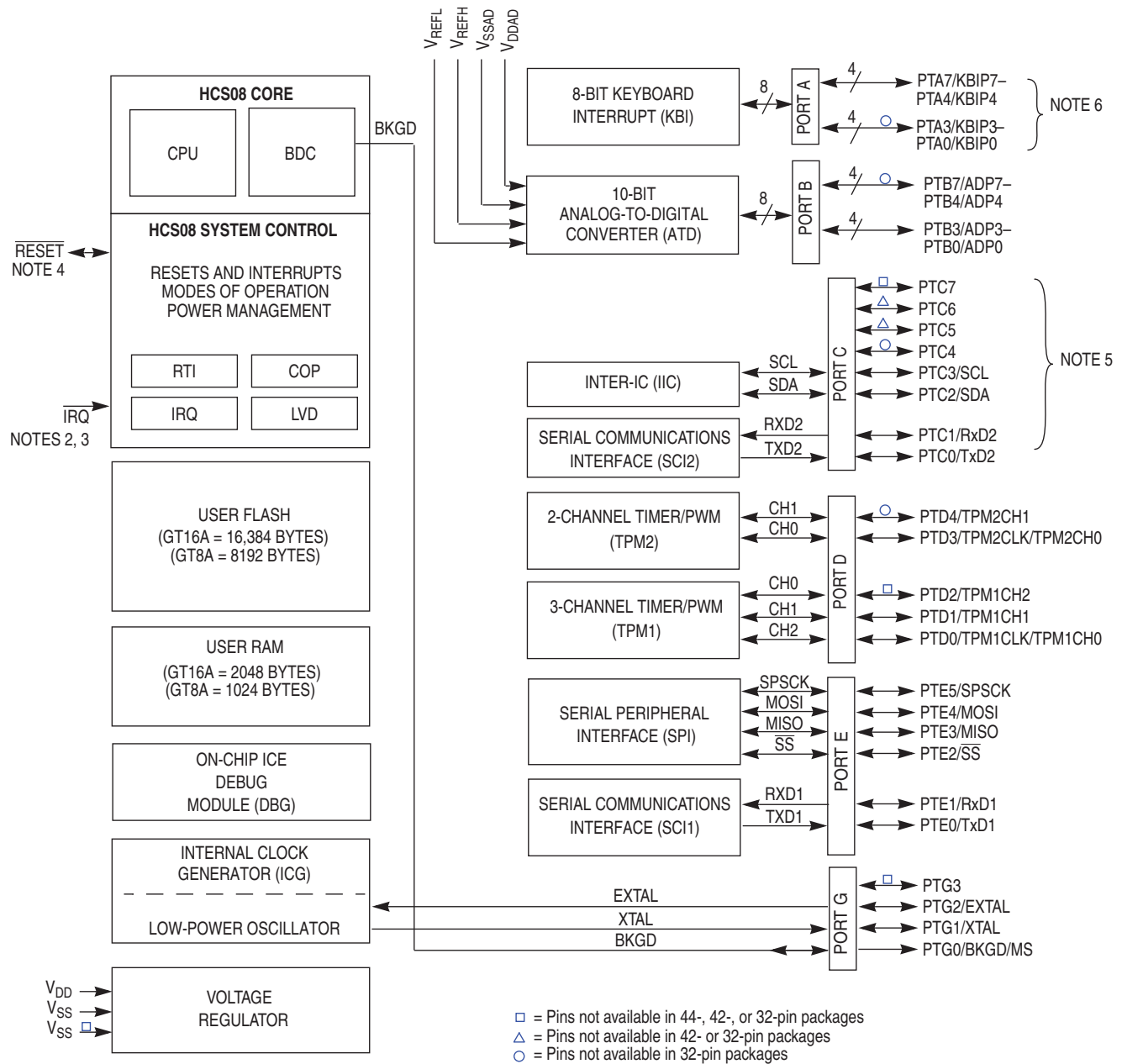
Development Support

- Background debugging system
- Breakpoint capability to allow single breakpoint setting during in-circuit debugging (plus two more breakpoints in on-chip debug module)
- On-chip, in-circuit emulation (ICE) debug module with real-time bus capture. On-chip ICE debug module containing two comparators and nine trigger modes. Eight deep FIFO for storing change-of-flow addresses and event-only data.
- Single-wire background debug interface

Package Options

- 48-pin QFN
- 44-pin QFP
- 42-pin PSDIP
- 32-pin QFN

Device Overview



NOTES:

1. Port pins are software configurable with pullup device if input port.
2. Pin contains pullup/pulldown device if IRQ enabled (IRQPE = 1).
3. IRQ does not have a clamp diode to V_{DD}. IRQ should not be driven above V_{DD}.
4. Pin contains integrated pullup device.
5. High current drive
6. Pins PTA[7:4] contain both pullup and pulldown devices. Pulldown available when KBI enabled (KBIPn = 1).

Figure 1-1. MC9S08GT16A/GT8A Block Diagram

Table 1-2 lists the functional versions of the on-chip modules.

Table 1-2. Block Versions

Module	Version
Analog-to-Digital Converter (ATD)	3
Internal Clock Generator (ICG)	4
Inter-Integrated Circuit (IIC)	1
Keyboard Interrupt (KBI)	1
Serial Communications Interface (SCI)	1
Serial Peripheral Interface (SPI)	3
Timer Pulse-Width Modulator (TPM)	2
Central Processing Unit (CPU)	2

1.2 System Clock Distribution

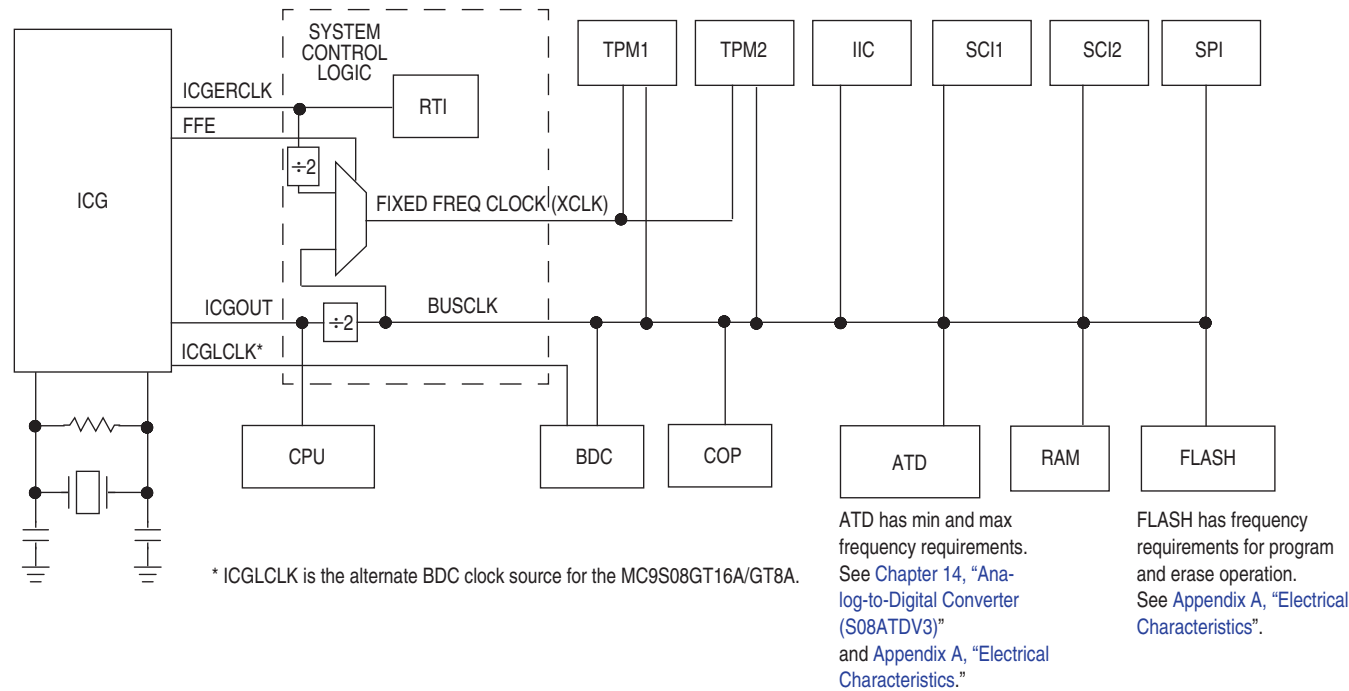


Figure 1-2. System Clock Distribution Diagram

Chapter 2

Pins and Connections

2.1 Introduction

This section describes signals that connect to package pins. It includes a pinout diagram, a table of signal properties, and detailed discussion of signals.

2.2 Device Pin Assignment

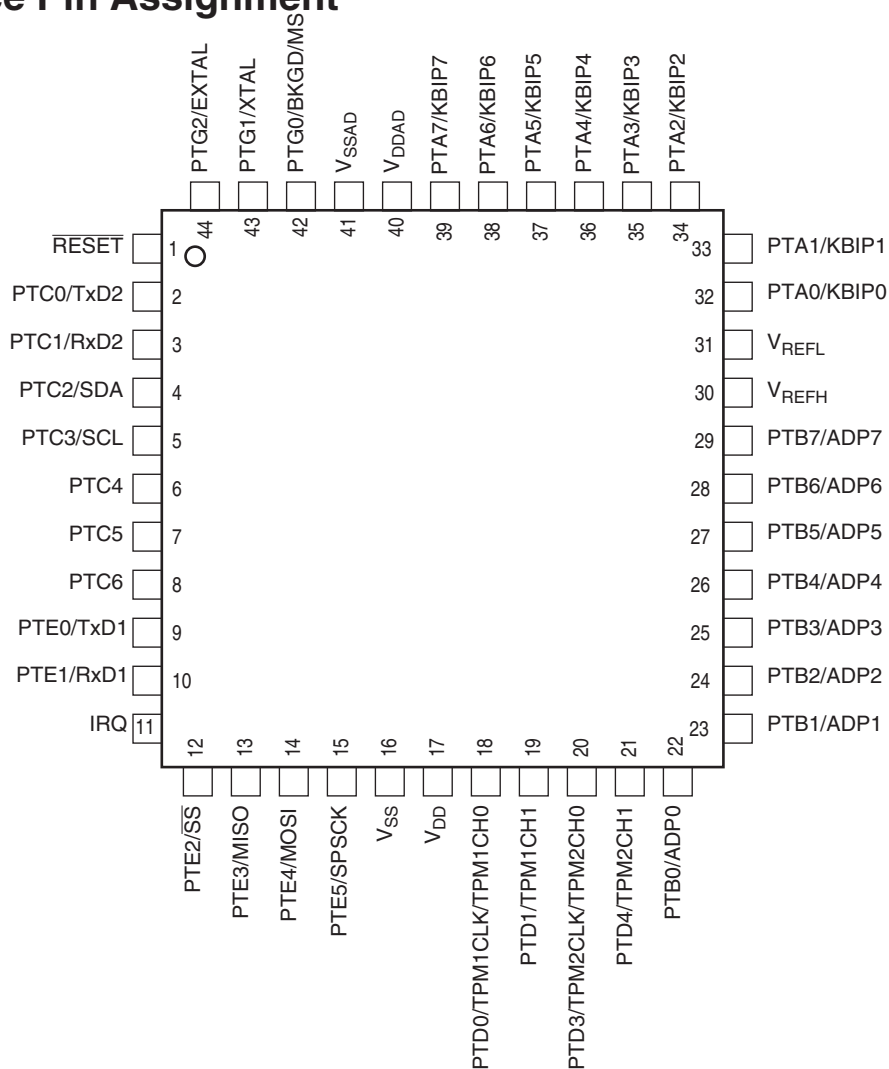


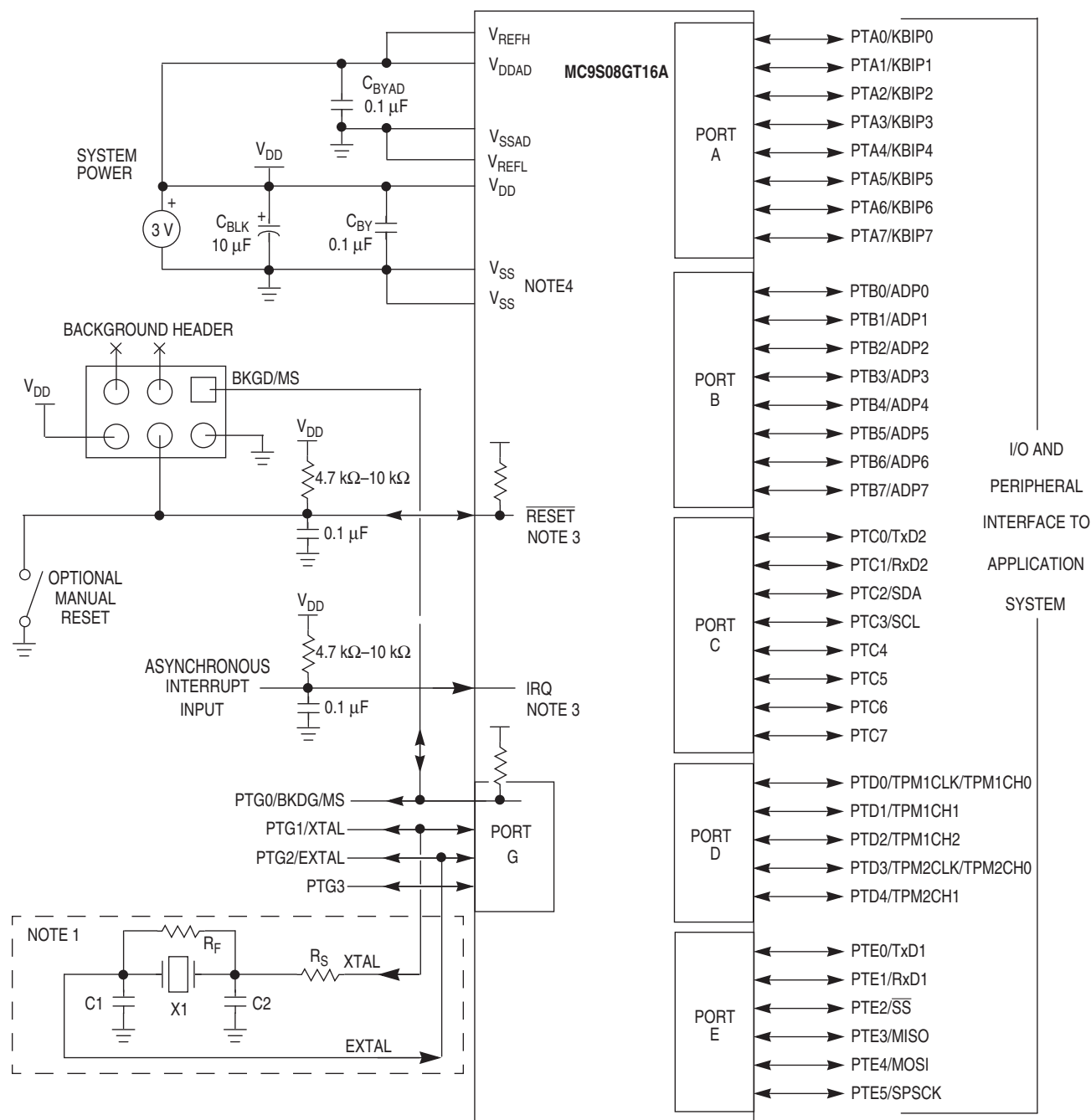
Figure 2-2. MC9S08GT16A/GT8A in 44-Pin QFP Package

MC9S08GT16A/GT8A Data Sheet, Rev. 1

2.3 Recommended System Connections

Pins and Connections

Figure 2-5 shows pin connections that are common to almost all MC9S08GT16A application systems. A more detailed discussion of system connections follows.



NOTES:

1. Not required if using the internal oscillator option.
2. The 48-pin QFN has 2 V_{SS} pins (V_{SS1} and V_{SS2}), both of which must be connected to GND.
3. RC filters on RESET and IRQ are recommended for EMC-sensitive applications and systems.

Figure 2-5. Basic System Connections

March 5, 2009
 Datasheet No – PD97381

International IOR Rectifier

IRS20957S Protected Digital Audio Driver

Features

- Floating PWM input enables easy half bridge implementation
- Programmable bidirectional over-current protection with self-reset function
- Programmable preset dead-time for improved THD performances
- High noise immunity
- $\pm 100V$ ratings deliver up to 500W in output power
- 3.3 V/ 5 V logic compatible input
- Operates up to 800kHz

Typical Applications

- Home theatre systems
- Mini component stereo systems
- Powered speaker systems
- General purpose audio power amplifiers

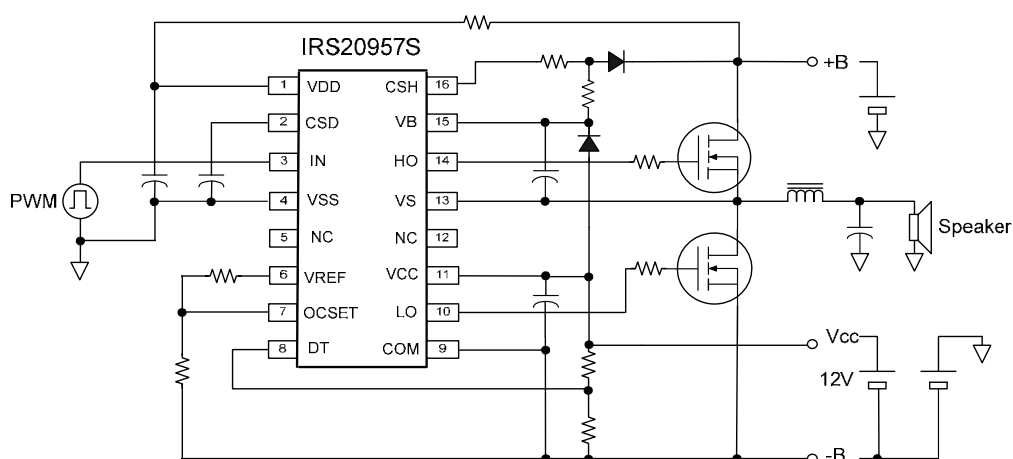
Product Summary

Topology	Half-Bridge
$V_{\text{OFFSET (max)}}$	+/- 100 V
I_{O+} & I_{O-} (typical)	1.0 A & 1.2 A
Selectable deadtime	15/25/35/80ns
Ton & toff (typical)	95ns & 80ns
OC protection delay	500ns (max)
Shutdown propagation delay	250ns (max)

Package



Typical Connection Diagram

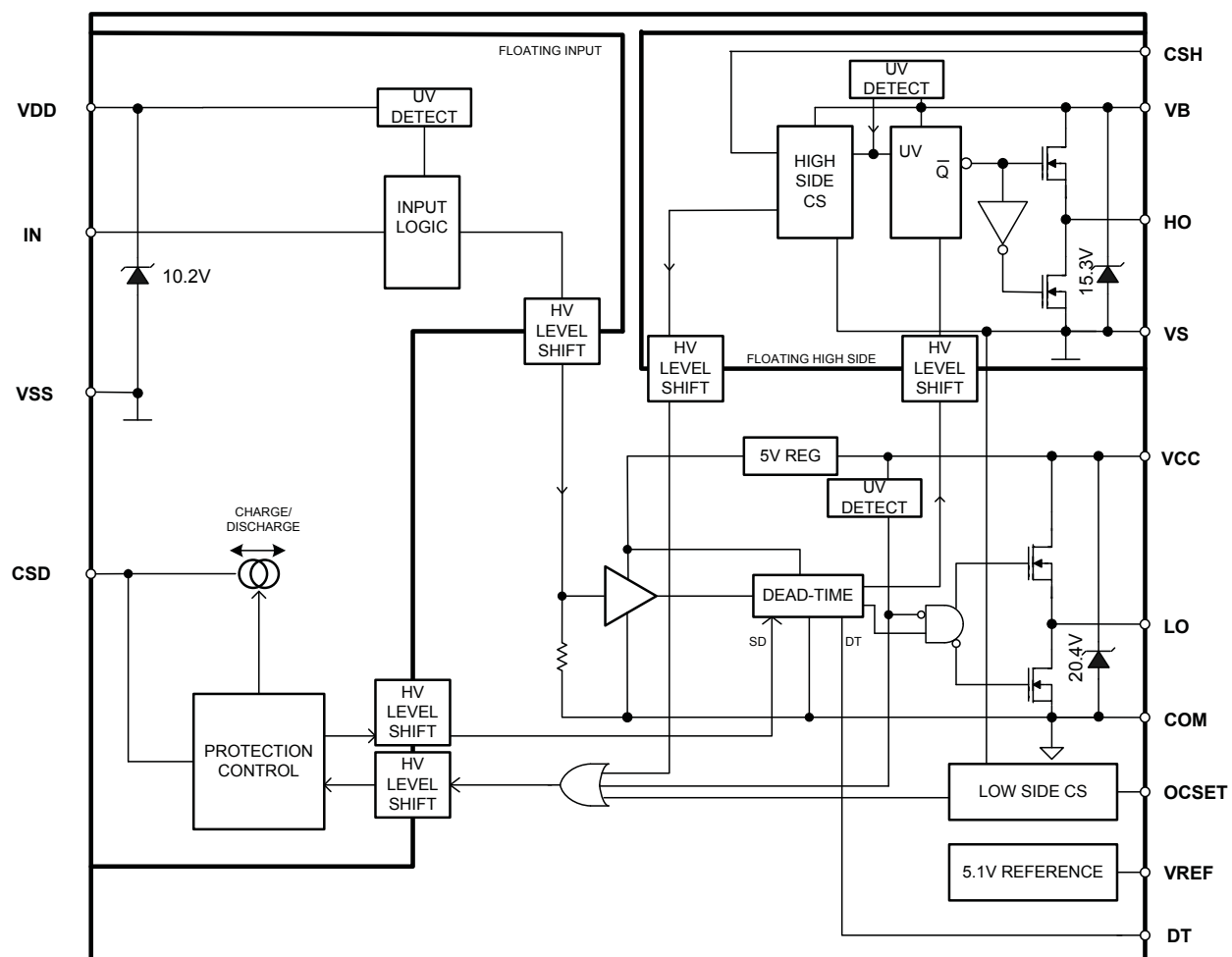


Note: Please refer to Lead Assignments for correct pin configuration. This diagram shows electrical connections only.

International
IR Rectifier

IRS20957S

Functional Block Diagram



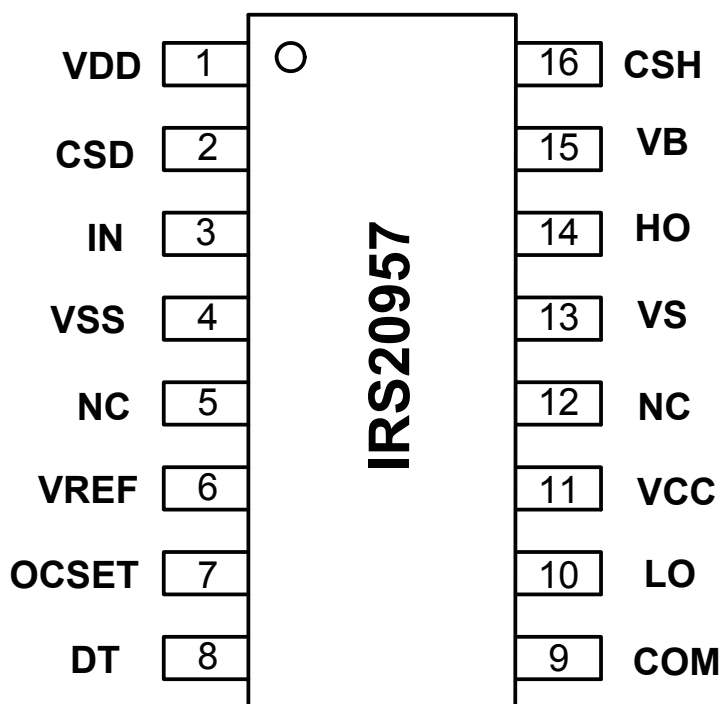
International
IR Rectifier

IRS20957S

Lead Definitions

Pin #	Symbol	Description
1	VDD	Floating input positive supply
2	CSD	Shutdown timing capacitor, referenced to VSS
3	IN	PWM non-inverting input, in phase with HO
4	VSS	Floating input supply return
5	NC	
6	VREF	5V reference output for setting OCSET
7	OCSET	Low side over current threshold setting, referenced to COM
8	DT	Input for programmable dead-time, referenced to COM
9	COM	Low side supply return
10	LO	Low side output
11	VCC	Low side logic supply
12	NC	
13	VS	High side floating supply return
14	HO	High side output
15	VB	High side floating supply
16	CSH	High side over current sensing input, referenced to VS

Lead Assignments





MIC2981/2982

High-Voltage High-Current Source Driver Array

General

The MIC2981/82 is an 8-channel, high-voltage, high-current source driver array ideal for switching high-power loads from logic-level TTL, CMOS, or PMOS control signals.

These drivers can manage multiple loads of up to 50V and 500mA, limited only by package power dissipation.

Micrel's MIC2981/82 features inputs compatible with 5V TTL and 5V to 15V CMOS or PMOS logic outputs. Micrel's dual-marked device replaces either UDN2981 or UDN2982 devices.

The MIC2981/82 is available in the 18-pin plastic DIP and 18-lead wide SOP package. Both devices operate in the industrial temperature range.

Features

- Output voltage to 50V
- Output current to 500mA
- Transient-protected outputs
- Integral clamp diodes
- TTL, CMOS, or PMOS compatible inputs

Applications

- Relay and solenoid switching
- Stepping motor
- LED and incandescent displays

Ordering Information

Reference	Part Number Manufacturing*	PbFree	Temperature Range	Package
MIC2981BN**	MIC2981/82BN	MIC2981/82YN	−40°C to +85°C	18-pin DIP
MIC2982BN**	MIC2981/82BN	MIC2981/82YN	−40°C to +85°C	18-pin DIP
MIC2981BWM**	MIC2981/82BWM	MIC2981/82YWM	−40°C to +85°C	18-pin wide SOP
MIC2982BWM**	MIC2981/82BWM	MIC2981/82YWM	−40°C to +85°C	18-pin wide SOP

* Order entry P/N.

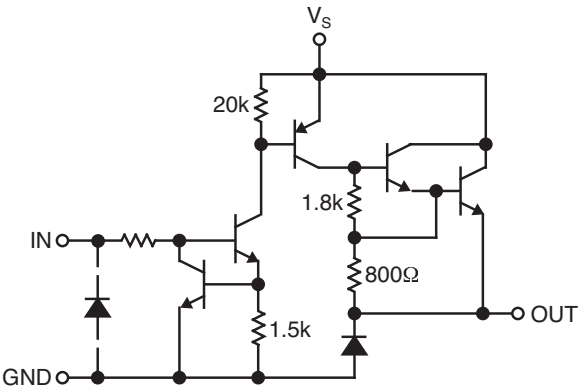
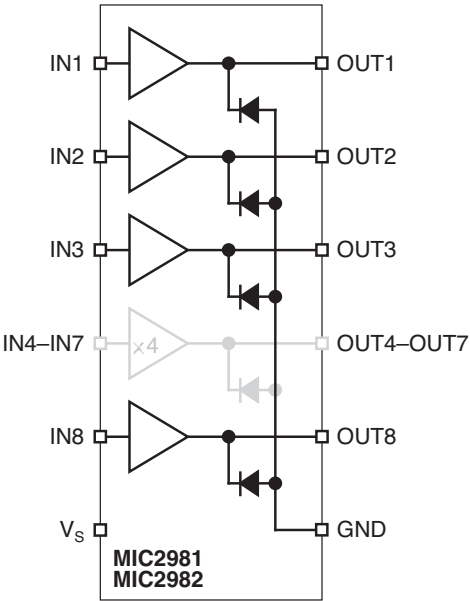
**Orders for MIC2981BN or MIC2982BN will be filled with dual-marked MIC2981/82BN.

**Orders for MIC2981YN or MIC2982YN will be filled with dual-marked MIC2981/82YN.

**Orders for MIC2981BWM or MIC2982BWM will be filled with dual-marked MIC2981/82BWM.

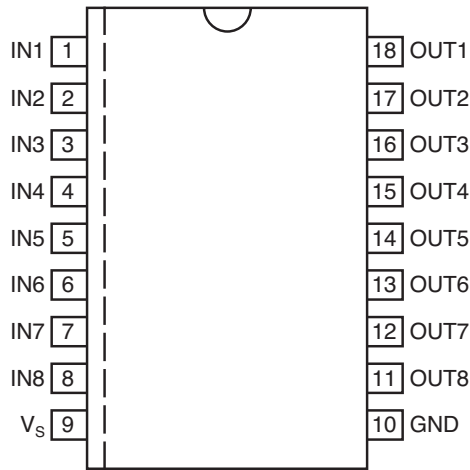
**Orders for MIC2981YWM or MIC2982YWM will be filled with dual-marked MIC2981/82YWM.

Functional Diagrams



Typical MIC2981/2982 Source Driver

Pin Configuration

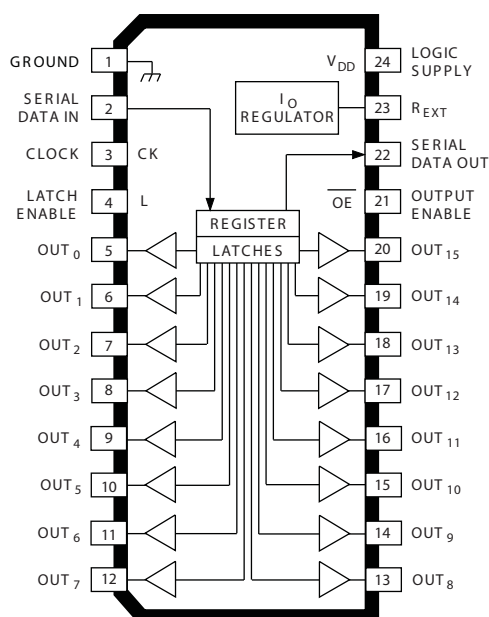


18-Pin DIP (N)
18-Pin Wide SOP (WM)

Pin Description

Pin No.	Pin No.	Pin Name	Pin Function
1–8	IN1–IN8	Input 1 through Input 8: Base drive to driver input transistor.	
9	V _s	Supply Input	
10	GND	Ground	
11–18	OUT8–OUT1	Output 8 through Output 1: Emitter of Darlington driver output.	

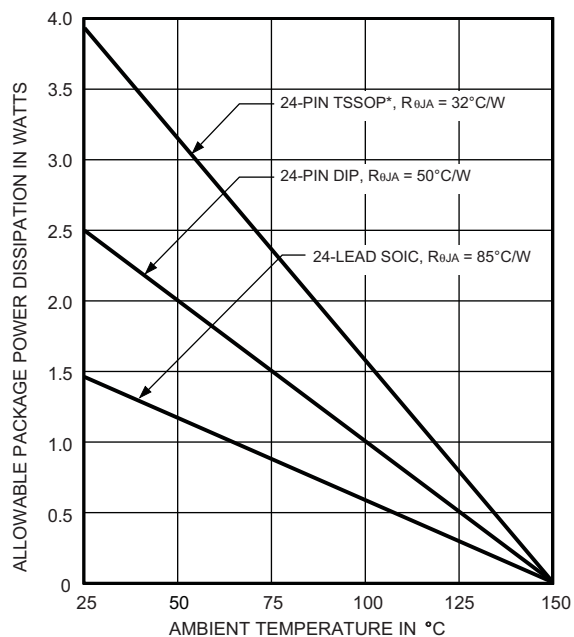
A6276ELW



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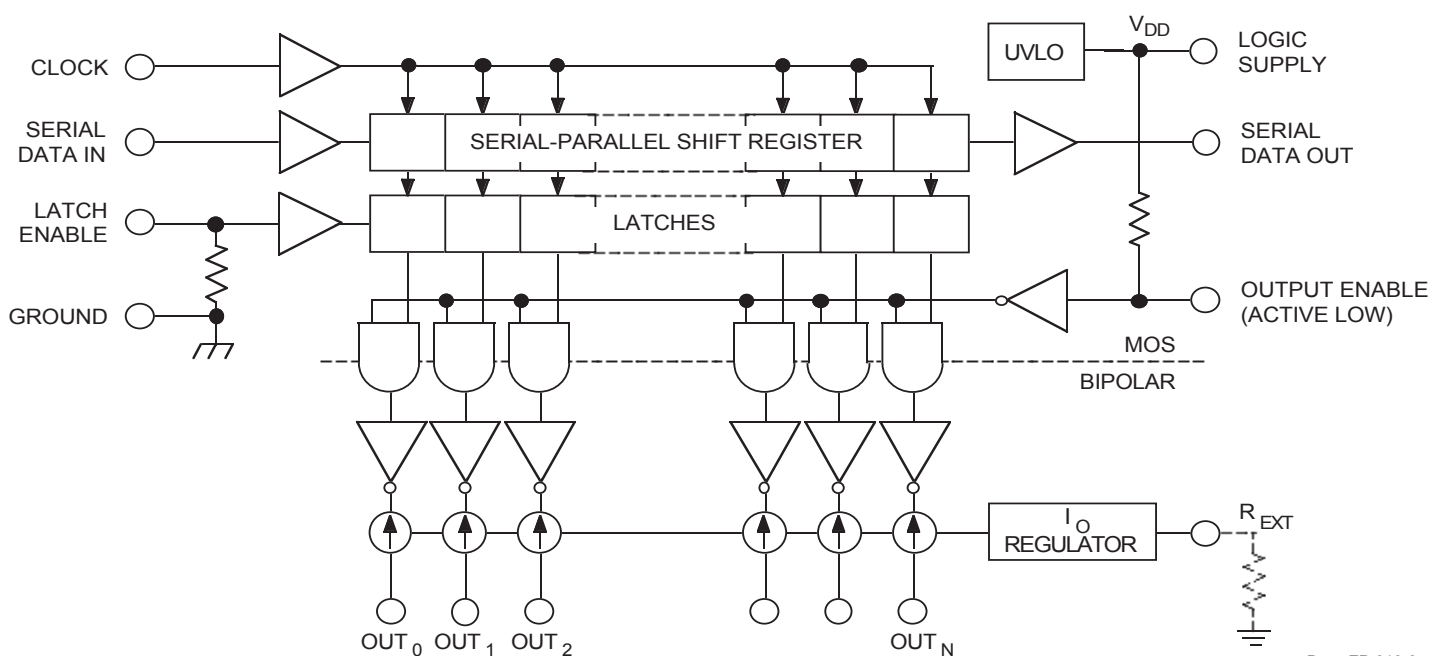
6276

16-BIT SERIAL-INPUT, CONSTANT-CURRENT LATCHED LED DRIVER



*Mounted on single-layer, two-sided PCB, with 3.8 in² copper each side;
additional information on Allegro Web site

FUNCTIONAL BLOCK DIAGRAM



Dwg. FP-013-3

3-to-8 line decoder/demultiplexer

74HC/HCT238

FEATURES

- Demultiplexing capability
- Multiple input enable for easy expansion
- Ideal for memory chip select decoding
- Active HIGH mutually exclusive outputs
- Output capability: standard
- I_{CC} category: MSI

GENERAL DESCRIPTION

The 74HC/HCT238 are high-speed Si-gate CMOS devices and are pin compatible with low power Schottky TTL (LSTTL). They are specified in compliance with JEDEC standard no. 7A.

The 74HC/HCT238 decoders accept three binary weighted address inputs (A₀, A₁, A₂) and when enabled,

provide 8 mutually exclusive active HIGH outputs (Y₀ to Y₇).

The “238” features three enable inputs: two active LOW ($\bar{E}_1$ and $\bar{E}_2$) and one active HIGH (E₃). Every output will be LOW unless $\bar{E}_1$ and $\bar{E}_2$ are LOW and E₃ is HIGH.

This multiple enable function allows easy parallel expansion of the “238” to a 1-of-32 (5 lines to 32 lines) decoder with just four “238” ICs and one inverter.

The “238” can be used as an eight output demultiplexer by using one of the active LOW enable inputs as the data input and the remaining enable inputs as strobes. Unused enable inputs must be permanently tied to their appropriate active HIGH or LOW state.

The “238” is identical to the “138” but has non-inverting outputs.

QUICK REFERENCE DATA

GND = 0 V; T_{amb} = 25 °C; t_r = t_f = 6 ns

SYMBOL	PARAMETER	CONDITIONS	TYPICAL		UNIT
			HC	HCT	
t _{PHL} / t _{PLH}	propagation delay	C _L = 15 pF; V _{CC} = 5 V			
	A _n to Y _n		14	18	ns
	E ₃ to Y _n		16	20	ns
	$\bar{E}_n$ to Y _n		17	21	ns
C _I	input capacitance		3.5	3.5	pF
C _{PD}	power dissipation capacitance per package	notes 1 and 2	72	76	pF

Notes

1. C_{PD} is used to determine the dynamic power dissipation (P_D in μW):

$$P_D = C_{PD} \times V_{CC}^2 \times f_i + \sum (C_L \times V_{CC}^2 \times f_o) \text{ where:}$$

f_i = input frequency in MHz

f_o = output frequency in MHz

∑ (C_L × V_{CC}² × f_o) = sum of outputs

C_L = output load capacitance in pF

V_{CC} = supply voltage in V

2. For HC the condition is V_I = GND to V_{CC}
For HCT the condition is V_I = GND to V_{CC} – 1.5 V

ORDERING INFORMATION

See “74HC/HCT/HCU/HCMOS Logic Package Information”.

3-to-8 line decoder/demultiplexer

74HC/HCT238

PIN DESCRIPTION

PIN NO.	SYMBOL	NAME AND FUNCTION
1, 2, 3	A ₀ to A ₂	address inputs
4, 5	$\bar{E}_1, \bar{E}_2$	enable inputs (active LOW)
6	E ₃	enable input (active HIGH)
8	GND	ground (0 V)
15, 14, 13, 12, 11, 10, 9, 7	Y ₀ to Y ₇	outputs (active HIGH)
16	V _{CC}	positive supply voltage

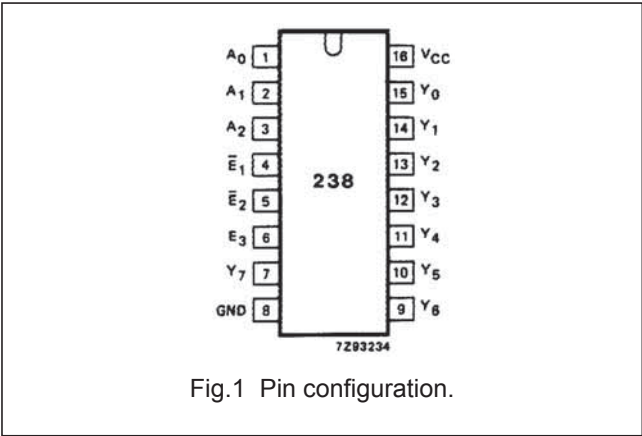


Fig.1 Pin configuration.

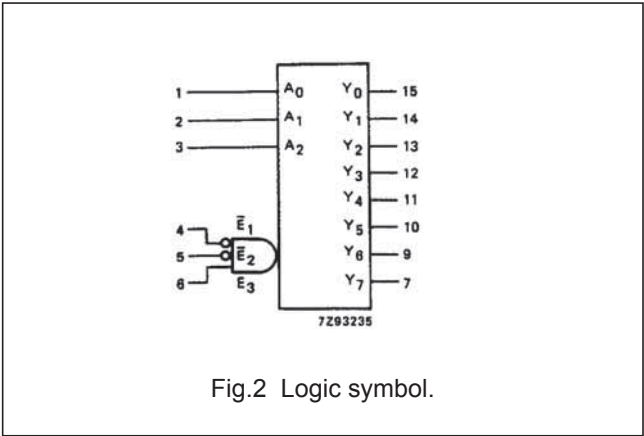


Fig.2 Logic symbol.

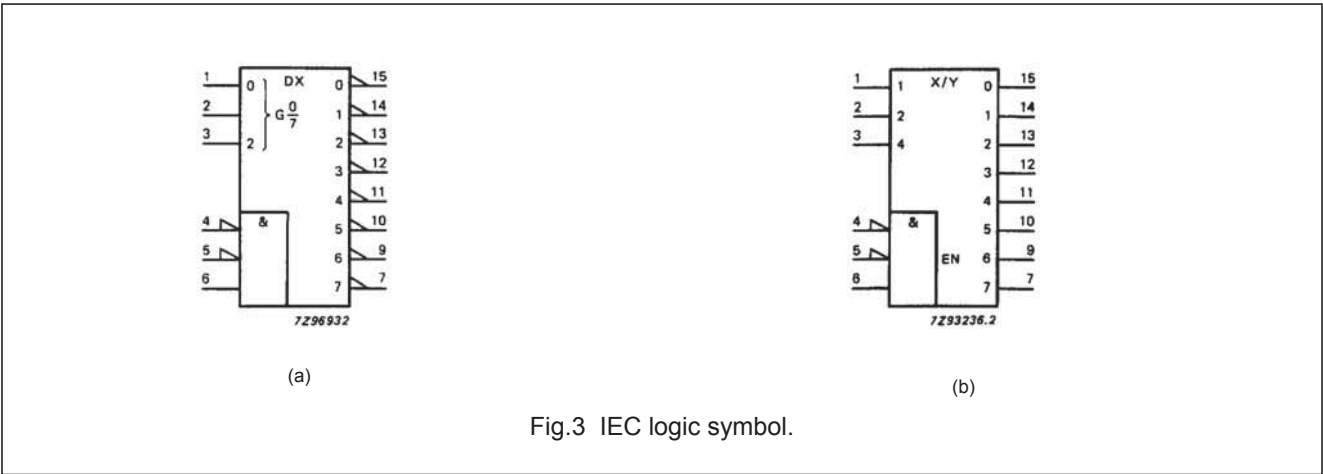


Fig.3 IEC logic symbol.

3-to-8 line decoder/demultiplexer

74HC/HCT238

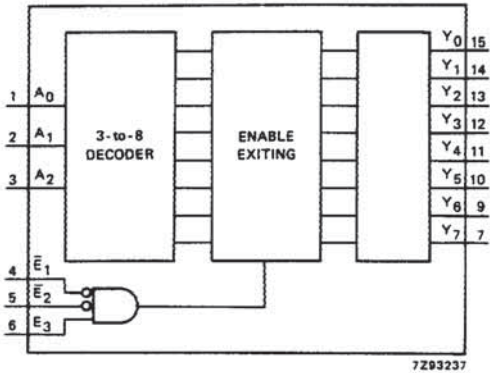


Fig.4 Functional diagram.

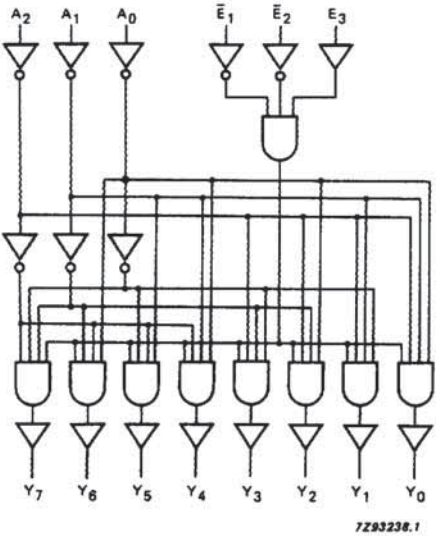


Fig.5 Logic diagram.

FUNCTION TABLE

INPUTS						OUTPUTS							
$\bar{E}_1$	$\bar{E}_2$	E_3	A_0	A_1	A_2	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
H	X	X	X	X	X	L	L	L	L	L	L	L	L
X	H	X	X	X	X	L	L	L	L	L	L	L	L
X	X	L	X	X	X	L	L	L	L	L	L	L	L
L	L	H	L	L	L	H	L	L	L	L	L	L	L
L	L	H	H	L	L	L	H	L	L	L	L	L	L
L	L	H	L	H	L	L	L	H	L	L	L	L	L
L	L	H	H	H	L	L	L	L	H	L	L	L	L
L	L	H	L	L	H	L	L	L	L	H	L	L	L
L	L	H	H	L	H	L	L	L	L	L	H	L	L
L	L	H	L	H	H	L	L	L	L	L	L	H	L
L	L	H	H	H	H	L	L	L	L	L	L	L	H

Note

- 1. H = HIGH voltage level
L = LOW voltage level
X = don't care



nRF24L01

Single Chip 2.4GHz Transceiver

Product Specification

Key Features

- Worldwide 2.4GHz ISM band operation
- Up to 2Mbps on air data rate
- Ultra low power operation
- 11.3mA TX at 0dBm output power
- 12.3mA RX at 2Mbps air data rate
- 900nA in power down
- 22µA in standby-I
- On chip voltage regulator
- 1.9 to 3.6V supply range
- Enhanced ShockBurst™
- Automatic packet handling
- Auto packet transaction handling
- 6 data pipe MultiCeiver™
- Air compatible with nRF2401A, 02, E1 and E2
- Low cost BOM
- ±60ppm 16MHz crystal
- 5V tolerant inputs
- Compact 20-pin 4x4mm QFN package

Applications

- Wireless PC Peripherals
- Mouse, keyboards and remotes
- 3-in-one desktop bundles
- Advanced Media center remote controls
- VoIP headsets
- Game controllers
- Sports watches and sensors
- RF remote controls for consumer electronics
- Home and commercial automation
- Ultra low power sensor networks
- Active RFID
- Asset tracing systems
- Toys

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July 2007

nRF24L01 Product Specification

1.2 Block diagram

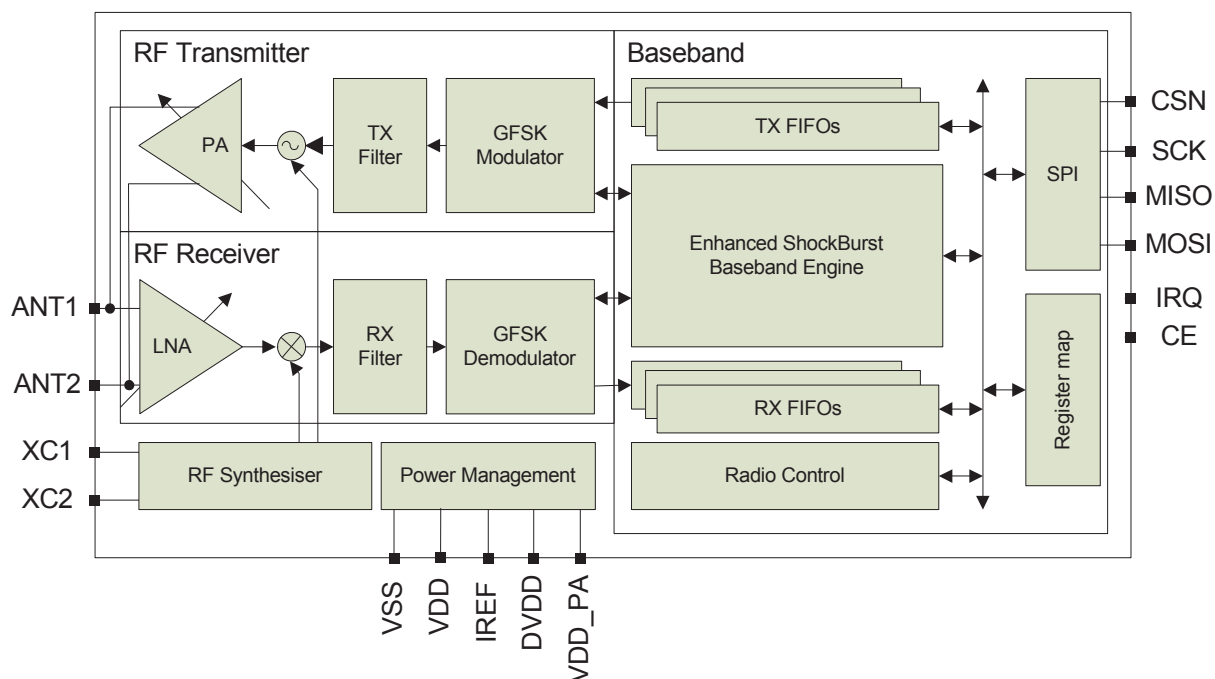


Figure 1. nRF24L01 block diagram

2.1 Pin assignment

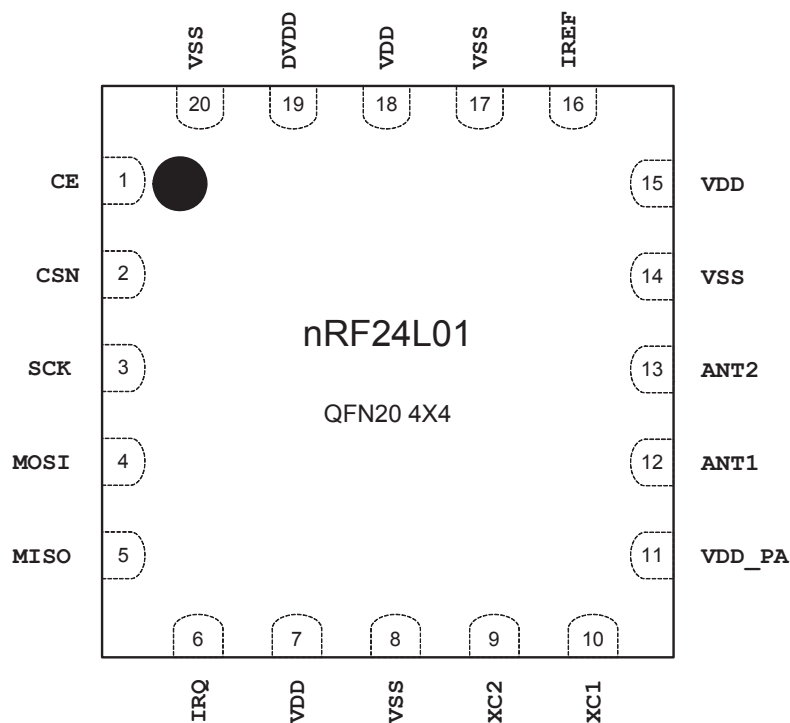


Figure 2. nRF24L01 pin assignment (top view) for the QFN20 4x4 package

nRF24L01 Product Specification



2.2 Pin functions

Pin	Name	Pin function	Description
1	CE	Digital Input	Chip Enable Activates RX or TX mode
2	CSN	Digital Input	SPI Chip Select
3	SCK	Digital Input	SPI Clock
4	MOSI	Digital Input	SPI Slave Data Input
5	MISO	Digital Output	SPI Slave Data Output, with tri-state option
6	IRQ	Digital Output	Maskable interrupt pin. Active low
7	VDD	Power	Power Supply (+1.9V - +3.6V DC)
8	VSS	Power	Ground (0V)
9	XC2	Analog Output	Crystal Pin 2
10	XC1	Analog Input	Crystal Pin 1
11	VDD_PA	Power Output	Power Supply Output(+1.8V) for the internal nRF24L01 Power Amplifier. Must be connected to ANT1 and ANT2 as shown in Figure 30 .
12	ANT1	RF	Antenna interface 1
13	ANT2	RF	Antenna interface 2
14	VSS	Power	Ground (0V)
15	VDD	Power	Power Supply (+1.9V - +3.6V DC)
16	IREF	Analog Input	Reference current. Connect a 22kΩ resistor to ground. See: Figure 30 .
17	VSS	Power	Ground (0V)
18	VDD	Power	Power Supply (+1.9V - +3.6V DC)
19	DVDD	Power Output	Internal digital supply output for de-coupling purposes. See: Figure 30 .
20	VSS	Power	Ground (0V)

Table 1. nRF24L01 pin function

ASAHI KASEI

[AK5384]



AK5384

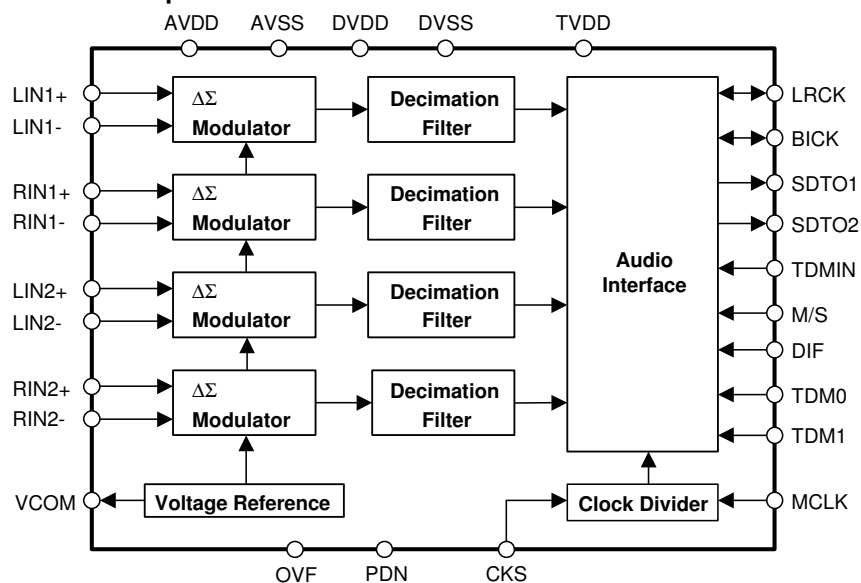
107dB 24-Bit 96kHz 4-Channel ADC

GENERAL DESCRIPTION

The AK5384 is a 4-channel A/D Converter with wide sampling rate of 8kHz ~ 96kHz and is suitable for Multi-channel audio system. The AK5384 achieves high accuracy and low cost by using Enhanced dual bit $\Delta\Sigma$ techniques. The AK5384 supports master mode and TDM format. Therefore, the AK5384 is suitable for multi-channel audio system.

FEATURES

- ☐ 4-Channel $\Delta\Sigma$ ADC
- ☐ Differential Inputs
- ☐ Digital HPF for DC-Offset Cancel
- ☐ S/(N+D): 100dB@5V for 48kHz
- ☐ DR: 107dB@5V for 48kHz
- ☐ S/N: 107dB@5V for 48kHz
- ☐ Sampling Rate Ranging from 8kHz to 96kHz
- ☐ Master Clock:
 - 256fs/384fs/512fs/768fs (~ 48kHz)
 - 256fs/384fs (~ 96kHz)
- ☐ TTL Digital Input Level
- ☐ Output format: 24bit MSB justified, I²S or TDM
- ☐ Cascade TDM Interface
- ☐ Master & Slave Mode
- ☐ Overflow Flag
- ☐ Power Supply: 4.75 to 5.25V
- ☐ Power Supply for output buffer: 3.0 to 5.25V
- ☐ Ta = -40 ~ 85°C
- ☐ 28pin VSOP



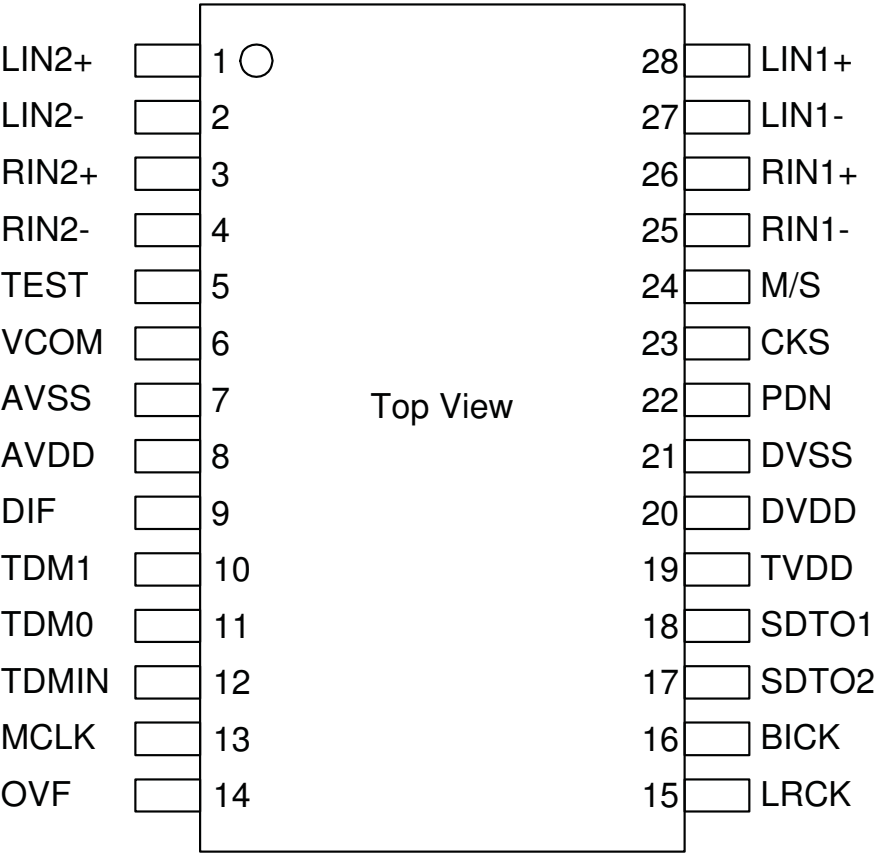
ASAHI KASEI

[AK5384]

■ Ordering Guide

AK5384VF	-40 ~ +85°C	28pin VSOP (0.65mm pitch)
AKD5384	Evaluation Board for AK5384	

■ Pin Layout



ASAHI KASEI

[AK5384]

PIN/FUNCTION			
No.	Pin Name	I/O	Function
1	LIN2+	I	ADC2 Lch Positive Analog Input Pin
2	LIN2–	I	ADC2 Lch Negative Analog Input Pin
3	RIN2+	I	ADC2 Rch Positive Analog Input Pin
4	RIN2–	I	ADC2 Rch Negative Analog Input Pin
5	TEST	I	Test Pin (Connected to AVSS)
6	VCOM	O	Common Voltage Output Pin, AVDD/2 Normally connected to AVSS with a 0.1μF ceramic capacitor in parallel with an electrolytic capacitor less than 2.2μF.
7	AVSS	-	Analog Ground Pin
8	AVDD	-	Analog Power Supply Pin, 4.75 ~ 5.25V
9	DIF	I	Audio Interface Format Pin “L” : 24bit MSB justified, “H” : 24bit I ² S Compatible
10	TDM1	I	TDM I/F BICK Frequency Select Pin “L” : 256fs, “H” : 128fs
11	TDM0	I	TDM I/F Format Enable Pin “L” : Normal Mode, “H” : TDM Mode
12	TDMIN	I	TDM Data Input Pin
13	MCLK	I	Master Clock Input Pin
14	OVF	O	Analog Input Overflow Detect Pin This pin goes to “H” if one of four analog inputs overflows.
15	LRCK	I/O	Output Channel Clock Pin “L” Output in Master Mode at Power-down mode.
16	BICK	I/O	Audio Serial Data Clock Pin “L” Output in Master Mode at Power-down mode.
17	SDTO2	O	ADC2 Audio Serial Data Output Pin “L” Output at Power-down mode.
18	SDTO1	O	ADC1 Audio Serial Data Output Pin “L” Output at Power-down mode.
19	TVDD	-	Output Buffer Power Supply Pin, 3.0 ~ 5.25V
20	DVDD	-	Digital Power Supply Pin, 4.75 ~ 5.25V
21	DVSS	-	Digital Ground Pin
22	PDN	I	Power-Down Mode Pin When “L”, the circuit is in power-down mode. The AK5384 should always be reset upon power-up.
23	CKS	I	Master Clock Select Pin “L” : 256fs, “H” : 512fs This pin is enabled in Master Mode.
24	M/S	I	Master / Slave Mode Pin “L” : Slave Mode, “H” : Master Mode
25	RIN1–	I	ADC1 Rch Negative Analog Input Pin
26	RIN1+	I	ADC1 Rch Positive Analog Input Pin
27	LIN1–	I	ADC1 Lch Negative Analog Input Pin
28	LIN1+	I	ADC1 Lch Positive Analog Input Pin

Note: All digital input pins should not be left floating.

ASAHI KASEI

[AK4382A]



AK4382A

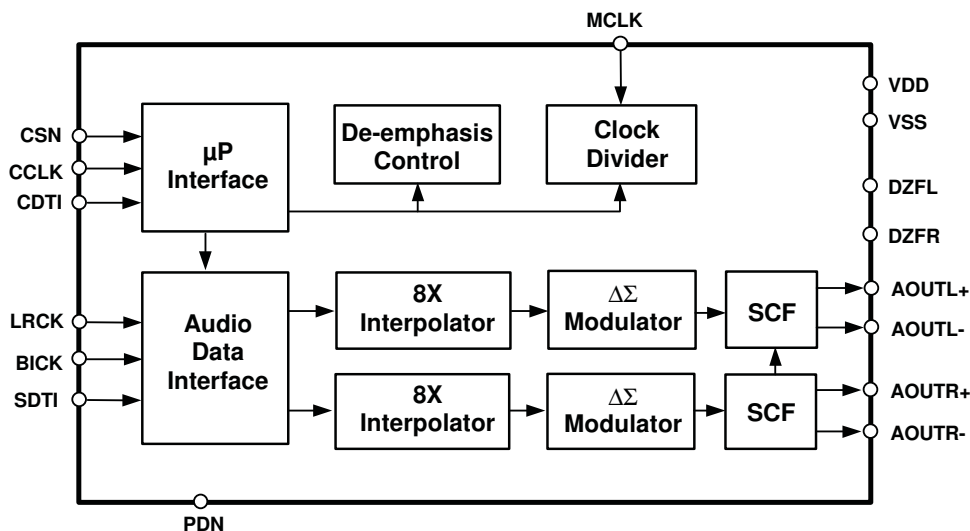
112dB 192kHz 24-Bit 2ch $\Delta\Sigma$ DAC

GENERAL DESCRIPTION

The AK4382A offers the perfect mix for cost and performance based audio systems. Using AKM's multi bit architecture for its modulator the AK4382A delivers a wide dynamic range while preserving linearity for improved THD+N performance. The AK4382A has full differential SCF outputs, removing the need for AC coupling capacitors and increasing performance for systems with excessive clock jitter. The 24 Bit word length and 192kHz sampling rate make this part ideal for a wide range of applications including DVD-Audio. The AK4382A is offered in a space saving 16pin TSSOP package.

FEATURES

- ☐ Sampling Rate Ranging from 8kHz to 192kHz
- ☐ 128 times Oversampling (Normal Speed Mode)
- ☐ 64 times Oversampling (Double Speed Mode)
- ☐ 32 times Oversampling (Quad Speed Mode)
- ☐ 24-Bit 8 times FIR Digital Filter
- ☐ On chip SCF
- ☐ Digital de-emphasis for 32k, 44.1k and 48kHz sampling
- ☐ Soft mute
- ☐ Digital Attenuator (256 steps)
- ☐ I/F format: 24-Bit MSB justified, 24/20/16-Bit LSB justified or I^2S
- ☐ Master clock: 256fs, 384fs, 512fs or 768fs (Normal Speed Mode)
128fs, 192fs, 256fs or 384fs (Double Speed Mode)
128fs, 192fs (Quad Speed Mode)
- ☐ THD+N: -94dB
- ☐ Dynamic Range: 112dB
- ☐ High Tolerance to Clock Jitter
- ☐ Power supply: 4.75 to 5.25V
- ☐ Very Small Package: 16pin TSSOP (6.4mm x 5.0mm)



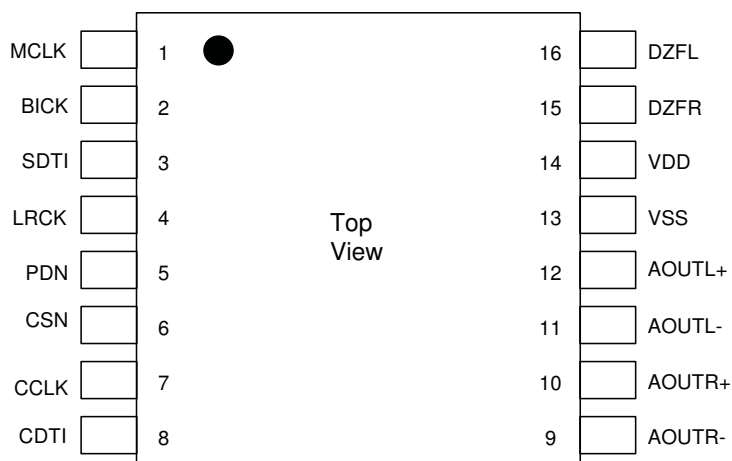
ASAHI KASEI

[AK4382A]

■ Ordering Guide

AK4382AVT -40 ~ +85°C 16pin TSSOP (0.65mm pitch)
 AKD4382 Evaluation Board for AK4382A

■ Pin Layout



PIN/FUNCTION			
No.	Pin Name	I/O	Function
1	MCLK	I	Master Clock Input Pin An external TTL clock should be input on this pin.
2	BICK	I	Audio Serial Data Clock Pin
3	SDTI	I	Audio Serial Data Input Pin
4	LRCK	I	L/R Clock Pin
5	PDN	I	Power-Down Mode Pin When at "L", the AK4382A is in the power-down mode and is held in reset. The AK4382A should always be reset upon power-up.
6	CSN	I	Chip Select Pin
7	CCLK	I	Control Data Input Pin
8	CDTI	I	Control Data Input Pin in serial mode
9	AOUTR-	O	Rch Negative Analog Output Pin
10	AOUTR+	O	Rch Positive Analog Output Pin
11	AOUTL-	O	Lch Negative Analog Output Pin
12	AOUTL+	O	Lch Positive Analog Output Pin
13	VSS	-	Ground Pin
14	VDD	-	Power Supply Pin
15	DZFR	O	Rch Data Zero Input Detect Pin
16	DZFL	O	Lch Data Zero Input Detect Pin

Note: All input pins should not be left floating.

Multiple voltage regulator with switch and ignition buffer

TDA3681

FEATURES

General

- Extremely low noise behaviour and good stability with very small output capacitors
- Second supply pin for regulators 3 and 4 to reduce power dissipation (e.g. via a DC-to-DC converter)
- Three V_P -state controlled regulators (regulators 1, 3 and 4) and a power switch
- Regulator 2, reset and ignition buffer operational during load dump and thermal shutdown
- Combined control pin for switching regulators 1 and 3
- Separate control pins for switching regulator 4 and the power switch
- Supply voltage range from -18 to $+50$ V
- Low quiescent current in standby mode (when regulators 1, 3 and 4 and power switch are switched off and ignition input is low)
- Hold output for low V_P (regulators 1, 3 and 4 and power switch off)
- Hold output when one of regulators 1 and 3 and/or 4 is out of regulation
- Hold output for foldback mode of power switch and regulators 1, 3 and 4
- Hold output for load dump and temperature protection
- Reset (push-pull output stage) for regulator 2 and hold output (open-collector output)
- Adjustable reset delay time
- High supply voltage ripple rejection
- Backup capacitor for regulator 2
- One independent ignition buffer (active HIGH).
- ESD protection on all pins
- Thermal protections
- Load dump protection
- Foldback current limit protection for regulators 1, 2, 3 and 4
- Delayed second current limit protection for the power switch (at short-circuit)
- The regulator outputs and the power switch are DC short-circuit safe to ground and supply (V_P).

GENERAL DESCRIPTION

The TDA3681 is a multiple output voltage regulator with a power switch and an ignition buffer. It is intended for use in car radios with or without a microcontroller. The TDA3681 contains the following:

- Four fixed voltage regulators with a foldback current protection (regulators 1, 2, 3 and 4). Regulator 2, which is intended to supply a microcontroller, also operates during load dump and thermal shutdown
- Regulators 3 and 4 have a second supply pin that can be connected to a lower supply voltage (>6.5 V) to reduce the power dissipation
- A power switch with protection, operated by a control input
- Reset and hold outputs that can be used to interface with the microcontroller; the reset signal can be used to call up the microcontroller
- Both supply pins can withstand load dump pulses and negative supply voltages
- Regulator 2, which is in regulation at a backup voltage above 6.5 V
- A provision for the use of a reserve supply capacitor that will hold enough energy for regulator 2 (5 V continuous) to allow a microcontroller to prepare for loss of voltage
- An ignition input Schmitt trigger with push-pull output stage.

Protections

- Reverse polarity safe (down to -18 V without high reverse current)
- Able to withstand voltages up to 18 V at the outputs (supply line may be short-circuited)

ORDERING INFORMATION

TYPE NUMBER	PACKAGE		
	NAME	DESCRIPTION	VERSION
TDA3681J	DBS17P	plastic DIL-bent-SIL power package; 17 leads (lead length 7.7 mm)	SOT243-3
TDA3681JR	DBS17P	plastic DIL-bent-SIL (special bent) power package; 17 leads (lead length 12 mm)	SOT475-1
TDA3681TH	HSOP20	plastic, heatsink small outline package; 20 leads; low stand-off height	SOT418-2

Multiple voltage regulator with switch and ignition buffer

TDA3681

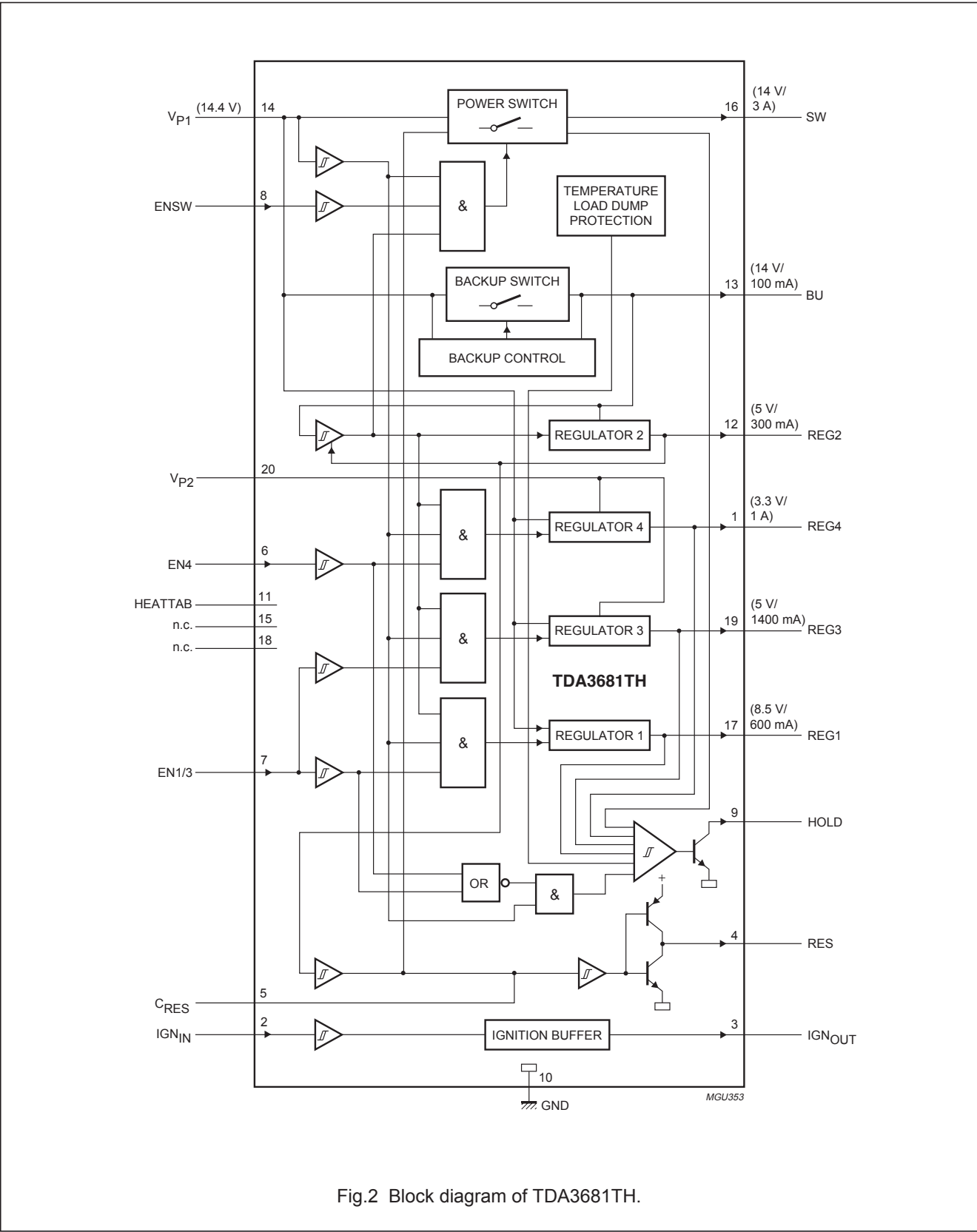


Fig.2 Block diagram of TDA3681TH.

Multiple voltage regulator with switch and ignition buffer

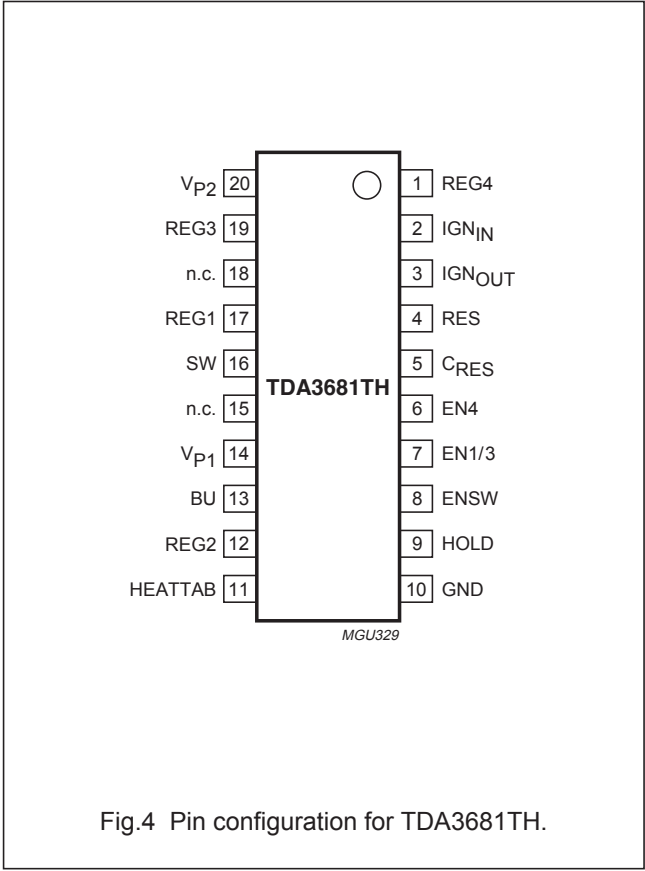
TDA3681

Pin description of TDA3681TH

SYMBOL	PIN	DESCRIPTION
REG4	1	regulator 4 output
IGN _{IN}	2	ignition input
IGN _{OUT}	3	ignition output (active HIGH)
RES	4	reset output (active LOW)
C _{RES}	5	reset delay capacitor
EN4	6	enable input for regulator 4
EN1/3	7	enable input for regulators 1 and 3
ENSW	8	enable input for power switch
HOLD	9	hold output (active LOW)
GND	10	ground
HEATTAB	11	heat tab connection; note 1
REG2	12	regulator 2 output
BU	13	backup switch output
V _{P1}	14	supply voltage
n.c.	15	not connected
SW	16	power switch output
REG1	17	regulator 1 output
n.c.	18	not connected
REG3	19	regulator 3 output
V _{P2}	20	second supply voltage

Note

1. The pin is used for final test purposes. In the application it should be connected directly to ground.



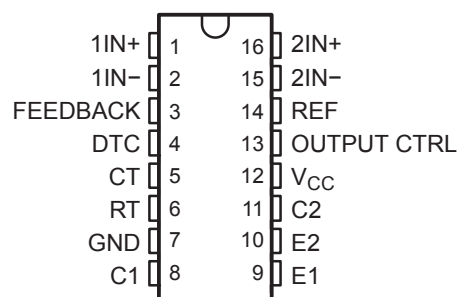
TL494 PULSE-WIDTH-MODULATION CONTROL CIRCUITS

SLVS074E–JANUARY 1983–REVISED FEBRUARY 2005

FEATURES

- Complete PWM Power-Control Circuitry
- Uncommitted Outputs for 200-mA Sink or Source Current
- Output Control Selects Single-Ended or Push-Pull Operation
- Internal Circuitry Prohibits Double Pulse at Either Output
- Variable Dead Time Provides Control Over Total Range
- Internal Regulator Provides a Stable 5-V Reference Supply With 5% Tolerance
- Circuit Architecture Allows Easy Synchronization

D, DB, N, NS, OR PW PACKAGE
(TOP VIEW)



DESCRIPTION

The TL494 incorporates all the functions required in the construction of a pulse-width-modulation (PWM) control circuit on a single chip. Designed primarily for power-supply control, this device offers the flexibility to tailor the power-supply control circuitry to a specific application.

The TL494 contains two error amplifiers, an on-chip adjustable oscillator, a dead-time control (DTC) comparator, a pulse-steering control flip-flop, a 5-V, 5%-precision regulator, and output-control circuits.

The error amplifiers exhibit a common-mode voltage range from -0.3 V to $V_{CC} - 2\text{ V}$. The dead-time control comparator has a fixed offset that provides approximately 5% dead time. The on-chip oscillator can be bypassed by terminating RT to the reference output and providing a sawtooth input to CT, or it can drive the common circuits in synchronous multiple-rail power supplies.

The uncommitted output transistors provide either common-emitter or emitter-follower output capability. The TL494 provides for push-pull or single-ended output operation, which can be selected through the output-control function. The architecture of this device prohibits the possibility of either output being pulsed twice during push-pull operation.

The TL494C is characterized for operation from 0°C to 70°C . The TL494I is characterized for operation from -40°C to 85°C .

AVAILABLE OPTIONS

T_A	PACKAGED DEVICES ⁽¹⁾				
	SMALL OUTLINE (D)	PLASTIC DIP (N)	SMALL OUTLINE (NS)	SHRINK SMALL OUTLINE (DB)	THIN SHRINK SMALL OUTLINE (PW)
0°C to 70°C	TL494CD	TL494CN	TL494CNS	TL494CDB	TL494CPW
-40°C to 85°C	TL494ID	TL494IN	—	—	—

(1) The D, DB, NS, and PW packages are available taped and reeled. Add the suffix R to device type (e.g., TL494CDR).

TL494
PULSE-WIDTH-MODULATION CONTROL CIRCUITS

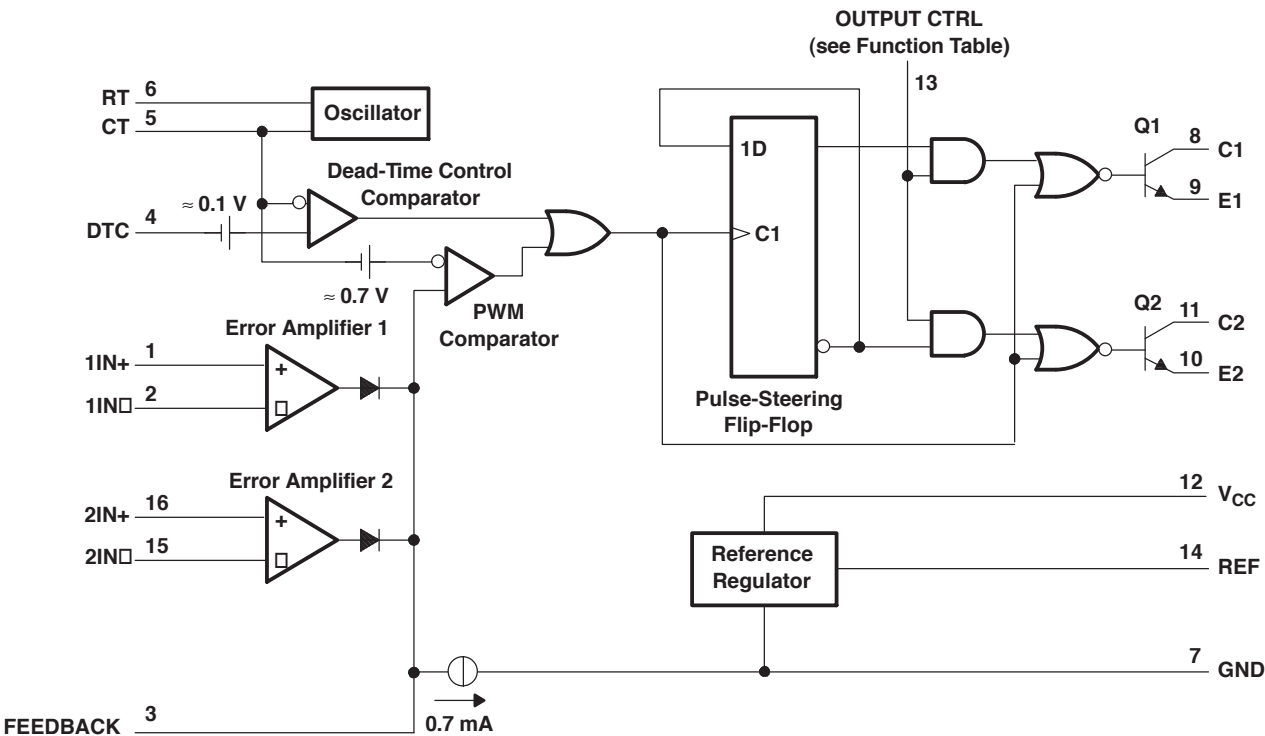
SLVS074E–JANUARY 1983–REVISED FEBRUARY 2005



FUNCTION TABLE

INPUT TO OUTPUT CTRL	OUTPUT FUNCTION
$V_I = \text{GND}$	Single-ended or parallel output
$V_I = V_{\text{ref}}$	Normal push-pull operation

FUNCTIONAL BLOCK DIAGRAM



BA50DD0WHFP HRP5

Standard Fixed Output LDO Regulators with Shutdown Switch

BA□□DD0 Series,BA□□CC0 Series,
BA□□DD0W Series,BA□□CC0W Series

Technical Note

●Block Diagrams

BA□□CC0WFP/ BA□□DD0WHFP/ BA□□CC0WT(V5)/ BA□□DD0WT

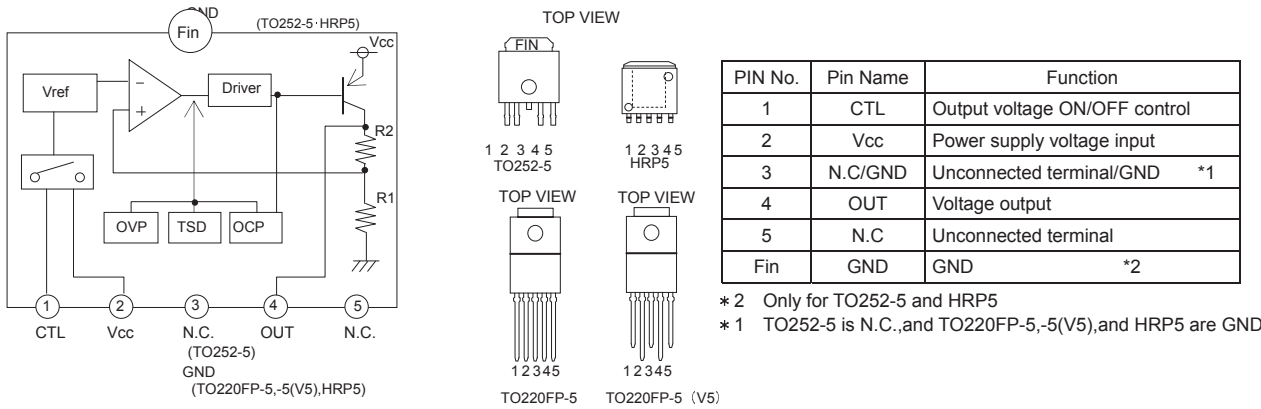


Fig.25

BA□□CC0T/ BA□□CC0FP/ BA□□DD0T

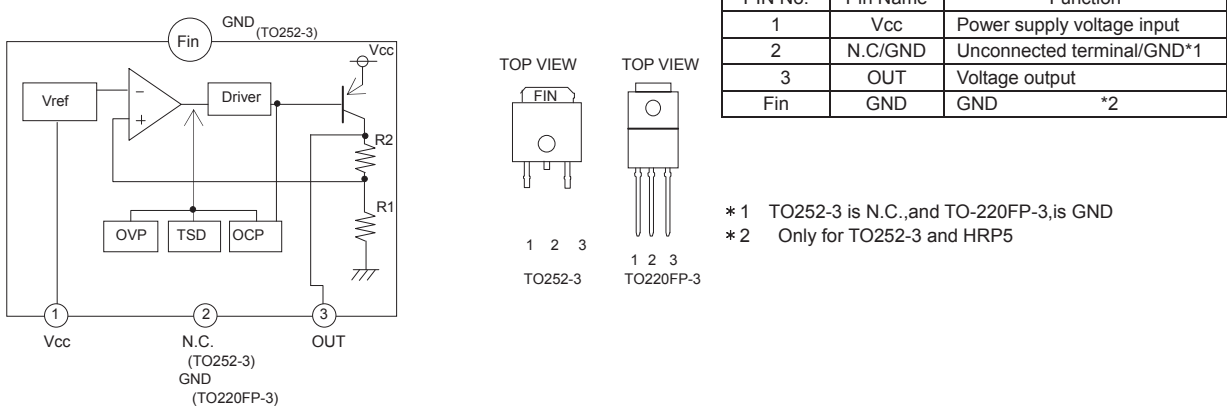


Fig.26

●Input / Output Equivalent Circuit Diagrams

< BA□□DD0 Series >

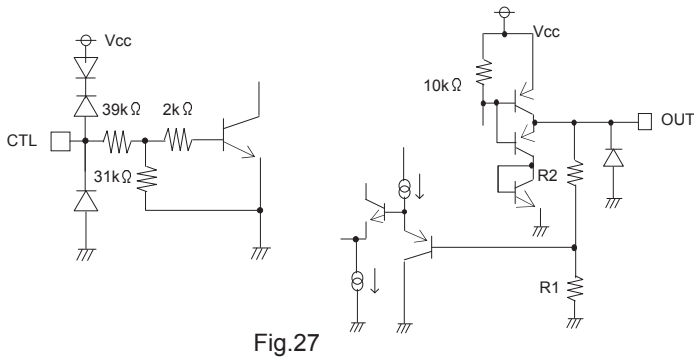


Fig.27

< BA□□CC0 Series >

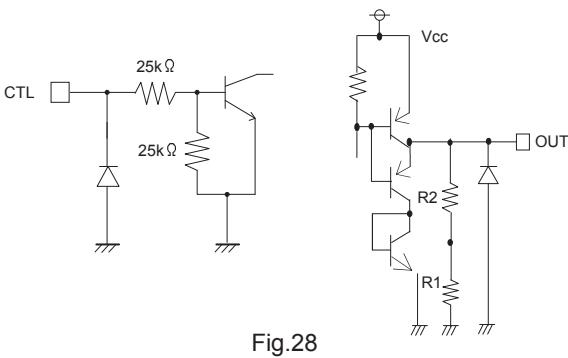
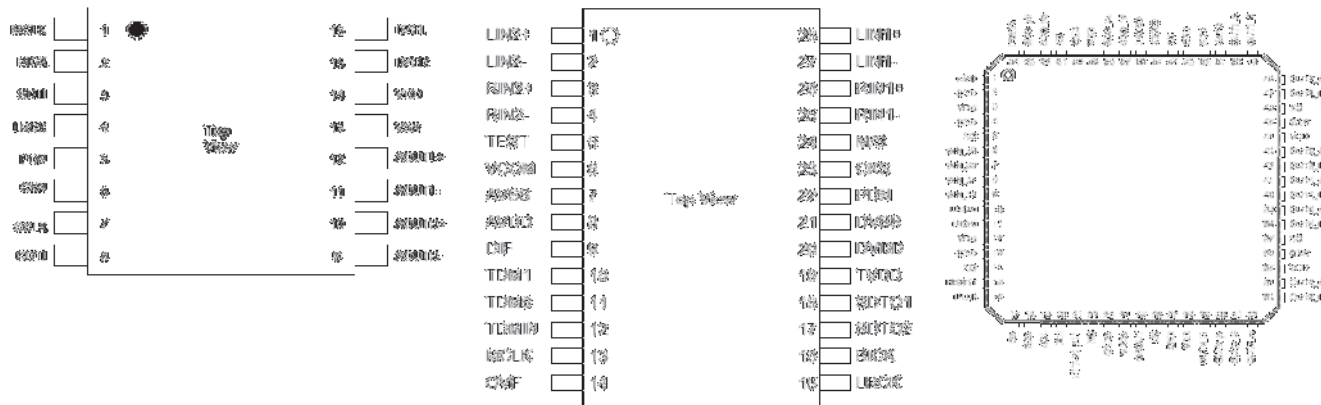


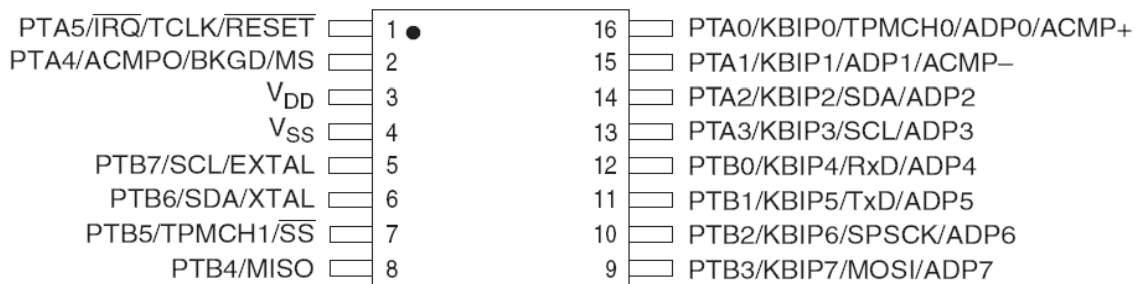
Fig.28

AK5384VF

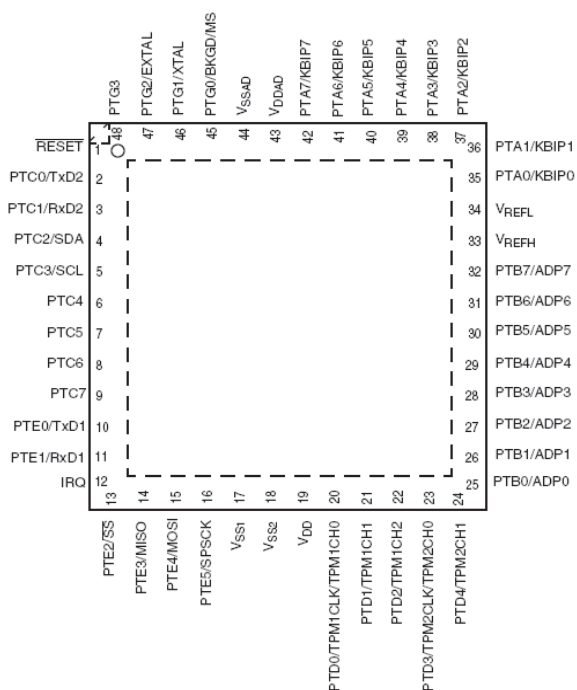
STA309A



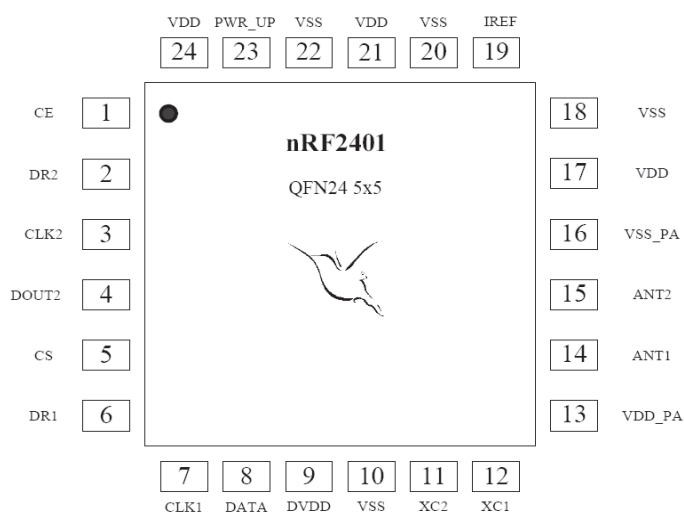
MC9S08QG8



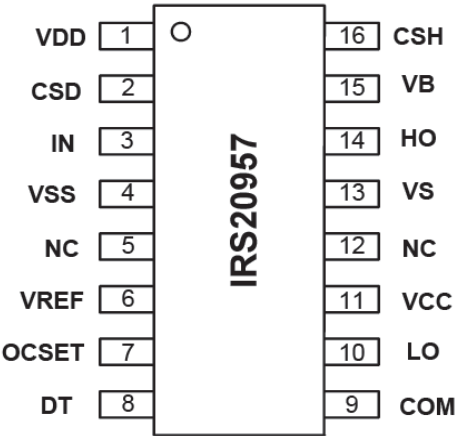
MC9S08GT16A



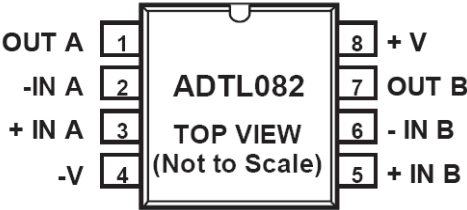
nRF2401+



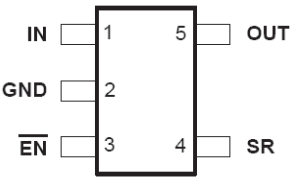
IRS20957



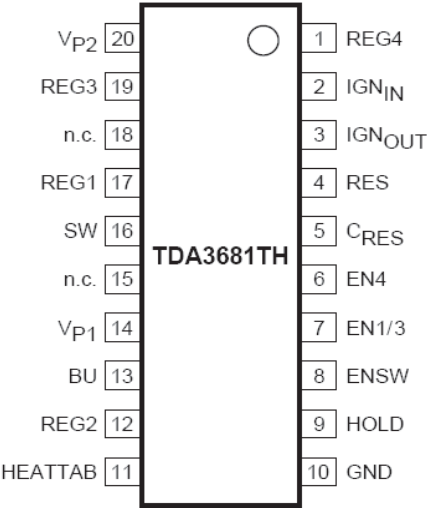
ADTL082



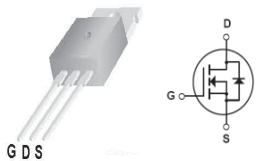
TPS78833



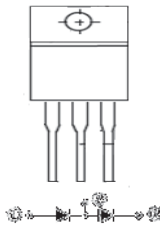
TDA3681TH



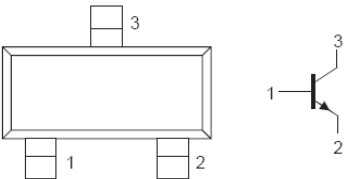
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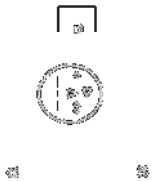
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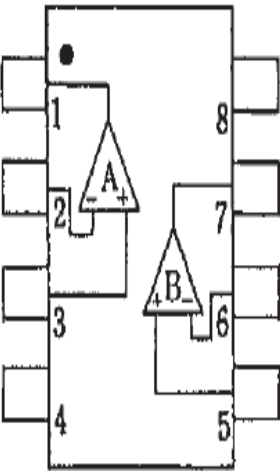
SOT23 Transistor



FDN358

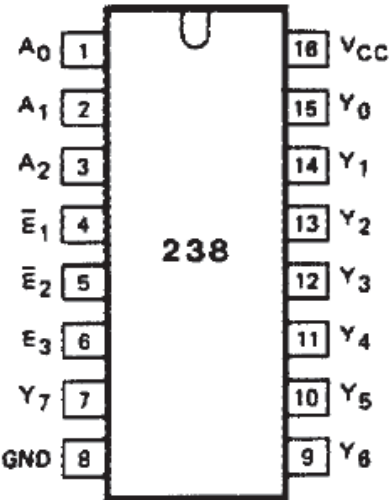


Op-Amp

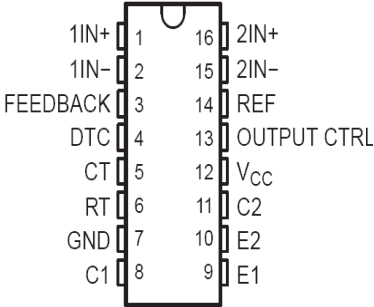


- PIN FUNCITON
- 1. A₊ OUTPUT
 - 2. A₋ INPUT
 - 3. A₊ INPUT
 - 4. V₋
 - 5. B₊ INPUT
 - 6. B₋ INPUT
 - 7. B₊ OUTPUT
 - 8. V₊

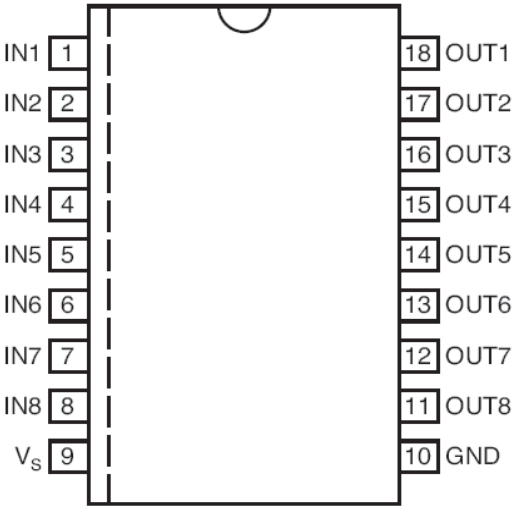
74HCT238



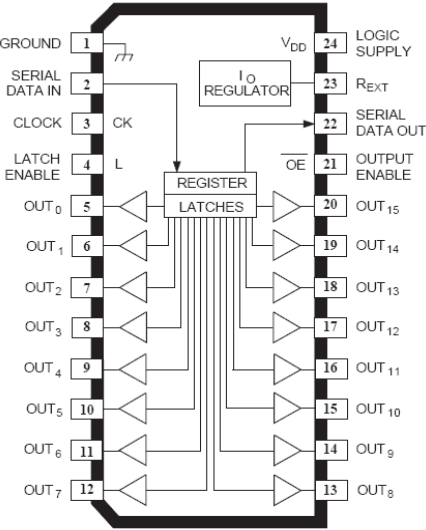
TL494 PWM



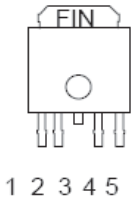
MIC2982



A6276ELW

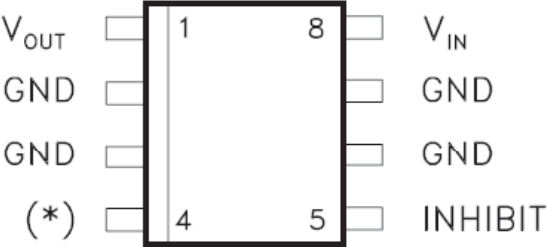


BA50DD0WHFP



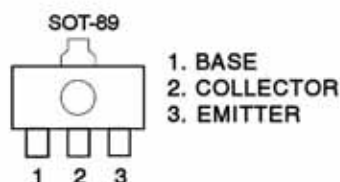
PIN No.	Pin Name	Function
1	CTL	Output voltage ON/OFF control
2	Vcc	Power supply voltage input
3	N.C/GND	Unconnected terminal/GND *1
4	OUT	Voltage output
5	N.C	Unconnected terminal
Fin	GND	GND *2

LM2931

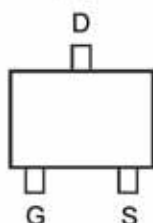


Semiconductor Pinouts

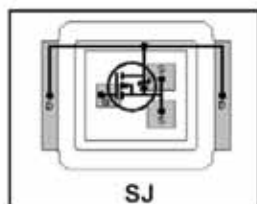
KTA1661Y
Q5,10, Q35



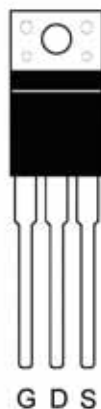
FDV301N MOSFET
Q8,25



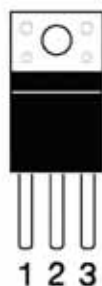
IRF6645TRBF
Q17-24



IRF3205 MOSFET
Q2,9,11,12,14,15



KIA7812
+12 REG
U2

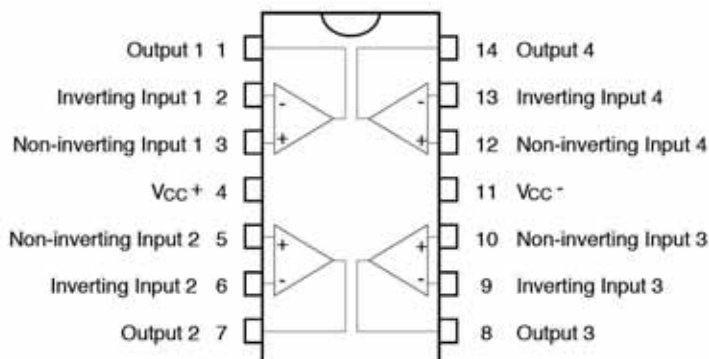


1. IN
2. GROUND
3. OUT

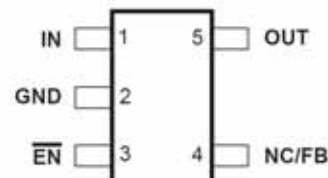
TL074ID LM224
QUAD OP-AMP

D, J, N, NS, OR PW PACKAGE
TL074... D, J, N, NS, PW,
OR W PACKAGE
(TOP VIEW)

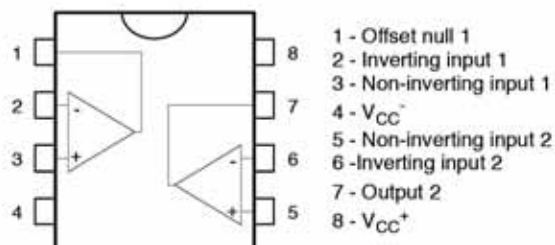
U3,U4,U5,U10,U11,U15



TPS77033
(LDO) Linear Regulator
SOT-23
VR4
(TOP VIEW)

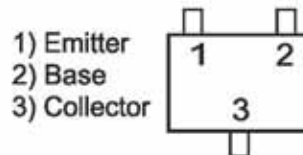


TL0724ID DUAL OP-AMP
ADTL082ARZ DUAL OP-AMP
U1,U2,U7,U12,U15,U16,U26,U27



MMBT3904
MMBT3906
KTA1504
KRC102S

Q1-10,11-16,17,18,27,34

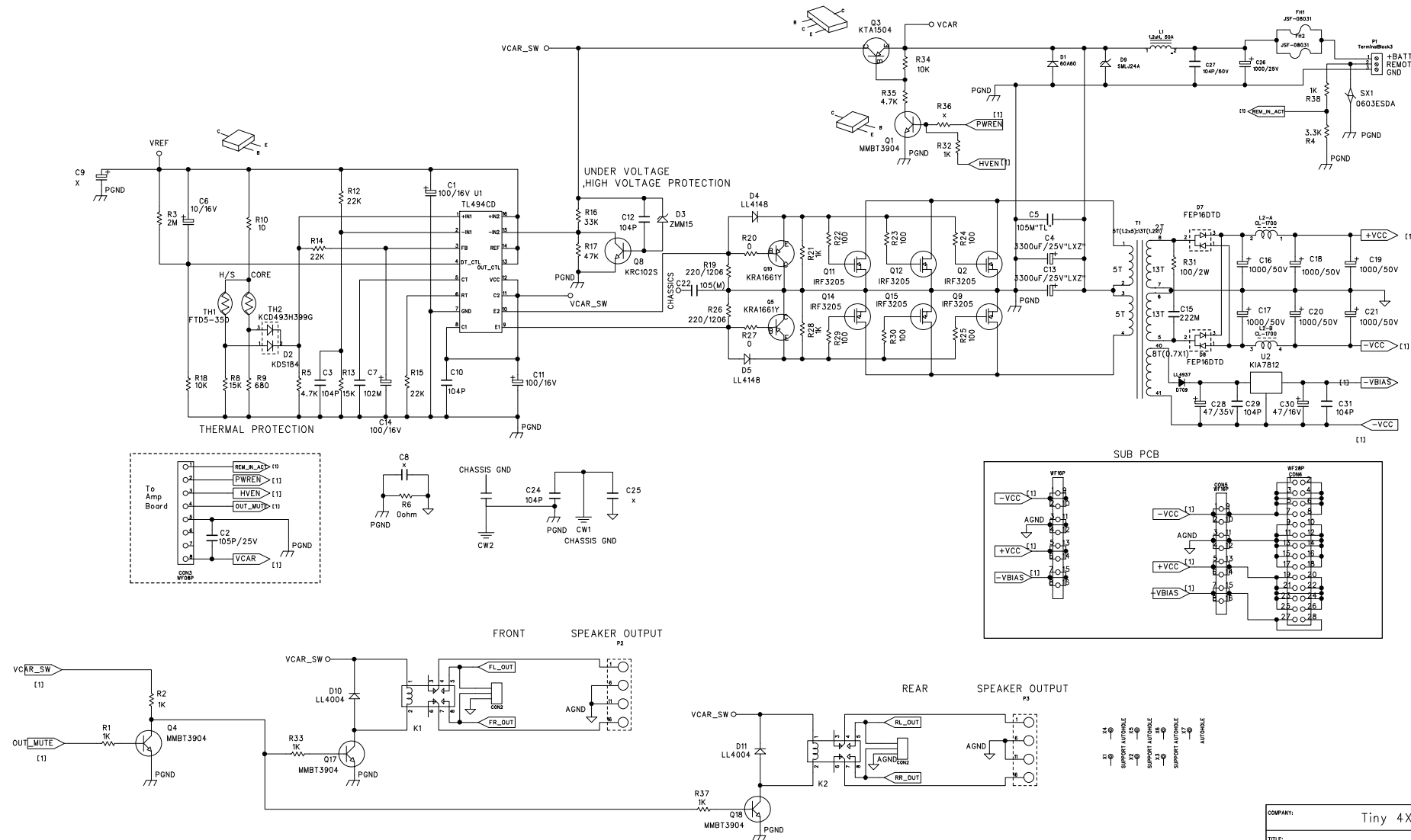


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NOTE: Unless otherwise specified all non-polarized capacitors are ceramic, 0603 package, 50V (or higher), and 10% tolerance. Unless otherwise specified, all resistors are 0603 package, 5% tolerance. All components have temperature ranges of -40 to +85 or wider.

REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE



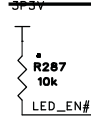
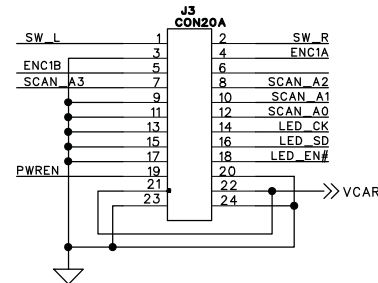
Harman Mobile and Communication Products Group			
DC-DC Boost Converter, Primary			
DOCUMENT NUMBER	Tiny Amp 4 x 100W Power Supply	REV	002
DATE	Saturday, March 18, 2006	SHEET	1 OF 4

DRAWN: Richard	DATED: Aug.07.2006
CHECKED:	DATED:
QUALITY CONTROL:	DATED:
RELEASED:	DATED:

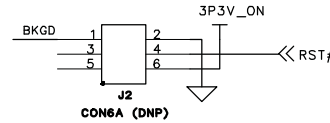
COMPANY: Tiny 4X100	
TITLE:	
CODE:	SIZE:
DRAWING NO:	REV:
SCALE: N/S	
SHEET: 1 OF 4	

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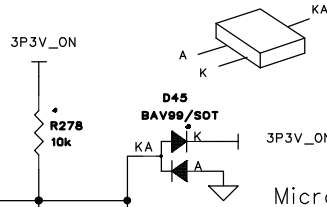
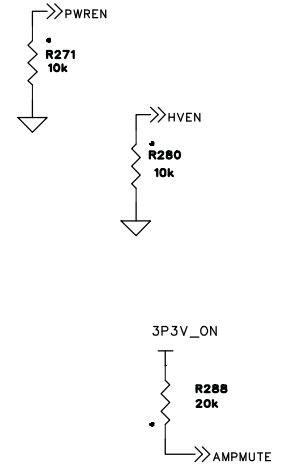
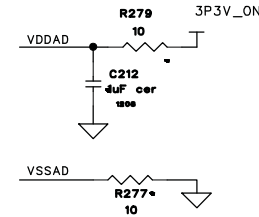
Data to
User
Interface
Board



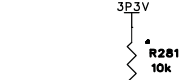
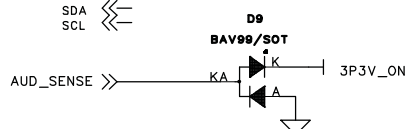
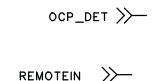
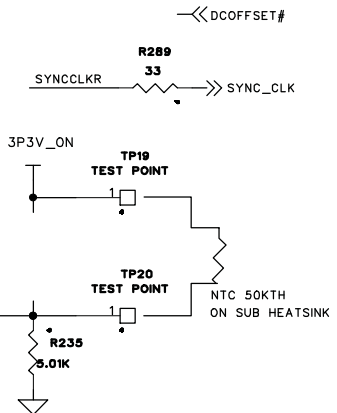
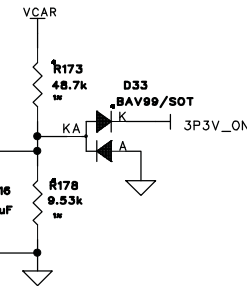
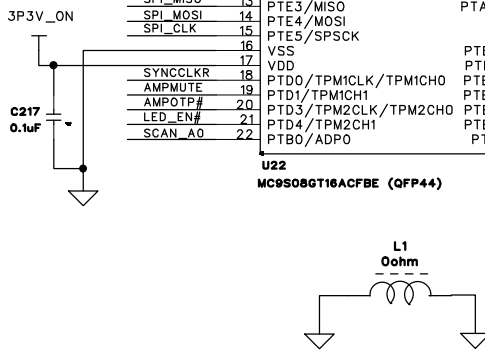
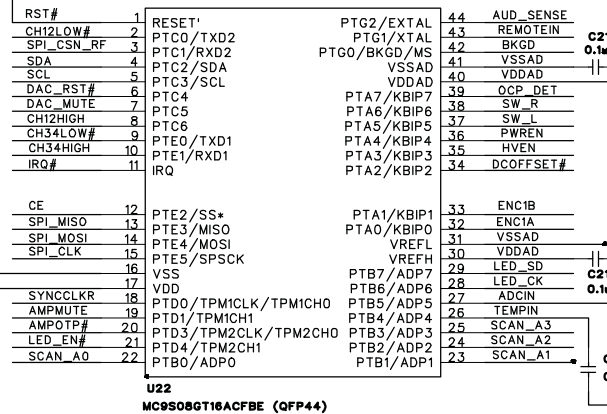
Connector for programming/debugging
the microcontroller



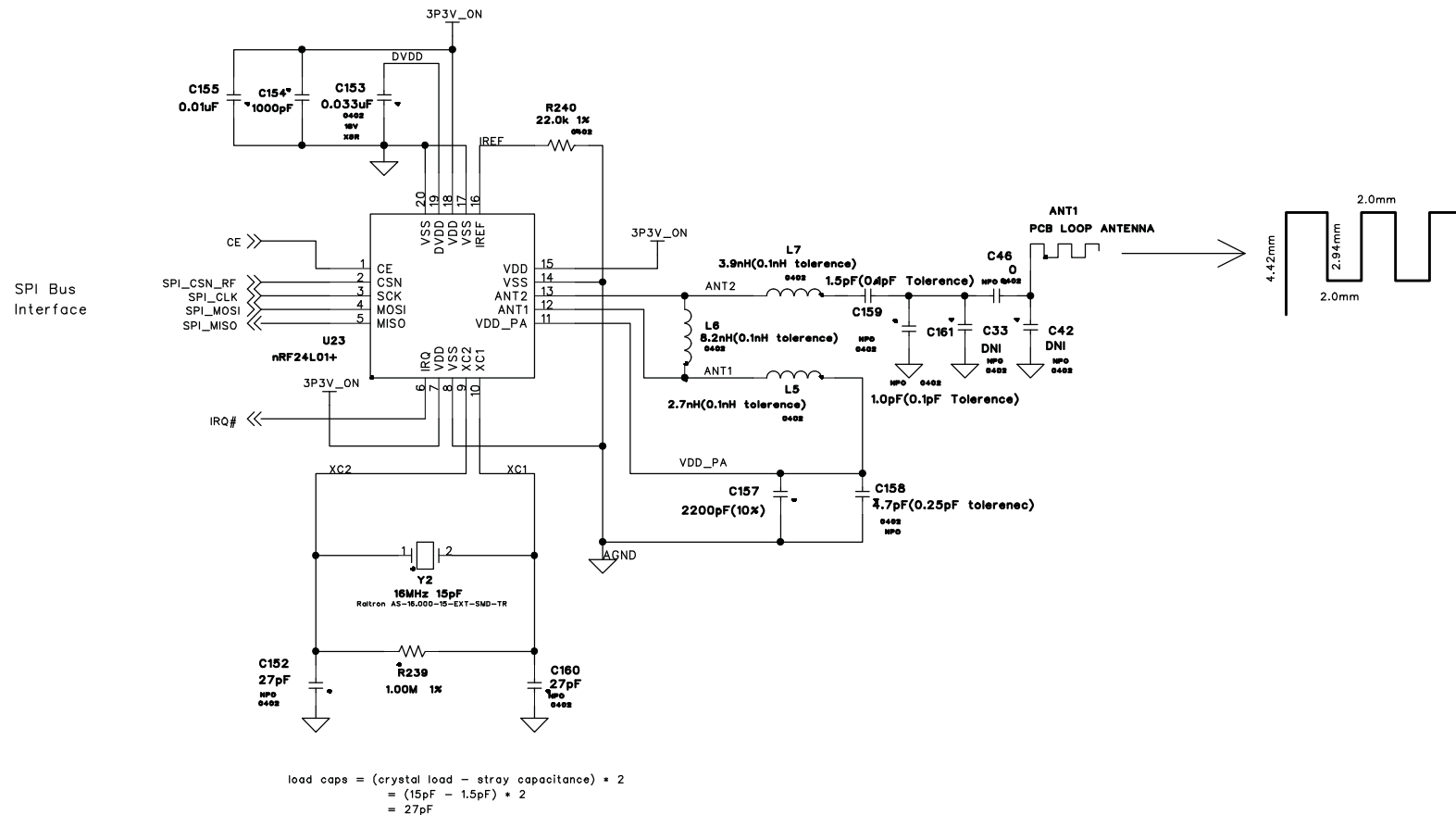
NOTE: The
microcontroller
is not 5V-tolerant



Microcontroller



Harman Mobile and Companion Products Group		
Title Microcontroller		
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Harman Mobile and Companion Products Group

 Title
 RF Transceiver

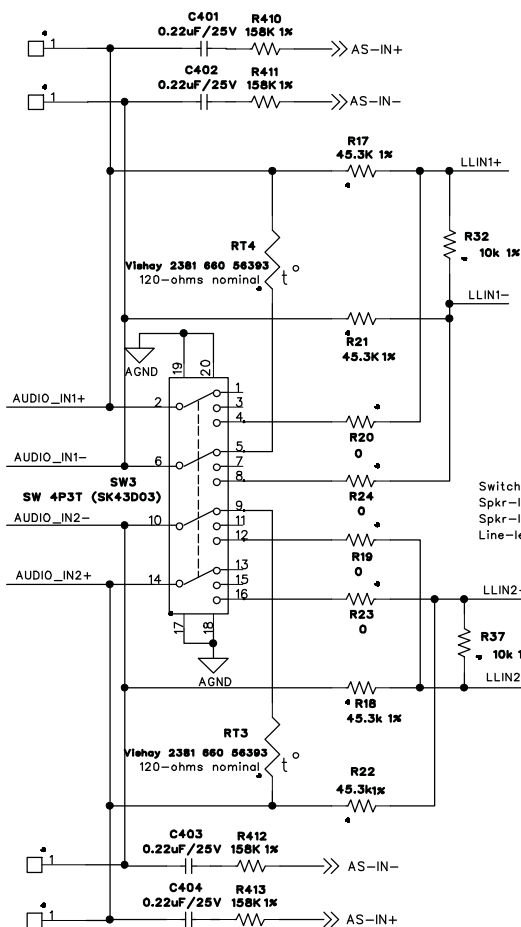
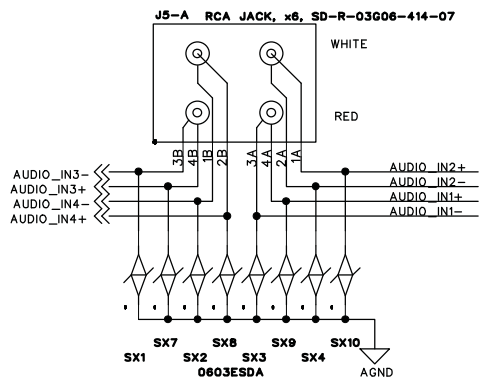
Size B Document Number

 Rev
 H

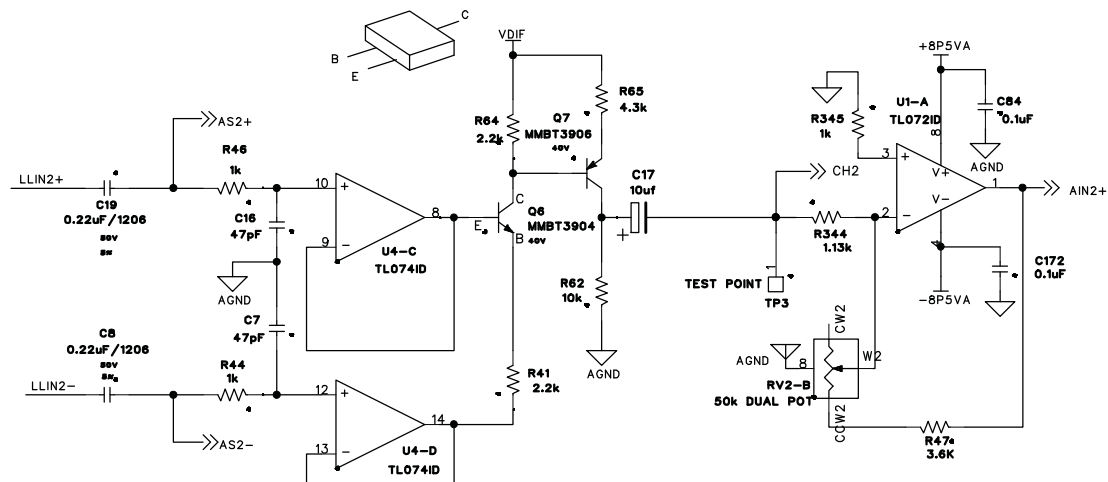
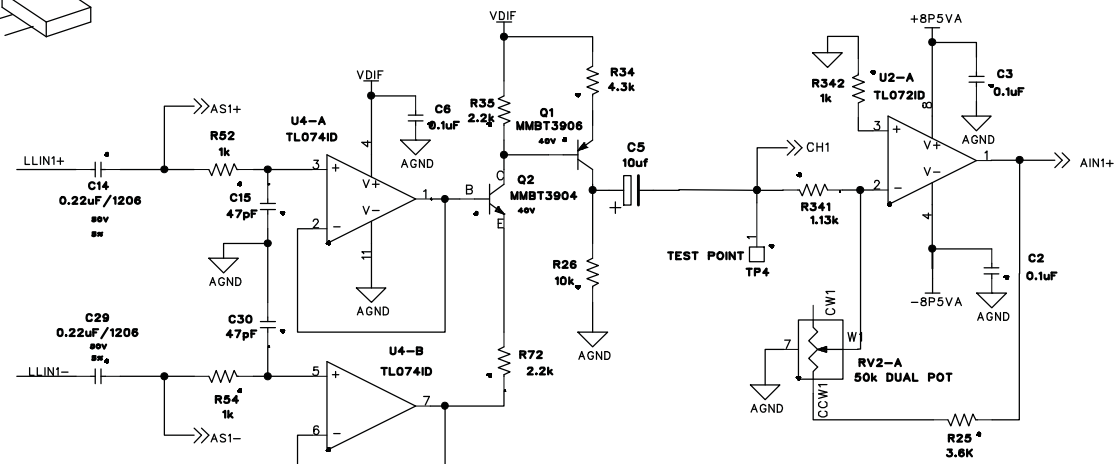
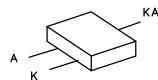
Date: Jan, 11, 2011

 Sheet 3 of 15
 1

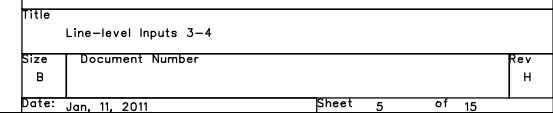
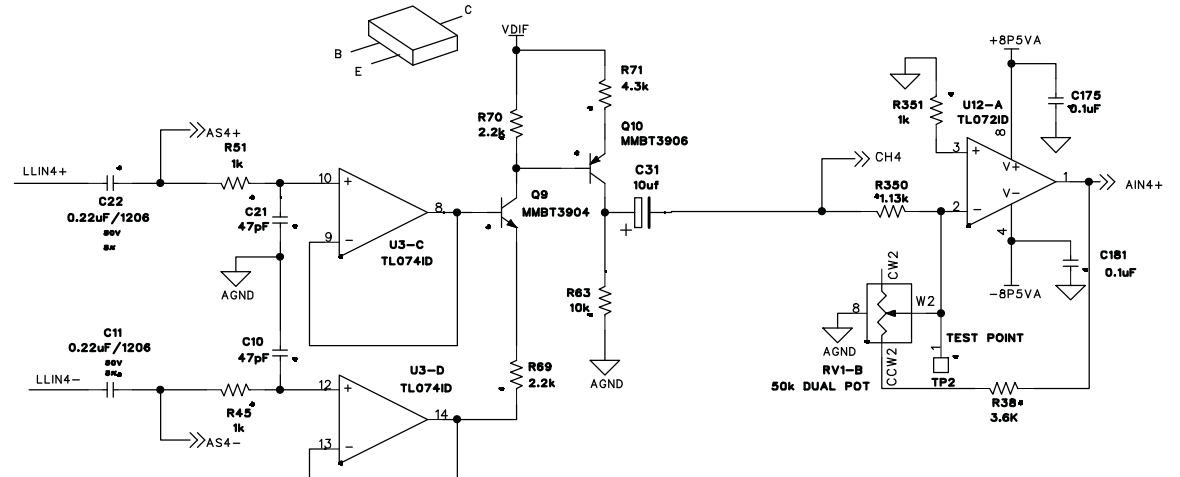
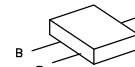
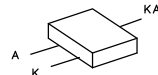
MS-A1004



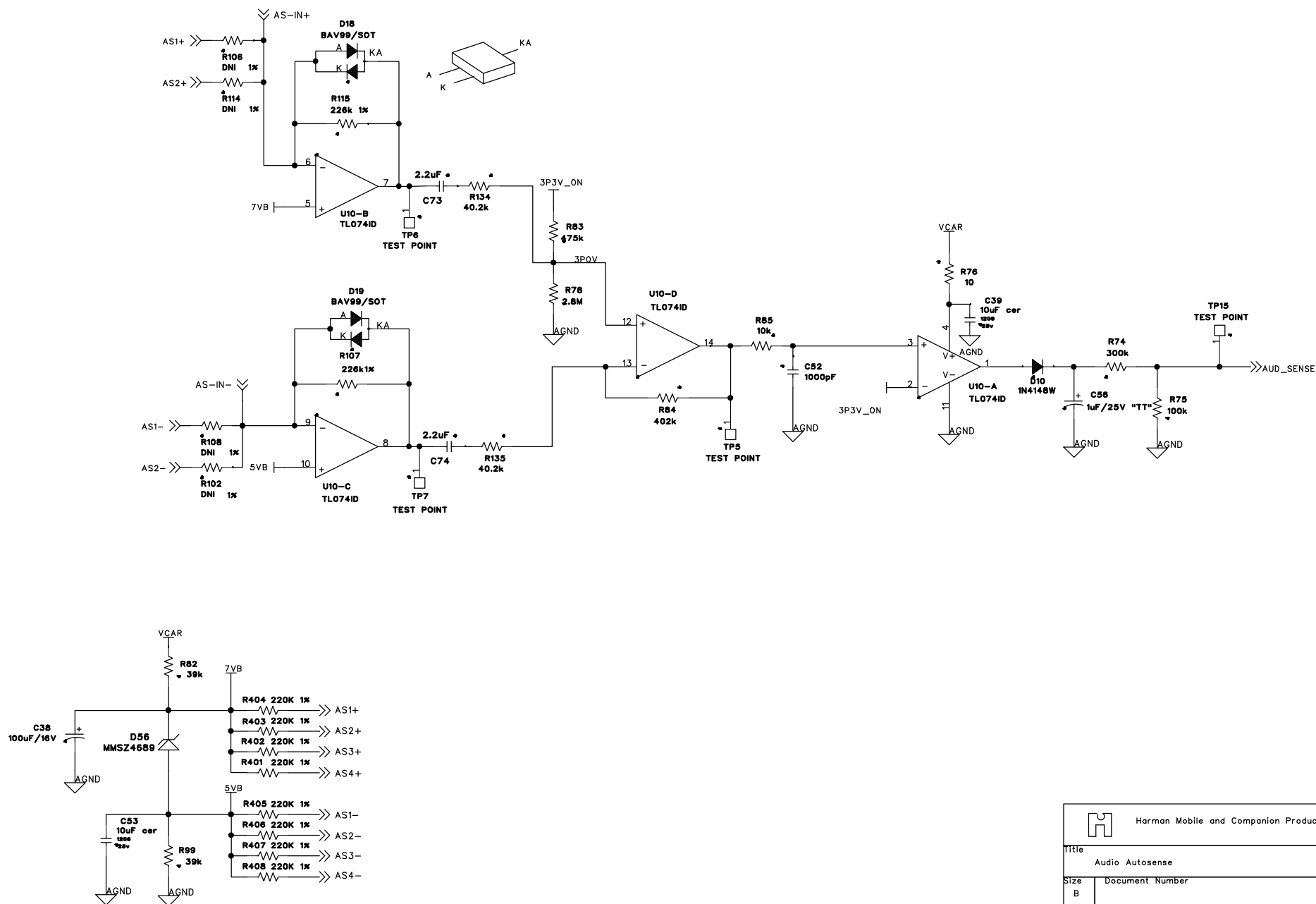
Switch positions
Spkr-level In, 120-ohm input impedance
Spkr-level In, 100k-ohm input impedance
Line-level In, 10k-ohm input impedance



 Harman Mobile and Companion Products Group		
Title Line-level Inputs 1-2		
Size B	Document Number	
Date: Jan, 11, 2011	Sheet 4	of 15



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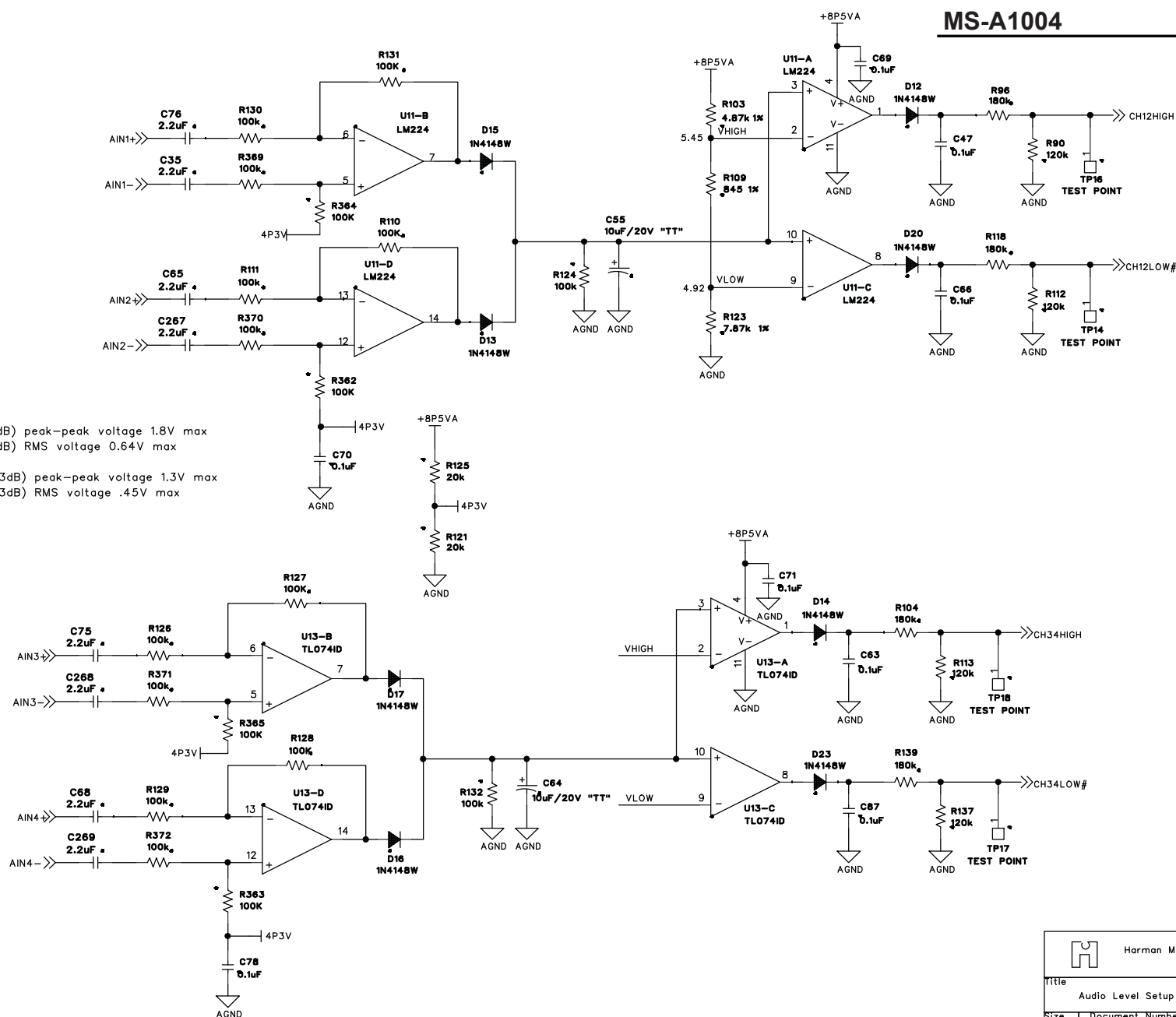
Harman Mobile and Companion Products Group		
Title Audio Autosense		
Size B	Document Number	Rev H
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Upper limit (0dB) peak-peak voltage 1.8V max
Upper limit (0dB) RMS voltage 0.64V max

Lower limit (-3dB) peak-peak voltage 1.3V max
Lower limit (-3dB) RMS voltage .45V max



Harman Mobile and Companion Products Group

Title
Audio Level Setup

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H

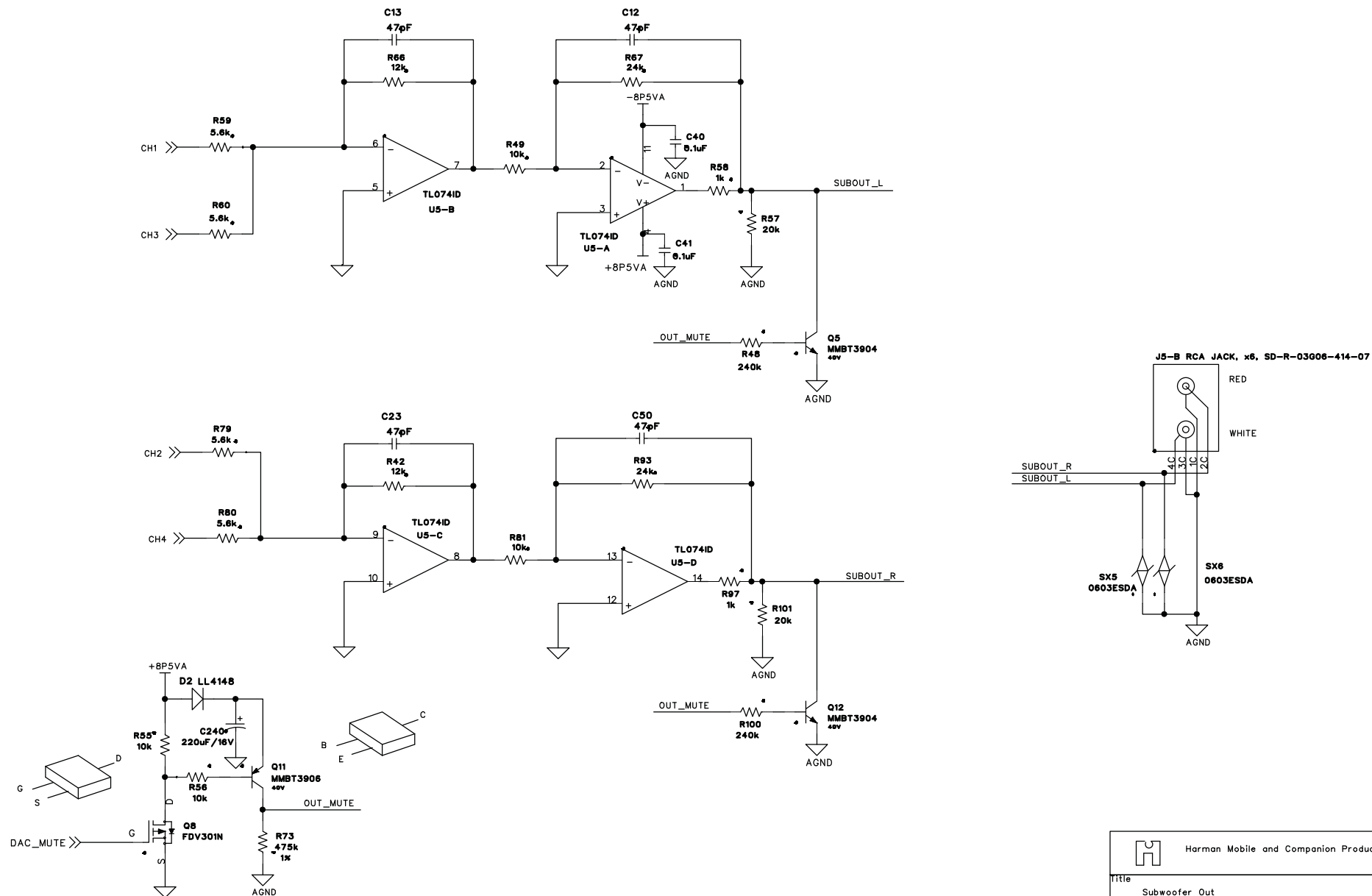


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Input = 0.63Vrms = 1.8Vp-p

Voltage gain (Gv) =
 $1 + 6.8k / 3.3k = 3.1 = +9.7dB$

Output = 2.0Vrms = 5.6Vp-p



Harman Mobile and Companion Products Group		
Title Subwoofer Out		
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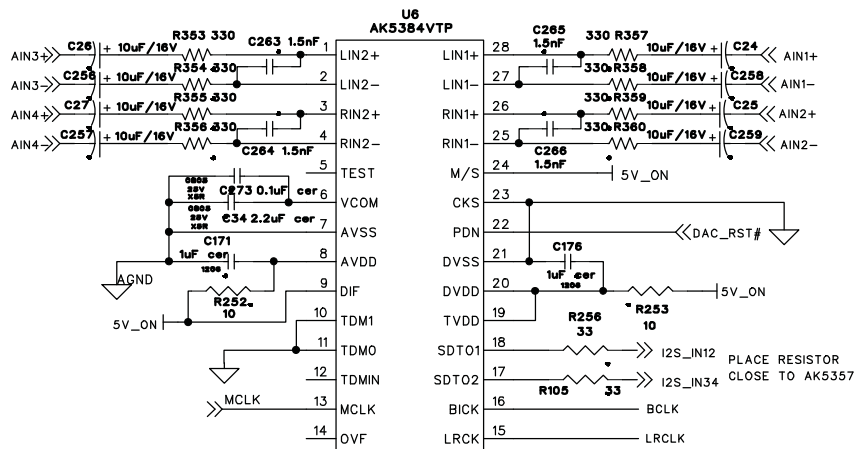
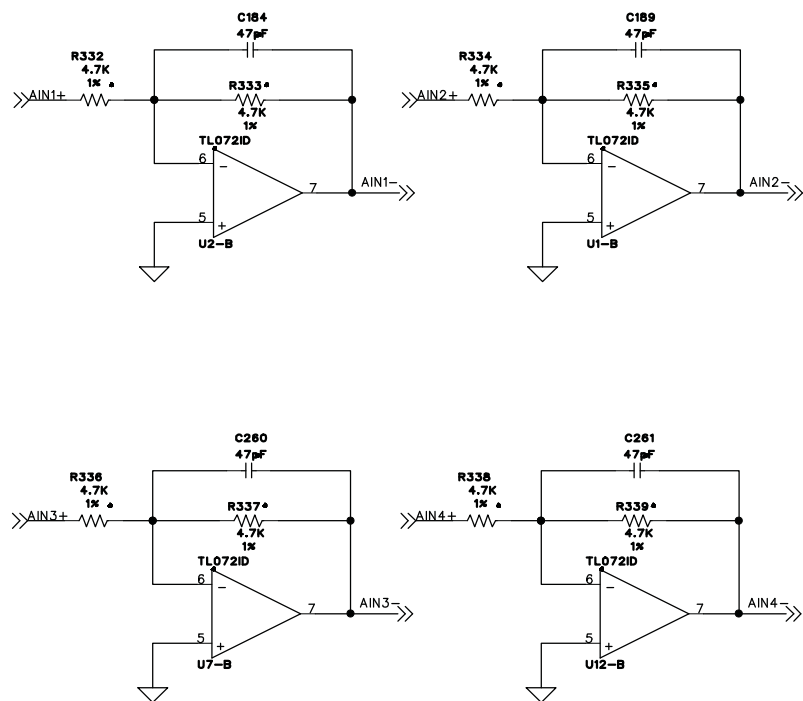


AK5384

Input peak-peak voltage $\pm 2.9V$ typ

Input RMS voltage $\pm 1.025V$ typ

Input impedance 18k min, 26k typ

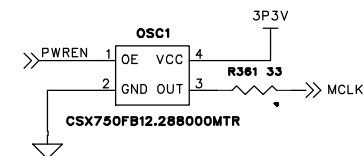


MCLK = 12.288 MHz $\rightarrow$ MCLK

BCLK = 3.072 MHz $\rightarrow$ BCLK

LRCLK = 48 kHz $\rightarrow$ LRCLK

USE DAISY CHAIN
ROUTING FOR
THESE TRACES



Harman Mobile and Companion Products Group

Title

ADCs

Size

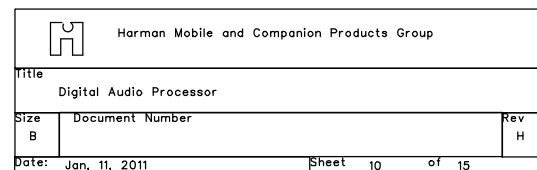
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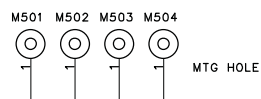
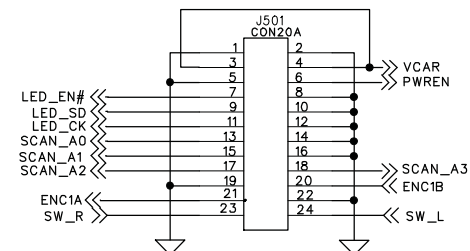
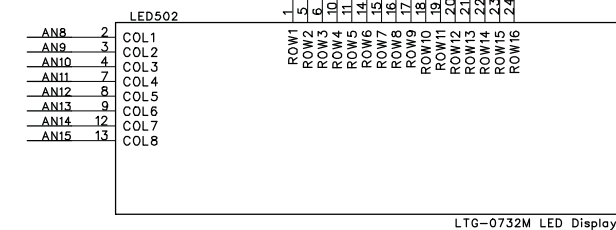
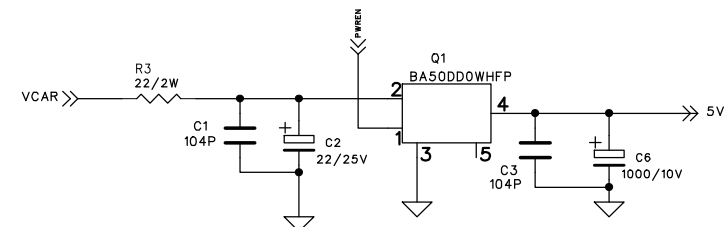
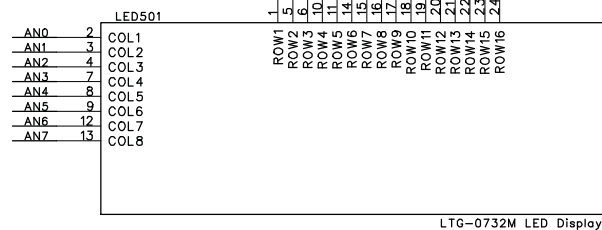
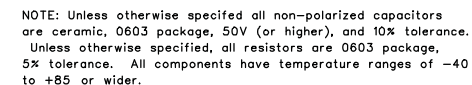
Rev

H

Date: Jan, 11, 2011

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P

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Title	User Interface
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Size	Document Number
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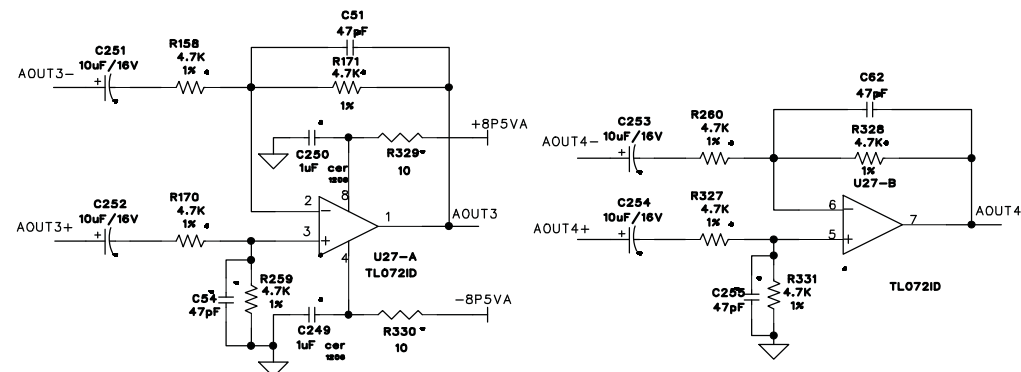
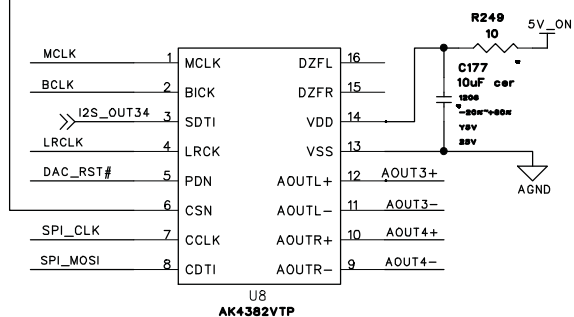
Rev	
C	

Date: Tuesday, February 26, 2008

Sheet 1 of 2



Output peak-peak voltage $\pm 2.55\text{V}$ min, $\pm 2.75\text{V}$ typ, $\pm 2.95\text{V}$ max
Output RMS voltage $\pm 1.8\text{V}$ min, $\pm 1.94\text{V}$ typ, $\pm 2.08\text{V}$ max
Idd is 20mA each in normal operation, 10uA in reset



—>> AOUT1
—>> AOUT2
—>> AOUT3
—>> AOUT4



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JBL

